

Furthermore, these results (along with results in [Baker and Selman, 1976]) can be used to make similar statements about the likelihood of answering questions about the polynomial hierarchy using currently available techniques, and much the same can be said about the P vs. PSPACE question. (In the case of space, however, there is some question as to what is an appropriate model for relativization results — see [Ladner and Lynch, 1976] and [Lynch, 1978].)

We conclude that substantially new proof techniques will probably be required to resolve the P vs. NP question, as well as the other related questions we have mentioned. That is, *if* we can resolve them. Hartmanis and Hopcroft [1976] use relativization techniques like those mentioned above to indicate at least the possibility that questions like this one might be independent of set theory, and hence not resolvable in our standard system of logic. Although we and other researchers in this area are not quite so pessimistic about the ultimate resolvability of the P vs. NP question, no one expects the answer to come very soon.

Appendix:

A List of NP-Complete Problems

It has now become more or less standard practice to address questions of NP-completeness (or NP-hardness) whenever a problem is studied from a computational point of view. As a consequence, many NP-completeness results have appeared in the technical literature, and perhaps an even larger number have been obtained but not published. In this appendix we collect together a large number of these results, presenting an extensive list of known NP-complete and NP-hard problems.

The list contains more than 300 main entries, and our comments on these extend the total number of results in the list to several times this many. A typical entry consists of four parts: (1) a problem name, (2) a problem definition, (3) a reference for the result, and (4) a section of additional comments.

The problem name is intended as a convenient shorthand for referring to the problem. For reasons of consistency, we have on occasion assigned names that differ slightly from those in previous usage. An asterisk in parentheses “(*)” following a name indicates that the problem is not known to belong to NP, so the claimed result should be taken as one of NP-hardness rather than NP-completeness.

Problem definitions are given in the format used throughout this book, by specifying a generic *instance* and a *question* asked about that instance. (We restrict our attention to decision problems for main entries, although other types of problems are sometimes mentioned in the comments.) In most cases we have been able to give a set-theoretic definition that describes the problem completely, although the need for conciseness has often led to some obscuring of the intuition behind the problem; the reference can be consulted in such cases. Certain terms that are widely known have not been defined, and lengthy definitions common to an entire

group of problems have been given only once, the first time the need for them arises. In a few cases the complete problem definition is so complex, or requires so much background information, that we have merely provided a sketchy definition using the appropriate “buzzwords,” explicitly referring the reader to the references for details.

The *reference* section includes both a bibliographical citation for the main result and an indication of an appropriate “known NP-complete (or NP-hard) problem” that can be used for proving the result. Problems mentioned in this respect have been chosen on the basis of our own view of the most natural proof, and often come from proofs cited in the comments or from unpublished proofs of our own, rather than from the main citation. Several results have been deemed so straightforward that no one has been assigned the credit (or blame) for them.

The *comments* section, present for most entries, elaborates on the main result. This normally takes the form of citing various results concerning the polynomial time solvability and NP-hardness of subproblems and variants of the main problem. Thus the list can serve as a useful source of information on what can and cannot be solved in polynomial time, assuming $P \neq NP$. However, we have not attempted to cite the “best” polynomial time algorithms for problems, but merely to justify our claim that they can be solved in polynomial time. Whenever appropriate, we also mention provably intractable and undecidable generalizations and variants of the main problem. For main entries not known to be in NP, we provide any additional complexity information that is known, such as whether the problem is PSPACE-complete or whether the corresponding enumeration problem is #P-complete.

The list itself is organized according to subject matter. The entries are divided into the following twelve categories:

- A1 Graph Theory
- A2 Network Design
- A3 Sets and Partitions
- A4 Storage and Retrieval
- A5 Sequencing and Scheduling
- A6 Mathematical Programming
- A7 Algebra and Number Theory
- A8 Games and Puzzles
- A9 Logic
- A10 Automata and Languages
- A11 Program Optimization
- A12 Miscellaneous

The problems within each category are numbered individually, each number being preceded by a two-letter abbreviation derived from the section heading (such as GT7 for the seventh problem under Graph Theory). A thirteenth section is devoted to a small collection of open problems, selected

primarily on the basis of our own judgment of their importance and difficulty. Additional open problems can be found scattered throughout the comments sections in the main list, and still more should be apparent from obvious gaps in the list.

Many of the problems in the list could have been placed in any one of a number of categories, so the reader is advised to make a general perusal of the list in order to get some idea of where particular types of problems can be found. One should also keep in mind that many results are mentioned only in the comments sections for other problems, so these should be examined carefully. We have attempted to provide helpful “navigational” information in the index.

Finally, it is appropriate that we make some mention of the standards and methods used in compiling this list. As with all such compilations, the collection is somewhat idiosyncratic, but we have attempted to be moderately thorough. In addition to plundering the published literature, we have sought, by word of mouth and by advertisements in appropriate periodicals and at relevant meetings, to make it widely known that we were compiling such a list and were interested in receiving any relevant unpublished results. However, no result has been included on the basis of an unsupported claim. Those that have not been published are backed up in our files by a manuscript or a proof sketch, checked either by ourselves or by other “trusted experts.” On occasion we have verified the result, rather than the proof, by coming up with our own alternative proof (this is often easier than checking someone else’s proof).

A certain amount of selectivity has been exercised in deciding what to exclude (although we may still be accused of including some rather obscure problems). For the most part, we have ignored problems with hopelessly convoluted descriptions unless they seem likely to be of interest to the relevant experts, although we usually do include pointers to the references where such results are presented. We also have felt free to exclude problems whose NP-completeness follows more or less immediately from that of other problems on the list, unless the problem is of substantial independent interest or has subproblems that we would like to comment on. Thus the list should not be regarded as an encyclopedia containing all known NP-completeness results. Rather, it is part annotated bibliography, serving as an access point to the literature on NP-completeness, and part data base, providing a large collection of known NP-complete problems that can be used for proving other problems NP-complete.

A1 GRAPH THEORY

A1.1 COVERING AND PARTITIONING

[GT1] VERTEX COVER

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is there a vertex cover of size K or less for G , i.e., a subset $V' \subseteq V$ with $|V'| \leq K$ such that for each edge $\{u, v\} \in E$ at least one of u and v belongs to V' ?

Reference: [Karp, 1972]. Transformation from 3SAT (see Chapter 3).

Comment: Equivalent complexity to INDEPENDENT SET with respect to restrictions on G . Variation in which the subgraph induced by V' is required to be connected is also NP-complete, even for planar graphs with no vertex degree exceeding 4 [Garey and Johnson, 1977a]. Easily solved in polynomial time if V' is required to be both a vertex cover and an independent set for G . The related EDGE COVER problem, in which one wants the smallest set $E' \subseteq E$ such that every $v \in V$ belongs to at least one $e \in E'$, can be solved in polynomial time by graph matching (e.g., see [Lawler, 1976a]).

[GT2] DOMINATING SET

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is there a dominating set of size K or less for G , i.e., a subset $V' \subseteq V$ with $|V'| \leq K$ such that for all $u \in V - V'$ there is a $v \in V'$ for which $\{u, v\} \in E$?

Reference: Transformation from VERTEX COVER.

Comment: Remains NP-complete for planar graphs with maximum vertex degree 3 and planar graphs that are regular of degree 4 [Garey and Johnson, —]. Variation in which the subgraph induced by V' is required to be connected is also NP-complete, even for planar graphs that are regular of degree 4 [Garey and Johnson, —]. Also NP-complete if V' is required to be both a dominating set and an independent set. Solvable in polynomial time for trees [Cockayne, Goodman, and Hedetniemi, 1975]. The related EDGE DOMINATING SET problem, where we ask for a set $E' \subseteq E$ of K or fewer edges such that every edge in E shares at least one endpoint with some edge in E' , is NP-complete, even for planar or bipartite graphs of maximum degree 3, but can be solved in polynomial time for trees [Yannakakis and Gavril, 1978], [Mitchell and Hedetniemi, 1977].

[GT3] DOMATIC NUMBER

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is the domatic number of G at least K , i.e., can V be partitioned into $k \geq K$ disjoint sets $V_1, V_2, \dots, V_k$ such that each V_i is a dominating set for G ?

Reference: [Garey, Johnson, and Tarjan, 1976b]. Transformation from 3SAT. The problem is discussed in [Cockayne and Hedetniemi, 1975].

Comment: Remains NP-complete for any fixed $K \geq 3$. (The domatic number is always at least 2 unless G contains an isolated vertex.)

[GT4] GRAPH K -COLORABILITY (CHROMATIC NUMBER)

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is G K -colorable, i.e., does there exist a function $f: V \rightarrow \{1, 2, \dots, K\}$ such that $f(u) \neq f(v)$ whenever $\{u, v\} \in E$?

Reference: [Karp, 1972]. Transformation from 3SAT.

Comment: Solvable in polynomial time for $K=2$, but remains NP-complete for all fixed $K \geq 3$ and, for $K=3$, for planar graphs having no vertex degree exceeding 4 [Garey, Johnson, and Stockmeyer, 1976]. Also remains NP-complete for $K=3$ if G is an intersection graph for straight line segments in the plane [Ehrlich, Even, and Tarjan, 1976]. For arbitrary K , the problem is NP-complete for circle graphs and circular arc graphs (even given their representation as families of arcs), although for circular arc graphs the problem is solvable in polynomial time for any fixed K (given their representation) [Garey, Johnson, Miller, and Papadimitriou, 1978]. The general problem can be solved in polynomial time for comparability graphs [Even, Pnueli, and Lempel, 1972], for chordal graphs [Gavril, 1972], for (3,1) graphs [Walsh and Burkhard, 1977], and for graphs having no vertex degree exceeding 3 [Brooks, 1941].

[GT5] ACHROMATIC NUMBER

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Does G have achromatic number K or greater, i.e., is there a partition of V into disjoint sets $V_1, V_2, \dots, V_k$, $k \geq K$, such that each V_i is an independent set for G (no two vertices in V_i are joined by an edge in E) and such that, for each pair of distinct sets V_i, V_j , $V_i \cup V_j$ is *not* an independent set for G ?

Reference: [Yannakakis and Gavril, 1978]. Transformation from MINIMUM MAXIMAL MATCHING.

Comment: Remains NP-complete even if G is the complement of a bipartite graph and hence has no independent set of more than two vertices.

[GT6] MONOCHROMATIC TRIANGLE

INSTANCE: Graph $G = (V, E)$.

QUESTION: Is there a partition of E into two disjoint sets E_1, E_2 such that neither $G_1 = (V, E_1)$ nor $G_2 = (V, E_2)$ contains a triangle?

Reference: [Burr, 1976]. Transformation from 3SAT.

Comment: Variants in which "triangle" is replaced by any larger fixed complete graph are also NP-complete [Burr, 1976]. Variants in which "triangle" is replaced by " k -star" (a single degree k vertex adjacent to k degree one vertices) is solvable in polynomial time [Burr, Erdős, and Lovasz, 1976].

[GT7] FEEDBACK VERTEX SET

INSTANCE: Directed graph $G = (V, A)$, positive integer $K \leq |V|$.

QUESTION: Is there a subset $V' \subseteq V$ with $|V'| \leq K$ such that V' contains at least one vertex from every directed cycle in G ?

Reference: [Karp, 1972]. Transformation from VERTEX COVER.

Comment: Remains NP-complete for digraphs having no in- or out-degree exceeding 2, for planar digraphs with no in- or out-degree exceeding 3 [Garey and

Johnson, —], and for edge digraphs [Gavril, 1977a], but can be solved in polynomial time for reducible graphs [Shamir, 1977]. The corresponding problem for undirected graphs is also NP-complete.

[GT8] FEEDBACK ARC SET

INSTANCE: Directed graph $G = (V, A)$, positive integer $K \leq |A|$.

QUESTION: Is there a subset $A' \subseteq A$ with $|A'| \leq K$ such that A' contains at least one arc from every directed cycle in G ?

Reference: [Karp, 1972]. Transformation from VERTEX COVER.

Comment: Remains NP-complete for digraphs in which no vertex has total in-degree and out-degree more than 3, and for edge digraphs [Gavril, 1977a]. Solvable in polynomial time for planar digraphs [Luchesi, 1976]. The corresponding problem for undirected graphs is trivially solvable in polynomial time.

[GT9] PARTIAL FEEDBACK EDGE SET

INSTANCE: Graph $G = (V, E)$, positive integers $K \leq |E|$ and $L \leq |V|$.

QUESTION: Is there a subset $E' \subseteq E$ with $|E'| \leq K$ such that E' contains at least one edge from every circuit of length L or less in G ?

Reference: [Yannakakis, 1978b]. Transformation from VERTEX COVER.

Comment: Remains NP-complete for any fixed $L \geq 3$ and for bipartite graphs (with fixed $L \geq 4$). However, if $L = |V|$, i.e., if we ask that E' contain an edge from every cycle in G , then the problem is trivially solvable in polynomial time.

[GT10] MINIMUM MAXIMAL MATCHING

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |E|$.

QUESTION: Is there a subset $E' \subseteq E$ with $|E'| \leq K$ such that E' is a maximal matching, i.e., no two edges in E' share a common endpoint and every edge in $E - E'$ shares a common endpoint with some edge in E' ?

Reference: [Yannakakis and Gavril, 1978]. Transformation from VERTEX COVER for cubic graphs.

Comment: Remains NP-complete for planar graphs and for bipartite graphs, in both cases even if no vertex degree exceeds 3. The problem of finding a maximum "maximal matching" is just the usual graph matching problem and is solvable in polynomial time (e.g., see [Lawler, 1976a]).

[GT11] PARTITION INTO TRIANGLES

INSTANCE: Graph $G = (V, E)$, with $|V| = 3q$ for some integer q .

QUESTION: Can the vertices of G be partitioned into q disjoint sets $V_1, V_2, \dots, V_q$, each containing exactly 3 vertices, such that for each $V_i = \{u_i, v_i, w_i\}$, $1 \leq i \leq q$, all three of the edges $\{u_i, v_i\}$, $\{u_i, w_i\}$, and $\{v_i, w_i\}$ belong to E ?

Reference: [Schaefer, 1974]. Transformation from 3DM (see Chapter 3).

Comment: See next problem for a generalization.

[GT12] PARTITION INTO ISOMORPHIC SUBGRAPHS

INSTANCE: Graphs $G = (V, E)$ and $H = (V', E')$ with $|V| = q|V'|$ for some $q \in \mathbb{Z}^+$.

QUESTION: Can the vertices of G be partitioned into q disjoint sets $V_1, V_2, \dots, V_q$ such that, for $1 \leq i \leq q$, the subgraph of G induced by V_i is isomorphic to H ?

Reference: [Kirkpatrick and Hell, 1978]. Transformation from 3DM.

Comment: Remains NP-complete for any fixed H that contains at least 3 vertices. The analogous problem in which the subgraph induced by V_i need only have the same number of vertices as H and contain a subgraph isomorphic to H is also NP-complete, for any fixed H that contains a connected component of three or more vertices. Both problems can be solved in polynomial time (by matching) for any H not meeting the stated restrictions.

[GT13] PARTITION INTO HAMILTONIAN SUBGRAPHS

INSTANCE: Directed graph $G = (V, A)$.

QUESTION: Can the vertices of G be partitioned into disjoint sets $V_1, V_2, \dots, V_k$, for some k , such that each V_i contains at least three vertices and induces a subgraph of G that contains a Hamiltonian circuit?

Reference: [Valiant, 1977a]. Transformation from 3SAT. (See also [Herrmann, 1973]).

Comment: Solvable in polynomial time by matching techniques if each V_i need only contain at least 2 vertices [Edmonds and Johnson, 1970]. The analogous problem for undirected graphs can be similarly solved, even with the requirement that $|V_i| \geq 3$. However, it becomes NP-complete if we require that $|V_i| \geq 6$ [Papadimitriou, 1978d] or if the instance includes an upper bound K on k .

[GT14] PARTITION INTO FORESTS

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Can the vertices of G be partitioned into $k \leq K$ disjoint sets $V_1, V_2, \dots, V_k$ such that, for $1 \leq i \leq k$, the subgraph induced by V_i contains no circuits?

Reference: [Garey and Johnson, —]. Transformation from GRAPH 3-COLORABILITY.

[GT15] PARTITION INTO CLIQUES

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Can the vertices of G be partitioned into $k \leq K$ disjoint sets $V_1, V_2, \dots, V_k$ such that, for $1 \leq i \leq k$, the subgraph induced by V_i is a complete graph?

Reference: [Karp, 1972] (there called CLIQUE COVER). Transformation from GRAPH K-COLORABILITY.

Comment: Remains NP-complete for edge graphs [Arjomandi, 1977], for graphs containing no complete subgraphs on 4 vertices (see construction for PARTITION INTO TRIANGLES in Chapter 3), and for all fixed $K \geq 3$. Solvable in polynomial time for $K \leq 2$, for graphs containing no complete subgraphs on 3 vertices (by

matching), for circular arc graphs (given their representations as families of arcs) [Gavril, 1974a], for chordal graphs [Gavril, 1972], and for comparability graphs [Golumbic, 1977].

[GT16] PARTITION INTO PERFECT MATCHINGS

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Can the vertices of G be partitioned into $k \leq K$ disjoint sets $V_1, V_2, \dots, V_k$ such that, for $1 \leq i \leq k$, the subgraph induced by V_i is a perfect matching (consists entirely of vertices with degree one)?

Reference: [Schaefer, 1978b]. Transformation from NOT-ALL-EQUAL 3SAT.

Comment: Remains NP-complete for $K=2$ and for planar cubic graphs.

[GT17] COVERING BY CLIQUES

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |E|$.

QUESTION: Are there $k \leq K$ subsets $V_1, V_2, \dots, V_k$ of V such that each V_i induces a complete subgraph of G and such that for each edge $\{u, v\} \in E$ there is some V_i that contains both u and v ?

Reference: [Kou, Stockmeyer, and Wong, 1978], [Orlin, 1976]. Transformation from PARTITION INTO CLIQUES.

[GT18] COVERING BY COMPLETE BIPARTITE SUBGRAPHS

INSTANCE: Bipartite graph $G = (V, E)$, positive integer $K \leq |E|$.

QUESTION: Are there $k \leq K$ subsets $V_1, V_2, \dots, V_k$ of V such that each V_i induces a complete bipartite subgraph of G and such that for each edge $\{u, v\} \in E$ there is some V_i that contains both u and v ?

Reference: [Orlin, 1976]. Transformation from PARTITION INTO CLIQUES.

A1.2 SUBGRAPHS AND SUPERGRAPHS

[GT19] CLIQUE

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Does G contain a clique of size K or more, i.e., a subset $V' \subseteq V$ with $|V'| \geq K$ such that every two vertices in V' are joined by an edge in E ?

Reference: [Karp, 1972]. Transformation from VERTEX COVER (see Chapter 3).

Comment: Solvable in polynomial time for graphs obeying any fixed degree bound d , for planar graphs, for edge graphs, for chordal graphs [Gavril, 1972], for comparability graphs [Even, Pnueli, and Lempel, 1972], for circle graphs [Gavril, 1973], and for circular arc graphs (given their representation as families of arcs) [Gavril, 1974a]. The variant in which, for a given r , $0 < r < 1$, we are asked whether G contains a clique of size $r|V|$ or more is NP-complete for any fixed value of r .

[GT20] INDEPENDENT SET

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Does G contain an independent set of size K or more, i.e., a subset

$V' \subseteq V$ such that $|V'| \geq K$ and such that no two vertices in V' are joined by an edge in E ?

Reference: Transformation from VERTEX COVER (see Chapter 3).

Comment: Remains NP-complete for cubic planar graphs [Garey, Johnson, and Stockmeyer, 1976], [Garey and Johnson, 1977a], [Maier and Storer, 1977], for edge graphs of directed graphs [Gavril, 1977a], for total graphs of bipartite graphs [Yannakakis and Gavril, 1978], and for graphs containing no triangles [Poljak, 1974]. Solvable in polynomial time for bipartite graphs (by matching, e.g., see [Harary, 1969]), for edge graphs (by matching), for graphs with no vertex degree exceeding 2, for chordal graphs [Gavril, 1972], for circle graphs [Gavril, 1973], for circular arc graphs (given their representation as families of arcs) [Gavril, 1974a], for comparability graphs [Golumbic, 1977], and for claw-free graphs [Minty, 1977].

[GT21] INDUCED SUBGRAPH WITH PROPERTY Π (*)

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is there a subset $V' \subseteq V$ with $|V'| \geq K$ such that the subgraph of G induced by V' has property Π (see comments for possible choices for Π)?

Reference: [Yannakakis, 1978a], [Yannakakis, 1978b], [Lewis, 1978]. Transformation from 3SAT.

Comment: NP-hard for any property Π that holds for arbitrarily large graphs, does not hold for all graphs, and is "hereditary," i.e., holds for all induced subgraphs of G whenever it holds for G . If in addition one can determine in polynomial time whether Π holds for a graph, then the problem is NP-complete. Examples of such properties Π include " G is a clique," " G is an independent set," " G is planar," " G is bipartite," " G is outerplanar," " G is an edge graph," " G is chordal," " G is a comparability graph," and " G is a forest." The same general results hold if G is restricted to planar graphs and Π satisfies the above constraints for planar graphs, or if G is restricted to acyclic directed graphs and Π satisfies the above constraints for such graphs. A weaker result holds when G is restricted to bipartite graphs [Yannakakis, 1978b].

[GT22] INDUCED CONNECTED SUBGRAPH WITH PROPERTY Π (*)

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is there a subset $V' \subseteq V$ with $|V'| \geq K$ such that the subgraph of G induced by V' is connected and has property Π (see comments for possible choices for Π)?

Reference: [Yannakakis, 1978b]. Transformation from 3SAT.

Comment: NP-hard for any hereditary property that holds for arbitrarily large connected graphs but not for all connected graphs. If, in addition, one can determine in polynomial time whether Π holds for a graph, then the problem is NP-complete. Examples include all the properties mentioned for the preceding problem except " G is an independent set". The related question "Is the maximum induced subgraph of G having property Π also connected?" is not in NP or co-NP unless NP = co-NP [Yannakakis, 1978b].

[GT23] INDUCED PATH

INSTANCE: Graph $G=(V,E)$, positive integer $K \leq |V|$.

QUESTION: Is there a subset $V' \subseteq V$ with $|V'| \geq K$ such that the subgraph induced by V' is a simple path on $|V'|$ vertices?

Reference: [Yannakakis, 1978c]. Transformation from 3SAT.

Comment: Note that this is not a hereditary property, so the result is not implied by either of the previous two results. Remains NP-complete if G is bipartite. The same result holds for the variant in which "simple path" is replaced by "simple cycle." The problems of finding the longest simple path or longest simple cycle (not necessarily induced) are also NP-complete.

[GT24] BALANCED COMPLETE BIPARTITE SUBGRAPH

INSTANCE: Bipartite graph $G=(V,E)$, positive integer $K \leq |V|$.

QUESTION: Are there two disjoint subsets $V_1, V_2 \subseteq V$ such that $|V_1|=|V_2|=K$ and such that $u \in V_1, v \in V_2$ implies that $\{u,v\} \in E$?

Reference: [Garey and Johnson, —]. Transformation from CLIQUE.

Comment: The related problem in which the requirement " $|V_1|=|V_2|=K$ " is replaced by " $|V_1|+|V_2|=K$ " is solvable in polynomial time for bipartite graphs (because of the connection between matchings and independent sets in such graphs, e.g., see [Harary, 1969]), but is NP-complete for general graphs [Yannakakis, 1978b].

[GT25] BIPARTITE SUBGRAPH

INSTANCE: Graph $G=(V,E)$, positive integer $K \leq |E|$.

QUESTION: Is there a subset $E' \subseteq E$ with $|E'| \geq K$ such that $G'=(V,E')$ is bipartite?

Reference: [Garey, Johnson, and Stockmeyer, 1976]. Transformation from MAXIMUM 2-SATISFIABILITY.

Comment: Remains NP-complete for graphs with no vertex degree exceeding 3 and no triangles and/or if we require that the subgraph be connected [Yannakakis, 1978b]. Solvable in polynomial time if G is planar [Hadlock, 1975], [Orlova and Dorfman, 1972], or if $K=|E|$.

[GT26] DEGREE-BOUNDED CONNECTED SUBGRAPH

INSTANCE: Graph $G=(V,E)$, non-negative integer $d \leq |V|$, positive integer $K \leq |E|$.

QUESTION: Is there a subset $E' \subseteq E$ with $|E'| \geq K$ such that the subgraph $G'=(V,E')$ is connected and has no vertex with degree exceeding d ?

Reference: [Yannakakis, 1978b]. Transformation from HAMILTONIAN PATH.

Comment: Remains NP-complete for any fixed $d \geq 2$. Solvable in polynomial time if G' is not required to be connected (by matching techniques, see [Edmonds and Johnson, 1970]). The corresponding induced subgraph problem, where we ask for a subset $V' \subseteq V$ with $|V'| \geq K$ such that the subgraph of G induced by V' has no vertex with degree exceeding d , is NP-complete for any fixed $d \geq 0$ [Lewis, 1976] and for any fixed $d \geq 2$ if we require that G' be connected [Yannakakis, 1978b].

[GT27] PLANAR SUBGRAPH

INSTANCE: Graph $G=(V,E)$, positive integer $K \leq |E|$.

QUESTION: Is there a subset $E' \subseteq E$ with $|E'| \geq K$ such that $G'=(V,E')$ is planar?

Reference: [Liu and Goldmacher, 1978]. Transformation from HAMILTONIAN PATH restricted to bipartite graphs.

Comment: Corresponding problem in which G' is the subgraph induced by a set V' of at least K vertices is also NP-complete [Krishnamoorthy and Deo, 1977a], [Yannakakis, 1978b]. The former can be solved in polynomial time when $K=|E|$, and the latter when $K=|V|$, since planarity testing can be done in polynomial time (e.g., see [Hopcroft and Tarjan, 1974]). The related problem in which we ask if G contains a connected "outerplanar" subgraph with K or more edges is also NP-complete [Yannakakis, 1978b].

[GT28] EDGE-SUBGRAPH

INSTANCE: Graph $G=(V,E)$, positive integer $K \leq |E|$.

QUESTION: Is there a subset $E' \subseteq E$ with $|E'| \geq K$ such that the subgraph $G'=(V,E')$ is an edge graph, i.e., there exists a graph $H=(U,F)$ such that G' is isomorphic to the graph having vertex set F and edge set consisting of all pairs $\{e,f\}$ such that the edges e and f share a common endpoint in H ?

Reference: [Yannakakis, 1978b]. Transformation from 3SAT.

Comment: Remains NP-complete even if G has no vertex with degree exceeding 4. If we require that the subgraph be connected, the degree bound for NP-completeness can be reduced to 3. Edge graphs can be recognized in polynomial time, e.g., see [Harary, 1969] (under the term "line graphs").

[GT29] TRANSITIVE SUBGRAPH

INSTANCE: Directed graph $G=(V,A)$, positive integer $K \leq |A|$.

QUESTION: Is there a subset $A' \subseteq A$ with $|A'| \geq K$ such that $G'=(V,A')$ is transitive, i.e., for all pairs $u,v \in V$, if there exists a $w \in V$ for which $(u,w), (w,v) \in A'$, then $(u,v) \in A'$?

Reference: [Yannakakis, 1978b]. Transformation from BIPARTITE SUBGRAPH with no triangles.

Comment: The variant in which G is undirected and we ask for a subgraph that is a "comparability graph," i.e., can be made into a transitive digraph by directing each of its edges in one of the two possible directions, is also NP-complete, even if G has no vertex with degree exceeding 3. For both problems, the variant in which we require the subgraph to be connected is also NP-complete.

[GT30] UNICONNECTED SUBGRAPH

INSTANCE: Directed graph $G=(V,A)$, positive integer $K \leq |A|$.

QUESTION: Is there a subset $A' \subseteq A$ with $|A'| \geq K$ such that $G'=(V,A')$ has at most one directed path between any pair of vertices?

Reference: [Maheshwari, 1976]. Transformation from VERTEX COVER.

Comment: Remains NP-complete for acyclic directed graphs.

[GT31] MINIMUM K-CONNECTED SUBGRAPH

INSTANCE: Graph $G = (V, E)$, positive integers $K \leq |V|$ and $B \leq |E|$.

QUESTION: Is there a subset $E' \subseteq E$ with $|E'| \leq B$ such that $G' = (V, E')$ is K -connected, i.e., cannot be disconnected by removing fewer than K vertices?

Reference: [Chung and Graham, 1977]. Transformation from HAMILTONIAN CIRCUIT.

Comment: Corresponding edge-connectivity problem is also NP-complete. Both problems remain NP-complete for any fixed $K \geq 2$ and can be solved trivially in polynomial time for $K = 1$.

[GT32] CUBIC SUBGRAPH

INSTANCE: Graph $G = (V, E)$.

QUESTION: Is there a nonempty subset $E' \subseteq E$ such that in the graph $G' = (V, E')$ every vertex has either degree 3 or degree 0?

Reference: [Chvátal, 1976]. Transformation from GRAPH 3-COLORABILITY.

[GT33] MINIMUM EQUIVALENT DIGRAPH

INSTANCE: Directed graph $G = (V, A)$, positive integer $K \leq |A|$.

QUESTION: Is there a subset $A' \subseteq A$ with $|A'| \leq K$ such that, for every ordered pair of vertices $u, v \in V$, the graph $G' = (V, A')$ contains a directed path from u to v if and only if G does?

Reference: [Sahni, 1974]. Transformation from DIRECTED HAMILTONIAN CIRCUIT for strongly connected graphs (see Chapter 3).

Comment: Corresponding problem in which $A' \subseteq V \times V$ instead of $A' \subseteq A$ (called TRANSITIVE REDUCTION) can be solved in polynomial time, e.g., see [Aho, Garey, and Ullman, 1972].

[GT34] HAMILTONIAN COMPLETION

INSTANCE: Graph $G = (V, E)$, non-negative integer $K \leq |V|$.

QUESTION: Is there a superset E' containing E such that $|E' - E| \leq K$ and the graph $G' = (V, E')$ has a Hamiltonian circuit?

Reference: Transformation from HAMILTONIAN CIRCUIT.

Comment: Remains NP-complete for any fixed $K \geq 0$. Corresponding "completion" versions of HAMILTONIAN PATH, DIRECTED HAMILTONIAN PATH, and DIRECTED HAMILTONIAN CIRCUIT are also NP-complete. HAMILTONIAN COMPLETION and HAMILTONIAN PATH COMPLETION can be solved in polynomial time if G is a tree [Boesch, Chen, and McHugh, 1974].

[GT35] INTERVAL GRAPH COMPLETION

INSTANCE: Graph $G = (V, E)$, non-negative integer K .

QUESTION: Is there a superset E' containing E such that $|E' - E| \leq K$ and the graph $G' = (V, E')$ is an interval graph?

Reference: [Garey, Gavril, and Johnson, 1977]. Transformation from OPTIMAL LINEAR ARRANGEMENT.

Comment: Remains NP-complete when G is restricted to be an edge graph. Solv-

able in polynomial time for $K = 0$ [Fulkerson and Gross, 1965], [Booth and Lueker, 1976].

[GT36] PATH GRAPH COMPLETION

INSTANCE: Graph $G = (V, E)$, non-negative integer K .

QUESTION: Is there a superset E' containing E such that $|E' - E| \leq K$ and the graph $G' = (V, E')$ is the intersection graph of a family of paths on an undirected tree?

Reference: [Gavril, 1977b]. Transformation from INTERVAL GRAPH COMPLETION.

Comment: Corresponding problem in which G' must be the intersection graph of a family of directed paths on an oriented tree (i.e., rooted, with all arcs directed away from the root) is also NP-complete.

A1.3 VERTEX ORDERING**[GT37] HAMILTONIAN CIRCUIT**

INSTANCE: Graph $G = (V, E)$.

QUESTION: Does G contain a Hamiltonian circuit?

Reference: [Karp, 1972]. Transformation from VERTEX COVER (see Chapter 3).

Comment: Remains NP-complete (1) if G is planar, cubic, 3-connected, and has no face with fewer than 5 edges [Garey, Johnson, and Tarjan, 1976a], (2) if G is bipartite [Krishnamoorthy, 1975], (3) if G is the square of a graph [Chvátal, 1976], and (4) if a Hamiltonian path for G is given as part of the instance [Papadimitriou and Stieglitz, 1976]. Solvable in polynomial time if G has no vertex with degree exceeding 2 or if G is an edge graph (e.g., see [Liu, 1968]). The cube of a non-trivial connected graph always has a Hamiltonian circuit [Karaganis, 1968].

[GT38] DIRECTED HAMILTONIAN CIRCUIT

INSTANCE: Directed graph $G = (V, A)$.

QUESTION: Does G contain a directed Hamiltonian circuit?

Reference: [Karp, 1972]. Transformation from VERTEX COVER (see Chapter 3).

Comment: Remains NP-complete if G is planar and has no vertex involved in more than three arcs [Plesnik, 1978]. Solvable in polynomial time if no in-degree (no out-degree) exceeds 1, if G is a tournament [Morrow and Goodman, 1976], or if G is an edge digraph (e.g., see [Liu, 1968]).

[GT39] HAMILTONIAN PATH

INSTANCE: Graph $G = (V, E)$.

QUESTION: Does G contain a Hamiltonian path?

Reference: Transformation from VERTEX COVER (see Chapter 3).

Comment: Remains NP-complete under restrictions (1) and (2) for HAMILTONIAN CIRCUIT and is polynomially solvable under the same restrictions as HC. Corresponding DIRECTED HAMILTONIAN PATH problem is also NP-complete, and the comments for DIRECTED HC apply to it as well. The variants in which ei-

ther the starting point or the ending point or both are specified in the instance are also NP-complete. DIRECTED HAMILTONIAN PATH can be solved in polynomial time for acyclic digraphs, e.g., see [Lawler, 1976a].

[GT40] BANDWIDTH

INSTANCE: Graph $G=(V,E)$, positive integer $K \leq |V|$.

QUESTION: Is there a linear ordering of V with bandwidth K or less, i.e., a one-to-one function $f: V \rightarrow \{1, 2, \dots, |V|\}$ such that, for all $\{u, v\} \in E$, $|f(u) - f(v)| \leq K$?

Reference: [Papadimitriou, 1976a]. Transformation from 3-PARTITION.

Comment: Remains NP-complete for trees with no vertex degree exceeding 3 [Garey, Graham, Johnson, and Knuth, 1978]. This problem corresponds to that of minimizing the "bandwidth" of a symmetric matrix by simultaneous row and column permutations.

[GT41] DIRECTED BANDWIDTH

INSTANCE: Directed graph $G=(V,A)$, positive integer $K \leq |V|$.

QUESTION: Is there a one-to-one function $f: V \rightarrow \{1, 2, \dots, |V|\}$ such that, for all $(u, v) \in A$, $f(u) < f(v)$ and $(f(v) - f(u)) \leq K$?

Reference: [Garey, Graham, Johnson, and Knuth, 1978]. Transformation from 3-PARTITION.

Comment: Remains NP-complete for rooted directed trees with maximum in-degree 1 and maximum out-degree at most 2. This problem corresponds to that of minimizing the "bandwidth" of an upper triangular matrix by simultaneous row and column permutations.

[GT42] OPTIMAL LINEAR ARRANGEMENT

INSTANCE: Graph $G=(V,E)$, positive integer K .

QUESTION: Is there a one-to-one function $f: V \rightarrow \{1, 2, \dots, |V|\}$ such that $\sum_{\{u,v\} \in E} |f(u) - f(v)| \leq K$?

Reference: [Garey, Johnson, and Stockmeyer, 1976]. Transformation from SIMPLE MAX CUT.

Comment: Remains NP-complete if G is bipartite [Even and Shiloach, 1975]. Solvable in polynomial time if G is a tree [Shiloach, 1976], [Gol'dberg and Klipker, 1976].

[GT43] DIRECTED OPTIMAL LINEAR ARRANGEMENT

INSTANCE: Directed graph $G=(V,A)$, positive integer K .

QUESTION: Is there a one-to-one function $f: V \rightarrow \{1, 2, \dots, |V|\}$ such that $f(u) < f(v)$ whenever $(u, v) \in A$ and such that $\sum_{(u,v) \in A} (f(v) - f(u)) \leq K$?

Reference: [Even and Shiloach, 1975]. Transformation from OPTIMAL LINEAR ARRANGEMENT.

Comment: Solvable in polynomial time if G is a tree, even if each edge has a given integer weight and the cost function is a weighted sum [Adolphson and Hu, 1973].

[GT44] MINIMUM CUT LINEAR ARRANGEMENT

INSTANCE: Graph $G=(V,E)$, positive integer K .

QUESTION: Is there a one-to-one function $f: V \rightarrow \{1, 2, \dots, |V|\}$ such that for all i , $1 < i < |V|$,

$$|\{ \{u, v\} \in E : f(u) \leq i < f(v) \}| \leq K ?$$

Reference: [Stockmeyer, 1974b], [Gavril, 1977a]. Transformation from SIMPLE MAX CUT.

[GT45] ROOTED TREE ARRANGEMENT

INSTANCE: Graph $G=(V,E)$, positive integer K .

QUESTION: Is there a rooted tree $T=(U,F)$, with $|U|=|V|$, and a one-to-one function $f: V \rightarrow U$ such that for every edge $\{u, v\} \in E$ there is a simple path from the root that includes both $f(u)$ and $f(v)$ and such that if $d(x, y)$ is the number of edges on the path from x to y in T , then $\sum_{\{u,v\} \in E} d(f(u), f(v)) \leq K$?

Reference: [Gavril, 1977a]. Transformation from OPTIMAL LINEAR ARRANGEMENT.

[GT46] DIRECTED ELIMINATION ORDERING

INSTANCE: Directed graph $G=(V,A)$, non-negative integer K .

QUESTION: Is there an elimination ordering for G with fill-in K or less, i.e., a one-to-one function $f: V \rightarrow \{1, 2, \dots, |V|\}$ such that there are at most K pairs of vertices $(u, v) \in (V \times V) - A$ with the property that G contains a directed path from u to v that only passes through vertices w satisfying $f(w) < \min\{f(u), f(v)\}$?

Reference: [Rose and Tarjan, 1978]. Transformation from 3SAT.

Comment: Problem arises in performing Gaussian elimination on sparse matrices. Solvable in polynomial time for $K=0$. The analogous problem for undirected graphs (symmetric matrices) is equivalent to CHORDAL GRAPH COMPLETION and is open as to complexity.

[GT47] ELIMINATION DEGREE SEQUENCE

INSTANCE: Graph $G=(V,E)$, sequence $\langle d_1, d_2, \dots, d_{|V|} \rangle$ of non-negative integers not exceeding $|V|-1$.

QUESTION: Is there a one-to-one function $f: V \rightarrow \{1, 2, \dots, |V|\}$ such that, for $1 \leq i \leq |V|$, if $f(v) = i$ then there are exactly d_i vertices u such that $f(u) > i$ and $\{u, v\} \in E$?

Reference: [Garey, Johnson, and Papadimitriou, 1977]. Transformation from EXACT COVER BY 3-SETS.

Comment: The variant in which it is required that f be such that, for $1 \leq i \leq |V|$, if $f(v) = i$ then there are exactly d_i vertices u such that $\{u, v\} \in E$, is trivially solvable in polynomial time.

A1.4 ISO- AND OTHER MORPHISMS

[GT48] SUBGRAPH ISOMORPHISM

INSTANCE: Graphs $G = (V_1, E_1)$, $H = (V_2, E_2)$.

QUESTION: Does G contain a subgraph isomorphic to H , i.e., a subset $V \subseteq V_1$ and a subset $E \subseteq E_1$ such that $|V| = |V_2|$, $|E| = |E_2|$, and there exists a one-to-one function $f: V_2 \rightarrow V$ satisfying $\{u, v\} \in E_2$ if and only if $\{f(u), f(v)\} \in E$?

Reference: [Cook, 1971a]. Transformation from CLIQUE.

Comment: Contains CLIQUE, COMPLETE BIPARTITE SUBGRAPH, HAMILTONIAN CIRCUIT, etc., as special cases. Can be solved in polynomial time if G is a forest and H is a tree [Edmonds and Matula, 1975] (see also [Reyner, 1977]), but remains NP-complete if G is a tree and H is a forest (see Chapter 4) or if G is a graph and H is a tree (HAMILTONIAN PATH). Variant for directed graphs is also NP-complete, even if G is acyclic and H is a directed tree [Aho and Sethi, 1977], but can be solved in polynomial time if G is a directed forest and H is a directed tree [Reyner, 1977]. If $|V_1| = |V_2|$ and $|E_1| = |E_2|$ we have the GRAPH ISOMORPHISM problem, which is open for both directed and undirected graphs.

[GT49] LARGEST COMMON SUBGRAPH

INSTANCE: Graphs $G = (V_1, E_1)$, $H = (V_2, E_2)$, positive integer K .

QUESTION: Do there exist subsets $E_1' \subseteq E_1$ and $E_2' \subseteq E_2$ with $|E_1'| = |E_2'| \geq K$ such that the two subgraphs $G' = (V_1, E_1')$ and $H' = (V_2, E_2')$ are isomorphic?

Reference: Transformation from CLIQUE.

Comment: Can be solved in polynomial time if both G and H are trees [Edmonds and Matula, 1975].

[GT50] MAXIMUM SUBGRAPH MATCHING

INSTANCE: Directed graphs $G = (V_1, A_1)$, $H = (V_2, A_2)$, positive integer K .

QUESTION: Is there a subset $R \subseteq V_1 \times V_2$ with $|R| \geq K$ such that, for all $\langle u, u' \rangle, \langle v, v' \rangle \in R$, $(u, v) \in A_1$ if and only if $(u', v') \in A_2$?

Reference: [Garey and Johnson, —]. Transformation from CLIQUE. Problem is discussed in [Barrow and Burstall, 1976].

[GT51] GRAPH CONTRACTABILITY

INSTANCE: Graphs $G = (V_1, E_1)$, $H = (V_2, E_2)$.

QUESTION: Can a graph isomorphic to H be obtained from G by a sequence of edge contractions, i.e., a sequence in which each step replaces two adjacent vertices u, v by a single vertex w adjacent to exactly those vertices that were previously adjacent to at least one of u and v ?

Reference: [Statman, 1976]. Transformation from 3SAT.

Comment: Can be solved in polynomial time if H is a triangle.

[GT52] GRAPH HOMOMORPHISM

INSTANCE: Graphs $G = (V_1, E_1)$, $H = (V_2, E_2)$.

QUESTION: Can a graph isomorphic to H be obtained from G by a sequence of

identifications of non-adjacent vertices, i.e., a sequence in which each step replaces two non-adjacent vertices u, v by a single vertex w adjacent to exactly those vertices that were previously adjacent to at least one of u and v ?

Reference: [Levin, 1973]. Transformation from GRAPH K-COLORABILITY.

Comment: Remains NP-complete for H fixed to be a triangle, but can be solved in polynomial time if H is just a single edge.

[GT53] DIGRAPH D-MORPHISM

INSTANCE: Directed graphs $G = (V_1, A_1)$, $H = (V_2, A_2)$.

QUESTION: Is there a D -morphism from G to H , i.e., a function $f: V_1 \rightarrow V_2$ such that for all $(u, v) \in A_1$ either $(f(u), f(v)) \in A_2$ or $(f(v), f(u)) \in A_2$ and such that for all $u \in V_1$ and $v' \in V_2$ if $(f(u), v') \in A_2$ then there exists a $v \in f^{-1}(v')$ for which $(u, v) \in A_1$?

Reference: [Fraenkel and Yesha, 1977]. Transformation from GRAPH GRUNDY NUMBERING.

A1.5 MISCELLANEOUS

[GT54] PATH WITH FORBIDDEN PAIRS

INSTANCE: Directed graph $G = (V, A)$, specified vertices $s, t \in V$, collection $C = \{(a_1, b_1), \dots, (a_n, b_n)\}$ of pairs of vertices from V .

QUESTION: Is there a directed path from s to t in G that contains at most one vertex from each pair in C ?

Reference: [Gabow, Maheshwari, and Osterweil, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete even if G is acyclic with no in- or out-degree exceeding 2. Variant in which the "forbidden pairs" are arcs instead of vertices is also NP-complete under the same restrictions. Both problems remain NP-complete even if all the given pairs are required to be disjoint.

[GT55] MULTIPLE CHOICE MATCHING

INSTANCE: Graph $G = (V, E)$, partition of E into disjoint sets $E_1, E_2, \dots, E_J$, positive integer K .

QUESTION: Is there a subset $E' \subseteq E$ with $|E'| \geq K$ such that no two edges in E' share a common vertex and such that E' contains at most one edge from each E_i , $1 \leq i \leq J$?

Reference: [Valiant, 1977c], [Itai and Rodeh, 1977a], [Itai, Rodeh, and Tanimota, 1978]. Transformation from 3SAT.

Comment: Remains NP-complete even if G is bipartite, each E_i contains at most 2 edges, and $K = |V|/2$. If each E_i contains only a single edge, this becomes the ordinary graph matching problem and is solvable in polynomial time.

[GT56] GRAPH GRUNDY NUMBERING

INSTANCE: Directed graph $G = (V, A)$.

QUESTION: Is there a function $f: V \rightarrow Z^+$ such that, for each $v \in V$, $f(v)$ is the

least non-negative integer not contained in the set $\{f(u): u \in V, (v, u) \in A\}$?

Reference: [van Leeuwen, 1976a]. Transformation from 3SAT.

Comment: Remains NP-complete when restricted to planar graphs in which no vertex has in- or out-degree exceeding 5 [Fraenkel and Yesha, 1977].

[GT57] KERNEL

INSTANCE: Directed graph $G = (V, A)$.

QUESTION: Does G have a kernel, i.e., a subset $V' \subseteq V$ such that no two vertices in V' are joined by an arc in A and such that for every vertex $v \in V - V'$ there is a vertex $u \in V'$ for which $(u, v) \in A$?

Reference: [Chvátal, 1973]. Transformation from 3SAT.

[GT58] K-CLOSURE

INSTANCE: Directed graph $G = (V, A)$, positive integer $K \leq |V|$.

QUESTION: Is there a subset $V' \subseteq V$ with $|V'| \leq K$ such that for all $(u, v) \in A$ either $u \in V'$ or $v \notin V'$?

Reference: [Queyranne, 1976]. Transformation from CLIQUE.

[GT59] INTERSECTION GRAPH BASIS

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |E|$.

QUESTION: Is G the intersection graph for a family of sets whose union has cardinality K or less, i.e., is there a K -element set S and for each $v \in V$ a subset $S[v] \subseteq S$ such that $\{u, v\} \in E$ if and only if $S[u]$ and $S[v]$ are not disjoint?

Reference: [Kou, Stockmeyer, and Wong, 1978]. Transformation from COVERING BY CLIQUES.

[GT60] PATH DISTINGUISHERS

INSTANCE: Acyclic directed graph $G = (V, A)$, specified vertices $s, t \in V$, positive integer $K \leq |A|$.

QUESTION: Is there a subset $A' \subseteq A$ with $|A'| \leq K$ such that, for any pair p_1, p_2 of paths from s to t in G , there is some arc in A' that is in one of p_1 and p_2 but not both?

Reference: [Maheshwari, 1976]. Transformation from VERTEX COVER.

[GT61] METRIC DIMENSION

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is there a metric basis for G of cardinality K or less, i.e., a subset $V' \subseteq V$ with $|V'| \leq K$ such that for each pair $u, v \in V$ there is a $w \in V'$ such that the length of the shortest path from u to w is different from the length of the shortest path from v to w ?

Reference: [Garey and Johnson, —]. Transformation from 3DM. The definition of metric dimension appears in [Harary and Melter, 1976].

[GT62] NESETRIL-RÖDL DIMENSION

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |E|$.

QUESTION: Is there a one-to-one function

$f: V \rightarrow \{(a_1, a_2, \dots, a_K): 1 \leq a_i \leq |V| \text{ for } 1 \leq i \leq K\}$ such that, for all $u, v \in V$, $\{u, v\} \in E$ if and only if $f(u)$ and $f(v)$ disagree in all K components?

Reference: [Nesetril and Pultr, 1977]. Transformation from GRAPH 3-COLORABILITY. The definition appears in [Nesetril and Rödl, 1977].

[GT63] THRESHOLD NUMBER

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |E|$.

QUESTION: Is there a partition of E into disjoint sets $E_1, E_2, \dots, E_K$ such that each of the graphs $G_i = (V, E_i)$, $1 \leq i \leq K$, is a "threshold graph"?

Reference: [Chvátal and Hammer, 1975]. Transformation from INDEPENDENT SET restricted to triangle free graphs.

Comment: Solvable in polynomial time for $K = 1$.

[GT64] ORIENTED DIAMETER

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Can the edges of G be directed in such a way that the resulting directed graph is strongly connected and has diameter no more than K ?

Reference: [Chvátal and Thomassen, 1978]. Transformation from SET SPLITTING.

Comment: The variation in which "diameter" is replaced by "radius" is also NP-complete. Both problems remain NP-complete for $K = 2$.

[GT65] WEIGHTED DIAMETER

INSTANCE: Graph $G = (V, E)$, collection C of $|E|$ not necessarily distinct non-negative integers, positive integer K .

QUESTION: Is there a one-to-one function $f: E \rightarrow C$ such that, if $f(e)$ is taken as the length of edge e , then G has diameter K or less, i.e., every pair of points $u, v \in V$ is joined by a path in G of length K or less.

Reference: [Perl and Zaks, 1978]. Transformation from 3-PARTITION.

Comment: NP-complete in the strong sense, even if G is a tree. The variant in which "diameter" is replaced by "radius" has the same complexity. If C consists entirely of 0's and 1's, then both the diameter and radius versions are solvable in polynomial time for trees, but are NP-complete for general graphs, even if K is fixed at 2 (diameter) or 1 (radius). The variant in which we ask for an assignment yielding diameter K or *greater* is NP-complete in the strong sense for general graphs, is solvable in polynomial time for trees in the diameter case, and is NP-complete for trees in the radius case.

A2 NETWORK DESIGN

A2.1 SPANNING TREES

[ND1] DEGREE CONSTRAINED SPANNING TREE

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is there a spanning tree for G in which no vertex has degree larger than K ?

Reference: Transformation from HAMILTONIAN PATH.

Comment: Remains NP-complete for any fixed $K \geq 2$.

[ND2] MAXIMUM LEAF SPANNING TREE

INSTANCE: Graph $G = (V, E)$, positive integer $K \leq |V|$.

QUESTION: Is there a spanning tree for G in which K or more vertices have degree 1?

Reference: [Garey and Johnson, —]. Transformation from DOMINATING SET.

Comment: Remains NP-complete if G is regular of degree 4 or if G is planar with no degree exceeding 4.

[ND3] SHORTEST TOTAL PATH LENGTH SPANNING TREE

INSTANCE: Graph $G = (V, E)$, integer bound $B \in \mathbb{Z}^+$.

QUESTION: Is there a spanning tree T for G such that the sum, over all pairs of vertices $u, v \in V$, of the length of the path in T from u to v is no more than K ?

Reference: [Johnson, Lenstra, and Rinnooy Kan, 1978]. Transformation from EXACT COVER BY 3-SETS.

[ND4] BOUNDED DIAMETER SPANNING TREE

INSTANCE: Graph $G = (V, E)$, weight $w(e) \in \mathbb{Z}^+$ for each $e \in E$, positive integer $D \leq |V|$, positive integer B .

QUESTION: Is there a spanning tree T for G such that the sum of the weights of the edges in T does not exceed B and such that T contains no simple path with more than D edges?

Reference: [Garey and Johnson, —]. Transformation from EXACT COVER BY 3-SETS.

Comment: Remains NP-complete for any fixed $D \geq 4$, even if all edge weights are either 1 or 2. Can be solved easily in polynomial time if $D \leq 3$, or if all edge weights are equal.

[ND5] CAPACITATED SPANNING TREE

INSTANCE: Graph $G = (V, E)$, specified vertex $v_0 \in V$, capacity $c(e) \in \mathbb{Z}_0^+$ and length $l(e) \in \mathbb{Z}_0^+$ for each $e \in E$, requirement $r(v) \in \mathbb{Z}_0^+$ for each $v \in V - \{v_0\}$, and a bound $B \in \mathbb{Z}_0^+$.

QUESTION: Is there a spanning tree T for G such that the sum of the lengths of the edges in T does not exceed B and such that for each edge e in T , if $U(e)$ is

the set of vertices whose path to v_0 in T contains e , then $\sum_{u \in U(e)} r(u) \leq c(e)$?

Reference: [Papadimitriou, 1976c]. Transformation from 3SAT.

Comment: NP-complete in the strong sense, even if all requirements are 1 and all capacities are equal to 3. Solvable in polynomial time by weighted matching techniques if all requirements are 1 and all capacities 2. Can also be solved in polynomial time (by minimum cost network flow algorithms, e.g., see [Edmonds and Karp, 1972]) if all capacities are 1 and all requirements are either 0 or 1, but remains NP-complete if all capacities are 2, all requirements 0 or 1, and all edge lengths 0 or 1 [Even and Johnson, 1977].

[ND6] GEOMETRIC CAPACITATED SPANNING TREE

INSTANCE: Set $P \subseteq \mathbb{Z} \times \mathbb{Z}$ of points in the plane, specified point $p_0 \in P$, requirement $r(p) \in \mathbb{Z}_0^+$ for each $p \in P - p_0$, capacity $c \in \mathbb{Z}^+$, bound $B \in \mathbb{Z}^+$.

QUESTION: Is there a spanning tree $T = (P, E')$ for the complete graph $G = (P, E)$ such that $\sum_{e \in E'} d(e) \leq B$, where $d((x_1, y_1), (x_2, y_2))$ is the discretized Euclidean distance $\lceil ((x_1 - x_2)^2 + (y_1 - y_2)^2)^{1/2} \rceil$, and such that for each $e \in E'$, if $U(e)$ is the set of vertices whose paths to p_0 pass through e , then $\sum_{u \in U(e)} r(u) \leq c$?

Reference: [Papadimitriou, 1976c]. Transformation from X3C.

Comment: Remains NP-complete even if all requirements are equal.

[ND7] OPTIMUM COMMUNICATION SPANNING TREE

INSTANCE: Complete graph $G = (V, E)$, weight $w(e) \in \mathbb{Z}_0^+$ for each $e \in E$, requirement $r(\{u, v\}) \in \mathbb{Z}_0^+$ for each pair $\{u, v\}$ of vertices from V , bound $B \in \mathbb{Z}_0^+$.

QUESTION: Is there a spanning tree T for G such that, if $W(\{u, v\})$ denotes the sum of the weights of the edges on the path joining u and v in T , then

$$\sum_{u, v \in V} \left[W(\{u, v\}) \cdot r(\{u, v\}) \right] \leq B?$$

Reference: [Johnson, Lenstra, and Rinnooy Kan, 1978]. Transformation from X3C.

Comment: Remains NP-complete even if all requirements are equal. Can be solved in polynomial time if all edge weights are equal [Hu, 1974].

[ND8] ISOMORPHIC SPANNING TREE

INSTANCE: Graph $G = (V, E)$, tree $T = (V_T, E_T)$.

QUESTION: Does G contain a spanning tree isomorphic to T ?

Reference: Transformation from HAMILTONIAN PATH.

Comment: Remains NP-complete even if (a) T is a path, (b) T is a full binary tree [Papadimitriou and Yannakakis, 1978], or if (c) T is a 3-star (that is, $V_T = \{v_0\} \cup \{u_i, v_i, w_i; 1 \leq i \leq n\}$, $E_T = \{\{v_0, u_i\}, \{u_i, v_i\}, \{v_i, w_i\}; 1 \leq i \leq n\}$) [Garey and Johnson, —]. Solvable in polynomial time by graph matching if G is a 2-star. For a classification of the complexity of this problem for other types of trees, see [Papadimitriou and Yannakakis, 1978].

[ND9] K^{th} BEST SPANNING TREE (*)

INSTANCE: Graph $G=(V,E)$, weight $w(e) \in \mathbb{Z}_0^+$ for each $e \in E$, positive integers K and B .

QUESTION: Are there K distinct spanning trees for G , each having total weight B or less?

Reference: [Johnson and Kashdan, 1976]. Turing reduction from HAMILTONIAN PATH.

Comment: Not known to be in NP. Can be solved in pseudo-polynomial time (polynomial in $|V|, K, \log B, \max\{\log w(e) : e \in E\}$) [Lawler, 1972], and hence in polynomial time for any fixed value of K . The corresponding enumeration problem is #P-complete. However, the unweighted case of the enumeration problem is solvable in polynomial time (e.g., see [Harary and Palmer, 1973]).

[ND10] BOUNDED COMPONENT SPANNING FOREST

INSTANCE: Graph $G=(V,E)$, weight $w(v) \in \mathbb{Z}_0^+$ for each $v \in V$, positive integers $K \leq |V|$ and B .

QUESTION: Can the vertices in V be partitioned into $k \leq K$ disjoint sets $V_1, V_2, \dots, V_k$ such that, for $1 \leq i \leq k$, the subgraph of G induced by V_i is connected and the sum of the weights of the vertices in V_i does not exceed B ?

Reference: [Hadlock, 1974]. Transformation from PARTITION INTO PATHS OF LENGTH 2.

Comment: Remains NP-complete even if all weights equal 1 and B is any fixed integer larger than 2 [Garey and Johnson, —]. Can be solved in polynomial time if G is a tree or if all weights equal 1 and $B=2$ [Hadlock, 1974].

[ND11] MULTIPLE CHOICE BRANCHING

INSTANCE: Directed graph $G=(V,A)$, a weight $w(a) \in \mathbb{Z}^+$ for each arc $a \in A$, a partition of A into disjoint sets $A_1, A_2, \dots, A_m$, and a positive integer K .

QUESTION: Is there a subset $A' \in A$ with $\sum_{a \in A'} w(a) \geq K$ such that no two arcs in A' enter the same vertex, A' contains no cycles, and A' contains at most one arc from each of the A_i , $1 \leq i \leq m$?

Reference: [Garey and Johnson, —]. Transformation from 3SAT.

Comment: Remains NP-complete even if G is strongly connected and all weights are equal. If all A_i have $|A_i|=1$, the problem becomes simply that of finding a "maximum weight branching," a 2-matroid intersection problem that can be solved in polynomial time (e.g., see [Tarjan, 1977]). (In a strongly connected graph, a maximum weight branching can be viewed as a maximum weight directed spanning tree.) Similarly, if the graph is symmetric, the problem becomes equivalent to the "multiple choice spanning tree" problem, another 2-matroid intersection problem that can be solved in polynomial time [Saurballe, 1975].

[ND12] STEINER TREE IN GRAPHS

INSTANCE: Graph $G=(V,E)$, a weight $w(e) \in \mathbb{Z}_0^+$ for each $e \in E$, a subset $R \subseteq V$, and a positive integer bound B .

QUESTION: Is there a subtree of G that includes all the vertices of R and such that the sum of the weights of the edges in the subtree is no more than B ?

Reference: [Karp, 1972]. Transformation from EXACT COVER BY 3-SETS.

Comment: Remains NP-complete if all edge weights are equal, even if G is a bipartite graph having no edges joining two vertices in R or two vertices in $V-R$ [Berlekamp, 1976] or G is planar [Garey and Johnson, 1977a].

[ND13] GEOMETRIC STEINER TREE

INSTANCE: Set $P \subseteq \mathbb{Z} \times \mathbb{Z}$ of points in the plane, positive integer K .

QUESTION: Is there a finite set $Q \subseteq \mathbb{Z} \times \mathbb{Z}$ such that there is a spanning tree of total weight K or less for the vertex set $P \cup Q$, where the weight of an edge $\{(x_1, y_1), (x_2, y_2)\}$ is the discretized Euclidean length $\lceil ((x_1 - x_2)^2 + (y_1 - y_2)^2)^{1/2} \rceil$?

Reference: [Garey, Graham, and Johnson, 1977]. Transformation from X3C.

Comment: NP-complete in the strong sense. Remains so if the distance measure is replaced by the L_1 "rectilinear" metric, $|x_1 - x_2| + |y_1 - y_2|$, [Garey and Johnson, 1977a] or the L_∞ metric, $\max\{|x_1 - x_2|, |y_1 - y_2|\}$, which is equivalent to L_1 under a 45° rotation. Problem remains NP-hard in the strong sense if the (nondiscretized) Euclidean metric $((x_1 - x_2)^2 + (y_1 - y_2)^2)^{1/2}$ is used, but is not known to be in NP [Garey, Graham, and Johnson, 1977]. Some polynomial time algorithms for special cases of the rectilinear case are presented in [Aho, Garey, and Hwang, 1977].

A2.2 CUTS AND CONNECTIVITY**[ND14] GRAPH PARTITIONING**

INSTANCE: Graph $G=(V,E)$, weights $w(v) \in \mathbb{Z}^+$ for each $v \in V$ and $l(e) \in \mathbb{Z}^+$ for each $e \in E$, positive integers K and J .

QUESTION: Is there a partition of V into disjoint sets $V_1, V_2, \dots, V_m$ such that $\sum_{v \in V_i} w(v) \leq K$ for $1 \leq i \leq m$ and such that if $E' \subseteq E$ is the set of edges that have their two endpoints in two different sets V_i , then $\sum_{e \in E'} l(e) \leq J$?

Reference: [Hyafil and Rivest, 1973]. Transformation from PARTITION INTO TRIANGLES.

Comment: Remains NP-complete for fixed $K \geq 3$ even if all vertex and edge weights are 1. Can be solved in polynomial time for $K=2$ by matching.

[ND15] ACYCLIC PARTITION

INSTANCE: Directed graph $G=(V,A)$, weight $w(v) \in \mathbb{Z}^+$ for each $v \in V$, cost $c(a) \in \mathbb{Z}^+$ for each $a \in A$, positive integers B and K .

QUESTION: Is there a partition of V into disjoint sets $V_1, V_2, \dots, V_m$ such that the directed graph $G'=(V',A')$, where $V'=\{V_1, V_2, \dots, V_m\}$, and $(V_i, V_j) \in A'$ if and only if $(v_i, v_j) \in A$ for some $v_i \in V_i$ and some $v_j \in V_j$, is acyclic, such that the sum of the weights of the vertices in each V_i does not exceed B , and such that the sum of the costs of all those arcs having their endpoints in different sets does not exceed K ?

Reference: [Garey and Johnson, —]. Transformation from X3C.

Comment: Remains NP-complete even if all $v \in V$ have $w(v)=1$ and all $a \in A$ have $c(a)=1$. Can be solved in polynomial time if G contains a Hamiltonian path (a

property that can be verified in polynomial time for acyclic digraphs) [Kernighan, 1971]. If G is a tree the general problem is NP-complete in the ordinary sense, but can be solved in pseudo-polynomial time [Lukes, 1974]. The tree problem can be solved in polynomial time if all edge weights are equal (see [Hadlock, 1974]) or if all vertex weights are equal [Garey and Johnson, —].

[ND16] MAX CUT

INSTANCE: Graph $G=(V,E)$, weight $w(e) \in \mathbb{Z}^+$ for each $e \in E$, positive integer K .

QUESTION: Is there a partition of V into disjoint sets V_1 and V_2 such that the sum of the weights of the edges from E that have one endpoint in V_1 and one endpoint in V_2 is at least K ?

Reference: [Karp, 1972]. Transformation from MAXIMUM 2-SATISFIABILITY.

Comment: Remains NP-complete if $w(e)=1$ for all $e \in E$ (the SIMPLE MAX CUT problem) [Garey, Johnson, and Stockmeyer, 1976], and if, in addition, no vertex has degree exceeding 3 [Yannakakis, 1978b]. Can be solved in polynomial time if G is planar [Hadlock, 1975], [Orlova and Dorfman, 1972].

[ND17] MINIMUM CUT INTO BOUNDED SETS

INSTANCE: Graph $G=(V,E)$, weight $w(e) \in \mathbb{Z}^+$ for each $e \in E$, specified vertices $s, t \in V$, positive integer $B \leq |V|$, positive integer K .

QUESTION: Is there a partition of V into disjoint sets V_1 and V_2 such that $s \in V_1$, $t \in V_2$, $|V_1| \leq B$, $|V_2| \leq B$, and such that the sum of the weights of the edges from E that have one endpoint in V_1 and one endpoint in V_2 is no more than K ?

Reference: [Garey, Johnson, and Stockmeyer, 1976]. Transformation from SIMPLE MAX CUT.

Comment: Remains NP-complete for $B=|V|/2$ and $w(e)=1$ for all $e \in E$. Can be solved in polynomial time for $B=|V|$ by standard network flow techniques.

[ND18] BICONNECTIVITY AUGMENTATION

INSTANCE: Graph $G=(V,E)$, weight $w(\{u,v\}) \in \mathbb{Z}^+$ for each unordered pair $\{u,v\}$ of vertices from V , positive integer B .

QUESTION: Is there a set E' of unordered pairs of vertices from V such that $\sum_{e \in E'} w(e) \leq B$ and such that the graph $G'=(V, E \cup E')$ is biconnected, i.e., cannot be disconnected by removing a single vertex?

Reference: [Eswaran and Tarjan, 1976]. Transformation from HAMILTONIAN CIRCUIT.

Comment: The related problem in which G' must be bridge connected, i.e., cannot be disconnected by removing a single edge, is also NP-complete. Both problems remain NP-complete if all weights are either 1 or 2 and E is empty. Both can be solved in polynomial time if all weights are equal.

[ND19] STRONG CONNECTIVITY AUGMENTATION

INSTANCE: Directed graph $G=(V,A)$, weight $w(u,v) \in \mathbb{Z}^+$ for each ordered pair $(u,v) \in V \times V$, positive integer B .

QUESTION: Is there a set A' of ordered pairs of vertices from V such that

$\sum_{a \in A'} w(a) \leq B$ and such that the graph $G'=(V, A \cup A')$ is strongly connected?

Reference: [Eswaran and Tarjan, 1976]. Transformation from HAMILTONIAN CIRCUIT.

Comment: Remains NP-complete if all weights are either 1 or 2 and A is empty. Can be solved in polynomial time if all weights are equal.

[ND20] NETWORK RELIABILITY (*)

INSTANCE: Graph $G=(V,E)$, subset $V' \subseteq V$, a rational "failure probability" $p(e)$, $0 \leq p(e) \leq 1$, for each $e \in E$, a positive rational number $q \leq 1$.

QUESTION: Assuming edge failures are independent of one another, is the probability q or greater that each pair of vertices in V' is joined by at least one path containing no failed edge?

Reference: [Rosenthal, 1974]. Transformation from STEINER TREE IN GRAPHS.

Comment: Not known to be in NP. Remains NP-hard even if $|V'|=2$ [Valiant, 1977b]. The related problem in which we want *two* disjoint paths between each pair of vertices in V' is NP-hard even if $V'=V$ [Ball, 1977b]. If G is directed and we ask for a directed path between each *ordered* pair of vertices in V' , the one-path problem is NP-hard for both $|V'|=2$ [Valiant, 1977b] and $V'=V$ [Ball, 1977a]. Many of the underlying subgraph enumeration problems are #P-complete (see [Valiant, 1977b]).

[ND21] NETWORK SURVIVABILITY (*)

INSTANCE: Graph $G=(V,E)$, a rational "failure probability" $p(x)$, $0 \leq p(x) \leq 1$, for each $x \in V \cup E$, a positive rational number $q \leq 1$.

QUESTION: Assuming all edge and vertex failures are independent of one another, is the probability q or greater that for all $\{u,v\} \in E$ at least one of u,v , or $\{u,v\}$ will fail?

Reference: [Rosenthal, 1974]. Transformation from VERTEX COVER.

Comment: Not known to be in NP.

A2.3 ROUTING PROBLEMS

[ND22] TRAVELING SALESMAN

INSTANCE: Set C of m cities, distance $d(c_i, c_j) \in \mathbb{Z}^+$ for each pair of cities $c_i, c_j \in C$, positive integer B .

QUESTION: Is there a tour of C having length B or less, i.e., a permutation $\langle c_{\pi(1)}, c_{\pi(2)}, \dots, c_{\pi(m)} \rangle$ of C such that

$$\left(\sum_{i=1}^{m-1} d(c_{\pi(i)}, c_{\pi(i+1)}) \right) + d(c_{\pi(m)}, c_{\pi(1)}) \leq B ?$$

Reference: Transformation from HAMILTONIAN CIRCUIT.

Comment: Remains NP-complete even if $d(c_i, c_j) \in \{1,2\}$ for all $c_i, c_j \in C$. Special cases that can be solved in polynomial time are discussed in [Gilmore and Gomory, 1964], [Garfinkel, 1977], and [Syslo, 1973]. The variant in which we ask for a tour

with "mean arrival time" of B or less is also NP-complete [Sahni and Gonzalez, 1976].

[ND23] GEOMETRIC TRAVELING SALESMAN

INSTANCE: Set $P \subseteq Z \times Z$ of points in the plane, positive integer B .

QUESTION: Is there a tour of length B or less for the TRAVELING SALESMAN instance with $C = P$ and $d((x_1, y_1), (x_2, y_2))$ equal to the discretized Euclidean distance $\lceil ((x_1 - x_2)^2 + (y_1 - y_2)^2)^{1/2} \rceil$?

Reference: [Papadimitriou, 1977] [Garey, Graham, and Johnson, 1976]. Transformation from X3C.

Comment: NP-complete in the strong sense. Remains NP-complete in the strong sense if the distance measure is replaced by the L_1 "rectilinear" metric [Garey, Graham, and Johnson, 1976] or the L_∞ metric, which is equivalent to L_1 under a 45° rotation. Problem remains NP-hard in the strong sense if the (nondiscretized) Euclidean metric is used, but is not known to be in NP [Garey, Graham, and Johnson, 1976].

[ND24] BOTTLENECK TRAVELING SALESMAN

INSTANCE: Set C of m cities, distance $d(c_i, c_j) \in Z^+$ for each pair of cities $c_i, c_j \in C$, positive integer B .

QUESTION: Is there a tour of C whose longest edge is no longer than B , i.e., a permutation $\langle c_{\pi(1)}, c_{\pi(2)}, \dots, c_{\pi(m)} \rangle$ of C such that $d(c_{\pi(i)}, c_{\pi(i+1)}) \leq B$ for $1 \leq i < m$ and such that $d(c_{\pi(m)}, c_{\pi(1)}) \leq B$?

Reference: Transformation from HAMILTONIAN CIRCUIT.

Comment: Remains NP-complete even if $d(c_i, c_j) \in \{1, 2\}$ for all $c_i, c_j \in C$. An important special case that is solvable in polynomial time can be found in [Gilmore and Gomory, 1964].

[ND25] CHINESE POSTMAN FOR MIXED GRAPHS

INSTANCE: Mixed graph $G = (V, A, E)$, where A is a set of directed edges and E is a set of undirected edges on V , length $l(e) \in Z_0^+$ for each $e \in A \cup E$, bound $B \in Z^+$.

QUESTION: Is there a cycle in G that includes each directed and undirected edge at least once, traversing directed edges only in the specified direction, and that has total length no more than B ?

Reference: [Papadimitriou, 1976b]. Transformation from 3SAT.

Comment: Remains NP-complete even if all edge lengths are equal, G is planar, and the maximum vertex degree is 3. Can be solved in polynomial time if either A or E is empty (i.e., if G is either a directed or an undirected graph) [Edmonds and Johnson, 1973].

[ND26] STACKER-CRANE

INSTANCE: Mixed graph $G = (V, A, E)$, length $l(e) \in Z_0^+$ for each $e \in A \cup E$, bound $B \in Z^+$.

QUESTION: Is there a cycle in G that includes each directed edge in A at least

once, traversing such edges only in the specified direction, and that has total length no more than B ?

Reference: [Frederickson, Hecht, and Kim, 1978]. Transformation from HAMILTONIAN CIRCUIT.

Comment: Remains NP-complete even if all edge lengths equal 1. The analogous path problem (with or without specified endpoints) is also NP-complete.

[ND27] RURAL POSTMAN

INSTANCE: Graph $G = (V, E)$, length $l(e) \in Z_0^+$ for each $e \in E$, subset $E' \subseteq E$, bound $B \in Z^+$.

QUESTION: Is there a circuit in G that includes each edge in E' and that has total length no more than B ?

Reference: [Lenstra and Rinnooy Kan, 1976]. Transformation from HAMILTONIAN CIRCUIT.

Comment: Remains NP-complete even if $l(e) = 1$ for all $e \in E$, as does the corresponding problem for directed graphs.

[ND28] LONGEST CIRCUIT

INSTANCE: Graph $G = (V, E)$, length $l(e) \in Z^+$ for each $e \in E$, positive integer K .

QUESTION: Is there a simple circuit in G of length K or more, i.e., whose edge lengths sum to at least K ?

Reference: Transformation from HAMILTONIAN CIRCUIT.

Comment: Remains NP-complete if $l(e) = 1$ for all $e \in E$, as does the corresponding problem for directed circuits in directed graphs. The directed problem with all $l(e) = 1$ can be solved in polynomial time if G is a "tournament" [Morrow and Goodman, 1976]. The analogous directed and undirected problems, which ask for a simple circuit of length K or less, can be solved in polynomial time (e.g., see [Itai and Rodeh, 1977b]), but are NP-complete if negative lengths are allowed.

[ND29] LONGEST PATH

INSTANCE: Graph $G = (V, E)$, length $l(e) \in Z^+$ for each $e \in E$, positive integer K , specified vertices $s, t \in V$.

QUESTION: Is there a simple path in G from s to t of length K or more, i.e., whose edge lengths sum to at least K ?

Reference: Transformation from HAMILTONIAN PATH BETWEEN TWO VERTICES.

Comment: Remains NP-complete if $l(e) = 1$ for all $e \in E$, as does the corresponding problem for directed paths in directed graphs. The general problem can be solved in polynomial time for acyclic digraphs, e.g., see [Lawler, 1976a]. The analogous directed and undirected "shortest path" problems can be solved for arbitrary graphs in polynomial time (e.g., see [Lawler, 1976a]), but are NP-complete if negative lengths are allowed.

[ND30] SHORTEST WEIGHT-CONSTRAINED PATH

INSTANCE: Graph $G=(V,E)$, length $l(e) \in \mathbb{Z}^+$, and weight $w(e) \in \mathbb{Z}^+$ for each $e \in E$, specified vertices $s, t \in V$, positive integers K, W .

QUESTION: Is there a simple path in G from s to t with total weight W or less and total length K or less?

Reference: [Megiddo, 1977]. Transformation from PARTITION.

Comment: Also NP-complete for directed graphs. Both problems are solvable in polynomial time if all weights are equal or all lengths are equal.

[ND31] K^{th} SHORTEST PATH (*)

INSTANCE: Graph $G=(V,E)$, length $l(e) \in \mathbb{Z}^+$ for each $e \in E$, specified vertices $s, t \in V$, positive integers B and K .

QUESTION: Are there K or more distinct simple paths from s to t in G , each having total length B or less?

Reference: [Johnson and Kashdan, 1976]. Turing reduction from HAMILTONIAN PATH.

Comment: Not known to be in NP. Corresponding K^{th} shortest circuit problem is also NP-hard. Both remain NP-hard if $l(e)=1$ for all $e \in E$, as do the corresponding problems for directed graphs. However, all versions can be solved in pseudo-polynomial time (polynomial in $|V|$, K , and $\log B$) and hence in polynomial time for any fixed value of K . The corresponding enumeration problems are #P-complete.

A2.4 FLOW PROBLEMS**[ND32] MINIMUM EDGE-COST FLOW**

INSTANCE: Directed graph $G=(V,A)$, specified vertices s and t , capacity $c(a) \in \mathbb{Z}^+$ and price $p(a) \in \mathbb{Z}_0^+$ for each $a \in A$, requirement $R \in \mathbb{Z}^+$, bound $B \in \mathbb{Z}^+$.

QUESTION: Is there a flow function $f: A \rightarrow \mathbb{Z}_0^+$ such that

- (1) $f(a) \leq c(a)$ for all $a \in A$,
- (2) for each $v \in V - \{s, t\}$, $\sum_{(u,v) \in A} f((u,v)) = \sum_{(v,u) \in A} f((v,u))$, i.e., flow is "conserved" at v ,
- (3) $\sum_{(u,t) \in A} f((u,t)) - \sum_{(t,u) \in A} f((t,u)) \geq R$, i.e., the net flow into t is at least R , and
- (4) if $A' = \{a \in A : f(a) \neq 0\}$, then $\sum_{a \in A'} p(a) \leq B$?

Reference: [Even and Johnson, 1977]. Transformation from X3C.

Comment: Remains NP-complete if $c(a)=2$ and $p(a) \in \{0,1\}$ for all $a \in A$. Solvable in polynomial time if $c(a)=1$ for all $a \in A$ [Even and Johnson, 1977] or if (4) is replaced by $\sum_{a \in A} p(a) \cdot f(a) \leq B$ (e.g., see [Lawler, 1976a]). However, becomes NP-complete once more if (4) is replaced by $\sum_{a \in A} (p_1(a)f(a)^2 + p_2(a)f(a)) \leq B$ [Herrmann, 1973].

[ND33] INTEGRAL FLOW WITH MULTIPLIERS

INSTANCE: Directed graph $G=(V,A)$, specified vertices s and t , multiplier $h(v) \in \mathbb{Z}^+$ for each $v \in V - \{s, t\}$, capacity $c(a) \in \mathbb{Z}^+$ for each $a \in A$, requirement $R \in \mathbb{Z}^+$.

QUESTION: Is there a flow function $f: A \rightarrow \mathbb{Z}_0^+$ such that

- (1) $f(a) \leq c(a)$ for all $a \in A$,
- (2) for each $v \in V - \{s, t\}$, $\sum_{(u,v) \in A} h(v) \cdot f((u,v)) = \sum_{(v,u) \in A} f((v,u))$, and
- (3) the net flow into t is at least R ?

Reference: [Sahni, 1974]. Transformation from PARTITION.

Comment: Can be solved in polynomial time by standard network flow techniques if $h(v)=1$ for all $v \in V - \{s, t\}$. Corresponding problem with non-integral flows allowed can be solved by linear programming.

[ND34] PATH CONSTRAINED NETWORK FLOW

INSTANCE: Directed graph $G=(V,A)$, specified vertices s and t , a capacity $c(a) \in \mathbb{Z}^+$ for each $a \in A$, a collection P of directed paths in G , and a requirement $R \in \mathbb{Z}^+$.

QUESTION: Is there a function $g: P \rightarrow \mathbb{Z}_0^+$ such that if $f: A \rightarrow \mathbb{Z}_0^+$ is the flow function defined by $f(a) = \sum_{p \in P(a)} g(p)$, where $P(a) \subseteq P$ is the set of all paths in P containing the arc a , then f is such that

- (1) $f(a) \leq c(a)$ for all $a \in A$,
- (2) for each $v \in V - \{s, t\}$, flow is conserved at v , and
- (3) the net flow into t is at least R ?

Reference: [Prömel, 1978]. Transformation from 3SAT.

Comment: Remains NP-complete even if all $c(a)=1$. The corresponding problem with non-integral flows is equivalent to LINEAR PROGRAMMING, but the question of whether the best rational flow fails to exceed the best integral flow is NP-complete.

[ND35] INTEGRAL FLOW WITH HOMOLOGOUS ARCS

INSTANCE: Directed graph $G=(V,A)$, specified vertices s and t , capacity $c(a) \in \mathbb{Z}^+$ for each $a \in A$, requirement $R \in \mathbb{Z}^+$, set $H \subseteq A \times A$ of "homologous" pairs of arcs.

QUESTION: Is there a flow function $f: A \rightarrow \mathbb{Z}_0^+$ such that

- (1) $f(a) \leq c(a)$ for all $a \in A$,
- (2) for each $v \in V - \{s, t\}$, flow is conserved at v ,
- (3) for all pairs $\langle a, a' \rangle \in H$, $f(a) = f(a')$, and
- (4) the net flow into t is at least R ?

Reference: [Sahni, 1974]. Transformation from 3SAT.

Comment: Remains NP-complete if $c(a)=1$ for all $a \in A$ (by modifying the construction in [Even, Itai, and Shamir, 1976]). Corresponding problem with non-integral flows is polynomially equivalent to LINEAR PROGRAMMING [Itai, 1977].

[ND36] INTEGRAL FLOW WITH BUNDLES

INSTANCE: Directed graph $G=(V,A)$, specified vertices s and t , "bundles" $I_1, I_2, \dots, I_k \subseteq A$ such that $\bigcup_{1 \leq j \leq k} I_j = A$, bundle capacities $c_1, c_2, \dots, c_k \in \mathbb{Z}^+$, requirement $R \in \mathbb{Z}^+$.

QUESTION: Is there a flow function $f: A \rightarrow \mathbb{Z}_0^+$ such that

- (1) for $1 \leq j \leq k$, $\sum_{a \in I_j} f(a) \leq c_j$,
- (2) for each $v \in V - \{s, t\}$, flow is conserved at v , and
- (3) the net flow into t is at least R ?

Reference: [Sahni, 1974]. Transformation from INDEPENDENT SET.

Comment: Remains NP-complete if all capacities are 1 and all bundles have two arcs. Corresponding problem with non-integral flows allowed can be solved by linear programming.

[ND37] UNDIRECTED FLOW WITH LOWER BOUNDS

INSTANCE: Graph $G=(V,E)$, specified vertices s and t , capacity $c(e) \in \mathbb{Z}^+$ and lower bound $l(e) \in \mathbb{Z}_0^+$ for each $e \in E$, requirement $R \in \mathbb{Z}^+$.

QUESTION: Is there a flow function $f: \{(u,v), (v,u): \{u,v\} \in E\} \rightarrow \mathbb{Z}_0^+$ such that

- (1) for all $\{u,v\} \in E$, either $f((u,v)) = 0$ or $f((v,u)) = 0$,
- (2) for each $e = \{u,v\} \in E$, $l(e) \leq \max\{f((u,v)), f((v,u))\} \leq c(e)$,
- (3) for each $v \in V - \{s, t\}$, flow is conserved at v , and
- (4) the net flow into t is at least R ?

Reference: [Itai, 1977]. Transformation from SATISFIABILITY.

Comment: Problem is NP-complete in the strong sense, even if non-integral flows are allowed. Corresponding problem for directed graphs can be solved in polynomial time, even if we ask that the total flow be R or less rather than R or more [Ford and Fulkerson, 1962] (see also [Lawler, 1976a]). The analogous DIRECTED M-COMMODITY FLOW WITH LOWER BOUNDS problem is polynomially equivalent to LINEAR PROGRAMMING for all $M \geq 2$ if non-integral flows are allowed [Itai, 1977].

[ND38] DIRECTED TWO-COMMODITY INTEGRAL FLOW

INSTANCE: Directed graph $G=(V,A)$, specified vertices s_1, s_2, t_1 , and t_2 , capacity $c(a) \in \mathbb{Z}^+$ for each $a \in A$, requirements $R_1, R_2 \in \mathbb{Z}^+$.

QUESTION: Are there two flow functions $f_1, f_2: A \rightarrow \mathbb{Z}_0^+$ such that

- (1) for each $a \in A$, $f_1(a) + f_2(a) \leq c(a)$,
- (2) for each $v \in V - \{s_1, t_1\}$ and $i \in \{1, 2\}$, flow f_i is conserved at v , and
- (3) for $i \in \{1, 2\}$, the net flow into t_i under flow f_i is at least R_i ?

Reference: [Even, Itai, and Shamir, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete even if $c(a) = 1$ for all $a \in A$ and $R_1 = 1$. Variant in which $s_1 = s_2, t_1 = t_2$, and arcs can be restricted to carry only one specified commodity is also NP-complete (follows from [Even, Itai, and Shamir, 1976]). Corresponding M-commodity problem with non-integral flows allowed is polynomially equivalent to LINEAR PROGRAMMING for all $M \geq 2$ [Itai, 1977].

[ND39] UNDIRECTED TWO-COMMODITY INTEGRAL FLOW

INSTANCE: Graph $G=(V,E)$, specified vertices s_1, s_2, t_1 , and t_2 , a capacity $c(e) \in \mathbb{Z}^+$ for each $e \in E$, requirements $R_1, R_2 \in \mathbb{Z}^+$.

QUESTION: Are there two flow functions $f_1, f_2: \{(u,v), (v,u): \{u,v\} \in E\} \rightarrow \mathbb{Z}_0^+$ such that

- (1) for all $\{u,v\} \in E$ and $i \in \{1, 2\}$, either $f_i((u,v)) = 0$ or $f_i((v,u)) = 0$,
- (2) for each $\{u,v\} \in E$,

$$\max\{f_1((u,v)), f_1((v,u))\} + \max\{f_2((u,v)), f_2((v,u))\} \leq c(\{u,v\}),$$
- (3) for each $v \in V - \{s_1, t_1\}$ and $i \in \{1, 2\}$, flow f_i is conserved at v , and
- (4) for $i \in \{1, 2\}$, the net flow into t_i under flow f_i is at least R_i ?

Reference: [Even, Itai, and Shamir, 1976]. Transformation from DIRECTED TWO-COMMODITY INTEGRAL FLOW.

Comment: Remains NP-complete even if $c(e) = 1$ for all $e \in E$. Solvable in polynomial time if $c(e)$ is even for all $e \in E$. Corresponding problem with non-integral flows allowed can be solved in polynomial time.

[ND40] DISJOINT CONNECTING PATHS

INSTANCE: Graph $G=(V,E)$, collection of disjoint vertex pairs $(s_1, t_1), (s_2, t_2), \dots, (s_k, t_k)$.

QUESTION: Does G contain k mutually vertex-disjoint paths, one connecting s_i and t_i for each $i, 1 \leq i \leq k$?

Reference: [Knuth, 1974c], [Karp, 1975a], [Lynch, 1974]. Transformation from 3SAT.

Comment: Remains NP-complete for planar graphs [Lynch, 1974], [Lynch, 1975]. Complexity is open for any fixed $k \geq 2$, but can be solved in polynomial time if $k = 2$ and G is planar or chordal [Perl and Shiloach, 1978]. (A polynomial time algorithm for the general 2 path problem has been announced in [Shiloach, 1978]). The directed version of this problem is also NP-complete in general and solvable in polynomial time when $k = 2$ and G is planar or acyclic [Perl and Shiloach, 1978].

[ND41] MAXIMUM LENGTH-BOUNDED DISJOINT PATHS

INSTANCE: Graph $G=(V,E)$, specified vertices s and t , positive integers $J, K \leq |V|$.

QUESTION: Does G contain J or more mutually vertex-disjoint paths from s to t , none involving more than K edges?

Reference: [Itai, Perl, and Shiloach, 1977]. Transformation from 3SAT.

Comment: Remains NP-complete for all fixed $K \geq 5$. Solvable in polynomial time for $K \leq 4$. Problem where paths need only be edge-disjoint is NP-complete for all fixed $K \geq 5$, polynomially solvable for $K \leq 3$, and open for $K = 4$. The same results hold if G is a directed graph and the paths must be directed paths. The problem of finding the maximum number of disjoint paths from s to t , under no length constraint, is solvable in polynomial time by standard network flow techniques in both the vertex-disjoint and edge-disjoint cases.

[ND42] MAXIMUM FIXED-LENGTH DISJOINT PATHS

INSTANCE: Graph $G=(V,E)$, specified vertices s and t , positive integers $J, K \leq |V|$.

QUESTION: Does G contain J or more mutually vertex-disjoint paths from s to t , each involving exactly K edges?

Reference: [Itai, Perl, and Shiloach, 1977]. Transformation from 3SAT.

Comment: Remains NP-complete for fixed $K \geq 4$. Solvable in polynomial time for $K \leq 3$. Corresponding problem for edge-disjoint paths is NP-complete for fixed $K \geq 4$, polynomially solvable for $K \leq 2$, and open for $K=3$. The same results hold for directed graphs and directed paths, except that the arc-disjoint version is polynomially solvable for $K=3$ and open for $K=4$.

A2.5 MISCELLANEOUS**[ND43] QUADRATIC ASSIGNMENT PROBLEM**

INSTANCE: Non-negative integer costs c_{ij} , $1 \leq i, j \leq n$, and distances d_{kl} , $1 \leq k, l \leq m$, bound $B \in \mathbb{Z}^+$.

QUESTION: Is there a one-to-one function $f: \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, m\}$ such that

$$\sum_{i=1}^n \sum_{j=1}^n c_{ij} d_{f(i)f(j)} \leq B?$$

Reference: [Sahni and Gonzalez, 1976]. Transformation from HAMILTONIAN CIRCUIT.

Comment: Special case in which each $d_{kl} = k - l$ and all $c_{ij} = c_{ij} \in \{0, 1\}$ is the NP-complete OPTIMAL LINEAR ARRANGEMENT problem. The general problem is discussed, for example, in [Garfinkel and Nemhauser, 1972].

[ND44] MINIMIZING DUMMY ACTIVITIES IN PERT NETWORKS

INSTANCE: Directed acyclic graph $G=(V,A)$ where vertices represent tasks and the arcs represent precedence constraints, and a positive integer $K \leq |V|$.

QUESTION: Is there a PERT network corresponding to G with K or fewer dummy activities, i.e., a directed acyclic graph $G'=(V',A')$ where $V'=\{v_i^-, v_i^+ : v_i \in V\}$ and $\{(v_i^-, v_i^+) : v_i \in V\} \subseteq A'$, and such that $|A'| \leq |V| + K$ and there is a path from v_i^+ to v_j^- in G' if and only if there is a path from v_i to v_j in G ?

Reference: [Krishnamoorthy and Deo, 1977b]. Transformation from VERTEX COVER.

[ND45] CONSTRAINED TRIANGULATION

INSTANCE: Graph $G=(V,E)$, coordinates $x(v), y(v) \in \mathbb{Z}$ for each $v \in V$.

QUESTION: Is there a subset $E' \subseteq E$, such that the set of line segments $\{(x(u), y(u)), (x(v), y(v))\} : \{u, v\} \in E'\}$ is a triangulation of the set of points $\{(x(v), y(v)) : v \in V\}$ in the plane?

Reference: [Lloyd, 1977].

Comment: NP-complete in the strong sense.

[ND46] INTERSECTION GRAPH FOR SEGMENTS ON A GRID

INSTANCE: Graph $G=(V,E)$, positive integers M, N .

QUESTION: Is G the intersection graph for a set of line segments on an $M \times N$ grid, i.e., is there a one-to-one function f that maps each $v \in V$ to a line segment $f(v) = [(x, y), (z, w)]$, where $1 \leq x \leq z \leq M$, $1 \leq y \leq w \leq N$, and either $x=z$ or $y=w$, such that $\{u, v\} \in E$ if and only if the line segments $f(u)$ and $f(v)$ intersect?

Reference: [Gavril, 1977a]. Transformation from 3-PARTITION.

Comment: The analogous problem, which asks if G is the intersection graph for a set of rectangles on an $M \times N$ grid, is also NP-complete [Gavril, 1977a].

[ND47] EDGE EMBEDDING ON A GRID

INSTANCE: Graph $G=(V,E)$, positive integers M, N .

QUESTION: Is there a one-to-one function $f: V \rightarrow \{1, 2, \dots, M\} \times \{1, 2, \dots, N\}$ such that if $\{u, v\} \in E$, $f(u) = (x_1, y_1)$, and $f(v) = (x_2, y_2)$, then either $x_1 = x_2$ or $y_1 = y_2$, i.e., $f(u)$ and $f(v)$ are both on the same "line" of the grid?

Reference: [Gavril, 1977a]. Transformation from 3-PARTITION.

[ND48] GEOMETRIC CONNECTED DOMINATING SET

INSTANCE: Set $P \subseteq \mathbb{Z} \times \mathbb{Z}$ of points in the plane, positive integers B and K .

QUESTION: Is there a subset $P' \subseteq P$ with $|P'| \leq K$ such that all points in $P - P'$ are within Euclidean distance B of some point in P' , and such that the graph $G=(P', E)$, with an edge between two points in P' if and only if they are within distance B of each other, is connected?

Reference: [Lichtenstein, 1977]. Transformation from PLANAR 3SAT.

Comment: Remains NP-complete if the Euclidean metric is replaced by the L_1 rectilinear metric or the L_∞ metric [Garey and Johnson, —].

[ND49] MINIMUM BROADCAST TIME

INSTANCE: Graph $G=(V,E)$, subset $V_0 \subseteq V$, and a positive integer K .

QUESTION: Can a message be "broadcast" from the base set V_0 to all other vertices in time K , i.e., is there a sequence $V_0, E_1, V_1, E_2, \dots, E_K, V_K$ such that each $V_i \subseteq V$, each $E_i \subseteq E$, $V_K = V$, and, for $1 \leq i \leq K$, (1) each edge in E_i has exactly one endpoint in V_{i-1} , (2) no two edges in E_i share a common endpoint, and (3) $V_i = V_{i-1} \cup \{v : \{u, v\} \in E_i\}$?

Reference: [Garey and Johnson, —]. Transformation from 3DM. For more on this problem, see [Farley, Hedetniemi, Mitchell, and Proskurowski, 1977].

Comment: Remains NP-complete for any fixed $K \geq 4$, but is solvable in polynomial time by matching if $K=1$. The special case where $|V_0|=1$ remains NP-complete, but is solvable in polynomial time for trees [Cockayne, Hedetniemi, and Slater, 1978].

[ND50] MIN-MAX MULTICENTER

INSTANCE: Graph $G=(V,E)$, weight $w(v) \in \mathbb{Z}_0^+$ for each $v \in V$, length $l(e) \in \mathbb{Z}_0^+$ for each $e \in E$, positive integer $K \leq |V|$, positive rational number B .

QUESTION: Is there a set P of K "points on G " (where a point on G can be either a vertex in V or a point on an edge $e \in E$, with e regarded as a line segment of length $l(e)$) such that if $d(v)$ is the length of the shortest path from v to the closest point in P , then $\max\{d(v) \cdot w(v) : v \in V\} \leq B$?

Reference: [Kariv and Hakimi, 1976a]. Transformation from DOMINATING SET.

Comment: Also known as the "p-center" problem. Remains NP-complete if $w(v)=1$ for all $v \in V$ and $l(e)=1$ for all $e \in E$. Solvable in polynomial time for any fixed K and for arbitrary K if G is a tree [Kariv and Hakimi, 1976a]. Variant in which we must choose a subset $P \subseteq V$ is also NP-complete but solvable for fixed K and for trees [Slater, 1976].

[ND51] MIN-SUM MULTICENTER

INSTANCE: Graph $G=(V,E)$, weight $w(v) \in Z_0^+$ for each $v \in V$, length $l(e) \in Z_0^+$ for each $e \in E$, positive integer $K \leq |V|$, positive rational number B .

QUESTION: Is there a set P of K "points on G " such that if $d(v)$ is the length of the shortest path from v to the closest point in P , then $\sum_{v \in V} d(v) \cdot w(v) \leq B$?

Reference: [Kariv and Hakimi, 1976b]. Transformation from DOMINATING SET.

Comment: Also known as the "p-median" problem. It can be shown that there is no loss of generality in restricting P to being a subset of V . Remains NP-complete if $w(v)=1$ for all $v \in V$ and $l(e)=1$ for all $e \in E$. Solvable in polynomial time for any fixed K and for arbitrary K if G is a tree.

A3 SETS AND PARTITIONS

A3.1 COVERING, HITTING, AND SPLITTING

[SP1] 3-DIMENSIONAL MATCHING (3DM)

INSTANCE: Set $M \subseteq W \times X \times Y$, where W , X , and Y are disjoint sets having the same number q of elements.

QUESTION: Does M contain a matching, i.e., a subset $M' \subseteq M$ such that $|M'|=q$ and no two elements of M' agree in any coordinate?

Reference: [Karp, 1972]. Transformation from 3SAT (see Section 3.1.2).

Comment: Remains NP-complete if M is "pairwise consistent," i.e., if for all elements a, b, c , whenever there exist elements w, z , and y such that $(a, b, w) \in M$, $(a, x, c) \in M$, and $(y, b, c) \in M$, then $(a, b, c) \in M$ (this follows from the proof of Theorem 3.1.2). Also remains NP-complete if no element occurs in more than three triples, but is solvable in polynomial time if no element occurs in more than two triples [Garey and Johnson, —]. The related 2-DIMENSIONAL MATCHING problem (where $M \subseteq W \times X$) is also solvable in polynomial time (e.g., see [Lawler, 1976a]).

[SP2] EXACT COVER BY 3-SETS (X3C)

INSTANCE: Set X with $|X|=3q$ and a collection C of 3-element subsets of X .

QUESTION: Does C contain an exact cover for X , i.e., a subcollection $C' \subseteq C$ such that every element of X occurs in exactly one member of C' ?

Reference: [Karp, 1972]. Transformation from 3DM.

Comment: Remains NP-complete if no element occurs in more than three subsets, but is solvable in polynomial time if no element occurs in more than two subsets [Garey and Johnson, —]. Related EXACT COVER BY 2-SETS problem is also solvable in polynomial time by matching techniques.

[SP3] SET PACKING

INSTANCE: Collection C of finite sets, positive integer $K \leq |C|$.

QUESTION: Does C contain at least K mutually disjoint sets?

Reference: [Karp, 1972]. Transformation from X3C.

Comment: Remains NP-complete even if all $c \in C$ have $|c| \leq 3$. Solvable in polynomial time by matching techniques if all $c \in C$ have $|c| \leq 2$.

[SP4] SET SPLITTING

INSTANCE: Collection C of subsets of a finite set S .

QUESTION: Is there a partition of S into two subsets S_1 and S_2 such that no subset in C is entirely contained in either S_1 or S_2 ?

Reference: [Lovasz, 1973]. Transformation from NOT-ALL-EQUAL 3SAT. The problem is also known as HYPERGRAPH 2-COLORABILITY.

Comment: Remains NP-complete even if all $c \in C$ have $|c| \leq 3$. Solvable in polynomial time if all $c \in C$ have $|c| \leq 2$ (becomes GRAPH 2-COLORABILITY).

[SP5] MINIMUM COVER

INSTANCE: Collection C of subsets of a finite set S , positive integer $K \leq |C|$.

QUESTION: Does C contain a cover for S of size K or less, i.e., a subset $C' \subseteq C$ with $|C'| \leq K$ such that every element of S belongs to at least one member of C' ?

Reference: [Karp, 1972]. Transformation from X3C.

Comment: Remains NP-complete even if all $c \in C$ have $|c| \leq 3$. Solvable in polynomial time by matching techniques if all $c \in C$ have $|c| \leq 2$.

[SP6] MINIMUM TEST SET

INSTANCE: Collection C of subsets of a finite set S , positive integer $K \leq |C|$.

QUESTION: Is there a subcollection $C' \subseteq C$ with $|C'| \leq K$ such that for each pair of distinct elements $u, v \in S$, there is some set $c \in C'$ that contains exactly one of u and v ?

Reference: [Garey and Johnson, —]. Transformation from 3DM.

Comment: Remains NP-complete if all $c \in C$ have $|c| \leq 3$, but is solvable in polynomial time if all $c \in C$ have $|c| \leq 2$. Variant in which C' can contain unions of subsets in C as well as subsets in C is also NP-complete [Ibaraki, Kameda, and Toida, 1977].

[SP7] SET BASIS

INSTANCE: Collection C of subsets of a finite set S , positive integer $K \leq |C|$.

QUESTION: Is there a collection B of subsets of S with $|B| = K$ such that, for each $c \in C$, there is a subcollection of B whose union is exactly c ?

Reference: [Stockmeyer, 1975]. Transformation from VERTEX COVER.

Comment: Remains NP-complete if all $c \in C$ have $|c| \leq 3$, but is trivial if all $c \in C$ have $|c| \leq 2$.

[SP8] HITTING SET

INSTANCE: Collection C of subsets of a finite set S , positive integer $K \leq |S|$.

QUESTION: Is there a subset $S' \subseteq S$ with $|S'| \leq K$ such that S' contains at least one element from each subset in C ?

Reference: [Karp, 1972]. Transformation from VERTEX COVER.

Comment: Remains NP-complete even if $|c| \leq 2$ for all $c \in C$.

[SP9] INTERSECTION PATTERN

INSTANCE: An $n \times n$ matrix $A = (a_{ij})$ with entries in Z_0^+ .

QUESTION: Is there a collection $C = \{C_1, C_2, \dots, C_n\}$ of sets such that for all i, j , $1 \leq i, j \leq n$, $a_{ij} = |C_i \cap C_j|$?

Reference: [Chvátal, 1978]. Transformation from GRAPH 3-COLORABILITY.

Comment: Remains NP-complete even if all $a_{ii} = 3$, $1 \leq i \leq m$ (and hence all C_i must have cardinality 3). If all $a_{ii} = 2$, it is equivalent to edge graph recognition and hence can be solved in polynomial time (e.g., see [Harary, 1969]).

[SP10] COMPARATIVE CONTAINMENT

INSTANCE: Two collections $R = \{R_1, R_2, \dots, R_k\}$ and $S = \{S_1, S_2, \dots, S_l\}$ of subsets of a finite set X , weights $w(R_i) \in Z^+$, $1 \leq i \leq k$, and $w(S_j) \in Z^+$, $1 \leq j \leq l$.

QUESTION: Is there a subset $Y \subseteq X$ such that

$$\sum_{Y \subseteq R_i} w(R_i) \geq \sum_{Y \subseteq S_j} w(S_j) ?$$

Reference: [Plaisted, 1976]. Transformation from VERTEX COVER.

Comment: Remains NP-complete even if all subsets in R and S have weight 1 [Garey and Johnson, —].

[SP11] 3-MATROID INTERSECTION

INSTANCE: Three matroids $(E, F_1), (E, F_2), (E, F_3)$, positive integer $K \leq |E|$. (A matroid (E, F) consists of a set E of elements and a non-empty family F of subsets of E such that (1) $S \in F$ implies all subsets of S are in F and (2) if two sets $S, S' \in F$ satisfy $|S| = |S'| + 1$, then there exists an element $e \in S - S'$ such that $(S' \cup \{e\}) \in F$.)

QUESTION: Is there a subset $E' \subseteq E$ such that $|E'| = K$ and $E' \in (F_1 \cap F_2 \cap F_3)$?

Reference: Transformation from 3DM.

Comment: The related 2-MATROID INTERSECTION problem can be solved in polynomial time, even if the matroids are described by giving polynomial time algorithms for recognizing their members, and even if each element $e \in E$ has a weight $w(e) \in Z^+$, with the goal being to find an $E' \in (F_1 \cap F_2)$ having maximum total weight (e.g., see [Lawler, 1976a]).

A3.2 WEIGHTED SET PROBLEMS**[SP12] PARTITION**

INSTANCE: Finite set A and a size $s(a) \in Z^+$ for each $a \in A$.

QUESTION: Is there a subset $A' \subseteq A$ such that $\sum_{a \in A'} s(a) = \sum_{a \in A - A'} s(a)$?

Reference: [Karp, 1972]. Transformation from 3DM (see Section 3.1.5).

Comment: Remains NP-complete even if we require that $|A'| = |A|/2$, or if the elements in A are ordered as $a_1, a_2, \dots, a_{2n}$ and we require that A' contain exactly one of a_{2i-1}, a_{2i} for $1 \leq i \leq n$. However, all these problems can be solved in pseudo-polynomial time by dynamic programming (see Section 4.2).

[SP13] SUBSET SUM

INSTANCE: Finite set A , size $s(a) \in Z^+$ for each $a \in A$, positive integer B .

QUESTION: Is there a subset $A' \subseteq A$ such that the sum of the sizes of the elements in A' is exactly B ?

Reference: [Karp, 1972]. Transformation from PARTITION.

Comment: Solvable in pseudo-polynomial time (see Section 4.2).

[SP14] SUBSET PRODUCT

INSTANCE: Finite set A , a size $s(a) \in \mathbb{Z}^+$ for each $a \in A$, and a positive integer B .

QUESTION: Is there a subset $A' \subseteq A$ such that the product of the sizes of the elements in A' is exactly B ?

Reference: [Yao, 1978b]. Transformation from X3C.

Comment: NP-complete in the strong sense.

[SP15] 3-PARTITION

INSTANCE: Set A of $3m$ elements, a bound $B \in \mathbb{Z}^+$, and a size $s(a) \in \mathbb{Z}^+$ for each $a \in A$ such that $B/4 < s(a) < B/2$ and such that $\sum_{a \in A} s(a) = mB$.

QUESTION: Can A be partitioned into m disjoint sets $A_1, A_2, \dots, A_m$ such that, for $1 \leq i \leq m$, $\sum_{a \in A_i} s(a) = B$ (note that each A_i must therefore contain exactly three elements from A)?

Reference: [Garey and Johnson, 1975]. Transformation from 3DM (see Section 4.2).

Comment: NP-complete in the strong sense.

[SP16] NUMERICAL 3-DIMENSIONAL MATCHING

INSTANCE: Disjoint sets W, X , and Y , each containing m elements, a size $s(a) \in \mathbb{Z}^+$ for each element $a \in W \cup X \cup Y$, and a bound $B \in \mathbb{Z}^+$.

QUESTION: Can $W \cup X \cup Y$ be partitioned into m disjoint sets $A_1, A_2, \dots, A_m$ such that each A_i contains exactly one element from each of W, X , and Y and such that, for $1 \leq i \leq m$, $\sum_{a \in A_i} s(a) = B$?

Reference: [Garey and Johnson, —]. Transformation from 3DM (see proof of Theorem 4.4).

Comment: NP-complete in the strong sense.

[SP17] NUMERICAL MATCHING WITH TARGET SUMS

INSTANCE: Disjoint sets X and Y , each containing m elements, a size $s(a) \in \mathbb{Z}^+$ for each element $a \in X \cup Y$, and a target vector $\langle B_1, B_2, \dots, B_m \rangle$ with positive integer entries.

QUESTION: Can $X \cup Y$ be partitioned into m disjoint sets $A_1, A_2, \dots, A_m$, each containing exactly one element from each of X and Y , such that, for $1 \leq i \leq m$, $\sum_{a \in A_i} s(a) = B_i$?

Reference: Transformation from NUMERICAL 3-DIMENSIONAL MATCHING.

Comment: NP-complete in the strong sense, but solvable in polynomial time if $B_1 = B_2 = \dots = B_m$.

[SP18] EXPECTED COMPONENT SUM

INSTANCE: Collection C of m -dimensional vectors $v = (v_1, v_2, \dots, v_m)$ with non-negative integer entries, positive integers K and B .

QUESTION: Is there a partition of C into disjoint sets $C_1, C_2, \dots, C_K$ such that

$$\sum_{i=1}^K \max_{1 \leq j \leq m} \left(\sum_{v \in C_i} v_j \right) \geq B?$$

Reference: [Garey and Johnson, —]. Transformation from X3C. The problem is due to [Witsenhausen, 1978] and corresponds to finding a partition that maximizes the expected value of the largest component sum, assuming all sets in the partition are equally likely.

Reference: NP-complete even if all entries are 0's and 1's. Solvable in polynomial time if K is fixed. The variant in which we ask for a partition with K non-empty sets that yields a sum of B or less is NP-complete even if K is fixed at 3 and all entries are 0's and 1's.

[SP19] MINIMUM SUM OF SQUARES

INSTANCE: Finite set A , a size $s(a) \in \mathbb{Z}^+$ for each $a \in A$, positive integers $K \leq |A|$ and J .

QUESTION: Can A be partitioned into K disjoint sets $A_1, A_2, \dots, A_K$ such that

$$\sum_{i=1}^K \left(\sum_{a \in A_i} s(a) \right)^2 \leq J?$$

Reference: Transformation from PARTITION or 3-PARTITION.

Comment: NP-complete in the strong sense. NP-complete in the ordinary sense and solvable in pseudo-polynomial time for any fixed K . Variants in which the bound K on the number of sets is replaced by a bound B on either the maximum set cardinality or the maximum total set size are also NP-complete in the strong sense [Wong and Yao, 1976]. In all these cases, NP-completeness is preserved if the exponent 2 is replaced by any fixed rational $\alpha > 1$.

[SP20] K^{th} LARGEST SUBSET (*)

INSTANCE: Finite set A , size $s(a) \in \mathbb{Z}^+$ for each $a \in A$, positive integers K and B .

QUESTION: Are there K or more distinct subsets $A' \subseteq A$ for which the sum of the sizes of the elements in A' does not exceed B ?

Reference: [Johnson and Kashdan, 1976]. Transformation from SUBSET SUM.

Comment: Not known to be in NP. Solvable in pseudo-polynomial time (polynomial in $K, |A|$, and $\log \sum s(a)$) [Lawler, 1972]. The corresponding enumeration problem is #P-complete.

[SP21] K^{th} LARGEST m -TUPLE (*)

INSTANCE: Sets $X_1, X_2, \dots, X_m \subseteq \mathbb{Z}^+$, a size $s(x) \in \mathbb{Z}^+$ for each $x \in X_i$, $1 \leq i \leq m$, and positive integers K and B .

QUESTION: Are there K or more distinct m -tuples $(x_1, x_2, \dots, x_m)$ in $X_1 \times X_2 \times \dots \times X_m$ for which $\sum_{i=1}^m s(x_i) \geq B$?

Reference: [Johnson and Mizoguchi, 1978]. Transformation from PARTITION.

Comment: Not known to be in NP. Solvable in polynomial time for fixed m , and in pseudo-polynomial time in general (polynomial in $K, \sum |X_i|$, and $\log \sum s(x)$). The corresponding enumeration problem is #P-complete.

A4 STORAGE AND RETRIEVAL

A4.1 DATA STORAGE

[SR1] BIN PACKING

INSTANCE: Finite set U of items, a size $s(u) \in \mathbb{Z}^+$ for each $u \in U$, a positive integer bin capacity B , and a positive integer K .

QUESTION: Is there a partition of U into disjoint sets $U_1, U_2, \dots, U_K$ such that the sum of the sizes of the items in each U_i is B or less?

Reference: Transformation from PARTITION, 3-PARTITION.

Comment: NP-complete in the strong sense. NP-complete and solvable in pseudo-polynomial time for each fixed $K \geq 2$. Solvable in polynomial time for any fixed B by exhaustive search.

[SR2] DYNAMIC STORAGE ALLOCATION

INSTANCE: Set A of items to be stored, each $a \in A$ having a size $s(a) \in \mathbb{Z}^+$, an arrival time $r(a) \in \mathbb{Z}_0^+$, and a departure time $d(a) \in \mathbb{Z}^+$, and a positive integer storage size D .

QUESTION: Is there a feasible allocation of storage for A , i.e., a function $\sigma: A \rightarrow \{1, 2, \dots, D\}$ such that for every $a \in A$ the allocated storage interval $I(a) = [\sigma(a), \sigma(a) + s(a) - 1]$ is contained in $[1, D]$ and such that, for all $a, a' \in A$, if $I(a) \cap I(a')$ is nonempty then either $d(a) \leq r(a')$ or $d(a') \leq r(a)$?

Reference: [Stockmeyer, 1976b]. Transformation from 3-PARTITION.

Comment: NP-complete in the strong sense, even if $s(a) \in \{1, 2\}$ for all $a \in A$. Solvable in polynomial time if all item sizes are the same, by interval graph coloring algorithms (e.g., see [Gavril, 1972]).

[SR3] PRUNED TRIE SPACE MINIMIZATION

INSTANCE: Finite set S , collection F of functions $f: S \rightarrow \mathbb{Z}^+$, and a positive integer K .

QUESTION: Is there a sequence $\langle f_1, f_2, \dots, f_m \rangle$ of distinct functions from F such that for every two elements $a, b \in S$ there is some i , $1 \leq i \leq m$, for which $f_i(a) \neq f_i(b)$ and such that, if $N(i)$ denotes the number of distinct i -tuples $X = (x_1, x_2, \dots, x_i)$ for which there is more than one $a \in S$ having $(f_1(a), f_2(a), \dots, f_i(a)) = X$, then $\sum_{i=1}^m N(i) \leq K$?

Reference: [Comer and Sethi, 1976]. Transformation from 3DM.

Comment: Remains NP-complete even if all $f \in F$ have range $\{0, 1\}$. Variants in which the "pruned trie" data structure abstracted above is replaced by "full trie," "collapsed trie," or "pruned 0-trie" are also NP-complete. The related "access time minimization" problem is also NP-complete for pruned tries, where we ask for a sequence $\langle f_1, f_2, \dots, f_m \rangle$ of functions from F that distinguishes every two elements from S as above and such that, if the access time $L(a)$ for $a \in S$ is defined to be the least i for which no other $b \in S$ has $(f_1(b), f_2(b), \dots, f_i(b))$ identical to $(f_1(a), f_2(a), \dots, f_i(a))$, then $\sum_{a \in S} L(a) \leq K$.

[SR4] EXPECTED RETRIEVAL COST

INSTANCE: Set R of records, rational probability $p(r) \in [0, 1]$ for each $r \in R$, with $\sum_{r \in R} p(r) = 1$, number m of sectors, and a positive integer K .

QUESTION: Is there a partition of R into disjoint subsets $R_1, R_2, \dots, R_m$ such that, if $p(R_i) = \sum_{r \in R_i} p(r)$ and the "latency cost" $d(i, j)$ is defined to be $j - i - 1$ if $1 \leq i < j \leq m$ and to be $m - i + j - 1$ if $1 \leq j \leq i \leq m$, then the sum over all ordered pairs i, j , $1 \leq i, j \leq m$, of $p(R_i) \cdot p(R_j) \cdot d(i, j)$ is at most K ?

Reference: [Cody and Coffman, 1976]. Transformation from PARTITION, 3-PARTITION.

Comment: NP-complete in the strong sense. NP-complete and solvable in pseudo-polynomial time for each fixed $m \geq 2$.

[SR5] ROOTED TREE STORAGE ASSIGNMENT

INSTANCE: Finite set X , collection $C = \{X_1, X_2, \dots, X_n\}$ of subsets of X , positive integer K .

QUESTION: Is there a collection $C' = \{X'_1, X'_2, \dots, X'_n\}$ of subsets of X such that $X_i \subseteq X'_i$ for $1 \leq i \leq n$, such that $\sum_{i=1}^n |X'_i - X_i| \leq K$, and such that there is a directed rooted tree $T = (X, A)$ in which the elements of each X'_i , $1 \leq i \leq n$, form a directed path?

Reference: [Gavril, 1977a]. Transformation from ROOTED TREE ARRANGEMENT.

[SR6] MULTIPLE COPY FILE ALLOCATION

INSTANCE: Graph $G = (V, E)$, for each $v \in V$ a usage $u(v) \in \mathbb{Z}^+$ and a storage cost $s(v) \in \mathbb{Z}^+$, and a positive integer K .

QUESTION: Is there a subset $V' \subseteq V$ such that, if for each $v \in V$ we let $d(v)$ denote the number of edges in the shortest path in G from v to a member of V' , we have

$$\sum_{v \in V'} s(v) + \sum_{v \in V} d(v) \cdot u(v) \leq K?$$

Reference: [Van Sickle and Chandy, 1977]. Transformation from VERTEX COVER.

Comment: NP-complete in the strong sense, even if all $v \in V$ have the same value of $u(v)$ and the same value of $s(v)$.

[SR7] CAPACITY ASSIGNMENT

INSTANCE: Set C of communication links, set $M \subseteq \mathbb{Z}^+$ of capacities, cost function $g: C \times M \rightarrow \mathbb{Z}^+$, delay penalty function $d: C \times M \rightarrow \mathbb{Z}^+$ such that, for all $c \in C$ and $i < j \in M$, $g(c, i) \leq g(c, j)$ and $d(c, i) \geq d(c, j)$, and positive integers K and J .

QUESTION: Is there an assignment $\sigma: C \rightarrow M$ such that the total cost $\sum_{c \in C} g(c, \sigma(c))$ does not exceed K and such that the total delay penalty $\sum_{c \in C} d(c, \sigma(c))$ does not exceed J ?

Reference: [Van Sickle and Chandy, 1977]. Transformation from SUBSET SUM.

Comment: Solvable in pseudo-polynomial time.

A4.2 COMPRESSION AND REPRESENTATION

[SR8] SHORTEST COMMON SUPERSEQUENCE

INSTANCE: Finite alphabet Σ , finite set R of strings from Σ^* , and a positive integer K .

QUESTION: Is there a string $w \in \Sigma^*$ with $|w| \leq K$ such that each string $x \in R$ is a subsequence of w , i.e., $w = w_0 x_1 w_1 x_2 w_2 \cdots x_k w_k$ where each $w_i \in \Sigma^*$ and $x = x_1 x_2 \cdots x_k$?

Reference: [Maier, 1978]. Transformation from VERTEX COVER.

Comment: Remains NP-complete even if $|\Sigma| = 5$. Solvable in polynomial time if $|R| = 2$ (by first computing the largest common subsequence) or if all $x \in R$ have $|x| \leq 2$.

[SR9] SHORTEST COMMON SUPERSTRING

INSTANCE: Finite alphabet Σ , finite set R of strings from Σ^* , and a positive integer K .

QUESTION: Is there a string $w \in \Sigma^*$ with $|w| \leq K$ such that each string $x \in R$ is a substring of w , i.e., $w = w_0 x w_1$ where each $w_i \in \Sigma^*$?

Reference: [Maier and Storer, 1977]. Transformation from VERTEX COVER for cubic graphs.

Comment: Remains NP-complete even if $|\Sigma| = 2$ or if all $x \in R$ have $|x| \leq 8$ and contain no repeated symbols. Solvable in polynomial time if all $x \in R$ have $|x| \leq 2$.

[SR10] LONGEST COMMON SUBSEQUENCE

INSTANCE: Finite alphabet Σ , finite set R of strings from Σ^* , and a positive integer K .

QUESTION: Is there a string $w \in \Sigma^*$ with $|w| \geq K$ such that w is a subsequence of each $x \in R$?

Reference: [Maier, 1978]. Transformation from VERTEX COVER.

Comment: Remains NP-complete even if $|\Sigma| = 2$. Solvable in polynomial time for any fixed K or for fixed $|R|$ (by dynamic programming, e.g., see [Wagner and Fischer, 1974]). The analogous LONGEST COMMON SUBSTRING problem is trivially solvable in polynomial time.

[SR11] BOUNDED POST CORRESPONDENCE PROBLEM

INSTANCE: Finite alphabet Σ , two sequences $a = (a_1, a_2, \dots, a_n)$ and $b = (b_1, b_2, \dots, b_n)$ of strings from Σ^* , and a positive integer $K \leq n$.

QUESTION: Is there a sequence $i_1, i_2, \dots, i_k$ of $k \leq K$ (not necessarily distinct) positive integers, each between 1 and n , such that the two strings $a_{i_1} a_{i_2} \cdots a_{i_k}$ and $b_{i_1} b_{i_2} \cdots b_{i_k}$ are identical?

Reference: [Constable, Hunt, and Sahni, 1974]. Generic transformation.

Comment: Problem is undecidable if no upper bound is placed on k , e.g., see [Hopcroft and Ullman, 1969].

[SR12] HITTING STRING

INSTANCE: Finite set A of strings over $\{0, 1, *\}$, all having the same length n .

QUESTION: Is there a string $x \in \{0, 1\}^*$ with $|x| = n$ such that for each string $a \in A$ there is some i , $1 \leq i \leq n$, for which the i^{th} symbol of a and the i^{th} symbol of x are identical?

Reference: [Fagin, 1974]. Transformation from 3SAT.

[SR13] SPARSE MATRIX COMPRESSION

INSTANCE: An $m \times n$ matrix A with entries $a_{ij} \in \{0, 1\}$, $1 \leq i \leq m$, $1 \leq j \leq n$, and a positive integer $K \leq mn$.

QUESTION: Is there a sequence $(b_1, b_2, \dots, b_{n+K})$ of integers b_i , each satisfying $0 \leq b_i \leq m$, and a function $s: \{1, 2, \dots, m\} \rightarrow \{1, 2, \dots, K\}$ such that, for $1 \leq i \leq m$ and $1 \leq j \leq n$, the entry $a_{ij} = 1$ if and only if $b_{s(i)+j-1} = i$?

Reference: [Even, Lichtenstein, and Shiloach, 1977]. Transformation from GRAPH 3-COLORABILITY.

Comment: Remains NP-complete for fixed $K = 3$.

[SR14] CONSECUTIVE ONES SUBMATRIX

INSTANCE: An $m \times n$ matrix A of 0's and 1's and a positive integer K .

QUESTION: Is there an $m \times K$ submatrix B of A that has the "consecutive ones" property, i.e., such that the columns of B can be permuted so that in each row all the 1's occur consecutively?

Reference: [Booth, 1975]. Transformation from HAMILTONIAN PATH.

Comment: The variant in which we ask instead that B have the "circular ones" property, i.e., that the columns of B can be permuted so that in each row either all the 1's or all the 0's occur consecutively, is also NP-complete. Both problems can be solved in polynomial time if $K = n$ (in which case we are asking if A has the desired property), e.g., see [Fulkerson and Gross, 1965], [Tucker, 1971], and [Booth and Lueker, 1976].

[SR15] CONSECUTIVE ONES MATRIX PARTITION

INSTANCE: An $m \times n$ matrix A of 0's and 1's.

QUESTION: Can the rows of A be partitioned into two groups such that the resulting $m_1 \times n$ and $m_2 \times n$ matrices ($m_1 + m_2 = m$) each have the consecutive ones property?

Reference: [Lipsky, 1978]. Transformation from HAMILTONIAN PATH for cubic graphs.

[SR16] CONSECUTIVE ONES MATRIX AUGMENTATION

INSTANCE: An $m \times n$ matrix A of 0's and 1's and a positive integer K .

QUESTION: Is there a matrix A' , obtained from A by changing K or fewer 0 entries to 1's, such that A' has the consecutive ones property?

Reference: [Booth, 1975], [Papadimitriou, 1976a]. Transformation from OPTIMAL LINEAR ARRANGEMENT.

Comment: Variant in which we ask instead that A' have the circular ones property is also NP-complete.

[SR17] CONSECUTIVE BLOCK MINIMIZATION

INSTANCE: An $m \times n$ matrix A of 0's and 1's and a positive integer K .

QUESTION: Is there a permutation of the columns of A that results in a matrix B having at most K blocks of consecutive 1's, i.e., having at most K entries b_{ij} such that $b_{ij} = 1$ and either $b_{i,j+1} = 0$ or $j = n$?

Reference: [Kou, 1977]. Transformation from HAMILTONIAN PATH.

Comment: Remains NP-complete if " $j = n$ " is replaced by " $j = n$ and $b_{i1} = 0$ " [Booth, 1975]. If K equals the number of rows of A that are not all 0, then these problems are equivalent to testing A for the consecutive ones property or the circular ones property, respectively, and can be solved in polynomial time.

[SR18] CONSECUTIVE SETS

INSTANCE: Finite alphabet Σ , collection $C = \{\Sigma_1, \Sigma_2, \dots, \Sigma_n\}$ of subsets of Σ , and a positive integer K .

QUESTION: Is there a string $w \in \Sigma^*$ with $|w| \leq K$ such that, for each i , the elements of Σ_i occur in a consecutive block of $|\Sigma_i|$ symbols of w ?

Reference: [Kou, 1977]. Transformation from HAMILTONIAN PATH.

Comment: The variant in which we ask only that the elements of each Σ_i occur in a consecutive block of $|\Sigma_i|$ symbols of the string ww (i.e., we allow blocks that circulate from the end of w back to its beginning) is also NP-complete [Booth, 1975]. If K is the number of distinct symbols in the Σ_i , then these problems are equivalent to determining whether a matrix has the consecutive ones property or the circular ones property and are solvable in polynomial time.

[SR19] 2-DIMENSIONAL CONSECUTIVE SETS

INSTANCE: Finite alphabet Σ , collection $C = \{\Sigma_1, \Sigma_2, \dots, \Sigma_n\}$ of subsets of Σ .

QUESTION: Is there a partition of Σ into disjoint sets $X_1, X_2, \dots, X_k$ such that each X_i has at most one element in common with each Σ_j and such that, for each $\Sigma_j \in C$, there is an index $l(j)$ such that Σ_j is contained in

$$X_{l(j)} \cup X_{l(j)+1} \cup \dots \cup X_{l(j)+|\Sigma_j|-1}?$$

Reference: [Lipsky, 1977b]. Transformation from GRAPH 3-COLORABILITY.

Comment: Remains NP-complete if all $\Sigma_j \in C$ have $|\Sigma_j| \leq 5$, but is solvable in polynomial time if all $\Sigma_j \in C$ have $|\Sigma_j| \leq 2$.

[SR20] STRING-TO-STRING CORRECTION

INSTANCE: Finite alphabet Σ , two strings $x, y \in \Sigma^*$, and a positive integer K .

QUESTION: Is there a way to derive the string y from the string x by a sequence of K or fewer operations of single symbol deletion or adjacent symbol interchange?

Reference: [Wagner, 1975]. Transformation from SET COVERING.

Comment: Solvable in polynomial time if the operation set is expanded to include the operations of changing a single character and of inserting a single character,

even if interchanges are not allowed (e.g., see [Wagner and Fischer, 1974]), or if the *only* operation is adjacent symbol interchange [Wagner, 1975]. See reference for related results for cases in which different operations can have different costs.

[SR21] GROUPING BY SWAPPING

INSTANCE: Finite alphabet Σ , string $x \in \Sigma^*$, and a positive integer K .

QUESTION: Is there a sequence of K or fewer adjacent symbol interchanges that converts x into a string y in which all occurrences of each symbol $a \in \Sigma$ are in a single block, i.e., y has no subsequences of the form aba for $a, b \in \Sigma$ and $a \neq b$?

Reference: [Howell, 1977]. Transformation from FEEDBACK EDGE SET.

[SR22] EXTERNAL MACRO DATA COMPRESSION

INSTANCE: Alphabet Σ , string $s \in \Sigma^*$, pointer cost $h \in \mathbb{Z}^+$, and a bound $B \in \mathbb{Z}^+$.

QUESTION: Are there strings D (dictionary string) and C (compressed string) in $(\Sigma \cup \{p_i; 1 \leq i \leq |s|\})^*$, where the symbols p_i are "pointers," such that

$$|D| + |C| + (h-1) \cdot (\text{number of occurrences of pointers in } D \text{ and } C) \leq B$$

and such that there is a way of identifying pointers with substrings of D so that S can be obtained from C by repeatedly replacing pointers in C by their corresponding substrings in D ?

Reference: [Storer, 1977], [Storer and Szymanski, 1978]. Transformation from VERTEX COVER.

Comment: Remains NP-complete even if h is any fixed integer 2 or greater. Many variants, including those in which D can contain no pointers and/or no pointers can refer to overlapping strings, are also NP-complete. If the alphabet size is fixed at 3 or greater, and the pointer cost is $\lceil h \cdot \log |s| \rceil$, the problem is also NP-complete. For further variants, including the case of "original pointers," see references.

[SR23] INTERNAL MACRO DATA COMPRESSION

INSTANCE: Alphabet Σ , string $s \in \Sigma^*$, pointer cost $h \in \mathbb{Z}^+$, and a bound $B \in \mathbb{Z}^+$.

QUESTION: Is there a single string $C \in (\Sigma \cup \{p_i; 1 \leq i \leq |s|\})^*$ such that

$$|C| + (h-1) \cdot (\text{number of occurrences of pointers in } C) \leq B$$

and such that there is a way of identifying pointers with substrings of C so that s can be obtained from C by using C as both compressed string and dictionary string in the manner indicated in the previous problem?

Reference: [Storer, 1977], [Storer and Szymanski, 1978]. Transformation from VERTEX COVER.

Comment: Remains NP-complete even if h is any fixed integer 2 or greater. For other NP-complete variants (as in the previous problem), see references.

[SR24] REGULAR EXPRESSION SUBSTITUTION

INSTANCE: Two finite alphabets $X = \{x_1, x_2, \dots, x_n\}$ and $Y = \{y_1, y_2, \dots, y_m\}$, a regular expression R over $X \cup Y$, regular expressions $R_1, R_2, \dots, R_n$ over Y , and a string $w \in Y^*$.

QUESTION: Is there a string z in the language determined by R and for each i ,

$1 \leq i \leq n$, a string w_i in the language determined by R_i such that, if each string w_i is substituted for every occurrence of the symbol x_i in z , then the resulting string is identical to w ?

Reference: [Aho and Ullman, 1977]. Transformation from X3C.

[SR25] RECTILINEAR PICTURE COMPRESSION

INSTANCE: An $n \times n$ matrix M of 0's and 1's, and a positive integer K .

QUESTION: Is there a collection of K or fewer rectangles that covers precisely those entries in M that are 1's, i.e., is there a sequence of quadruples (a_i, b_i, c_i, d_i) , $1 \leq i \leq K$, where $a_i \leq b_i$, $c_i \leq d_i$, $1 \leq i \leq K$, such that for every pair (i, j) , $1 \leq i, j \leq n$, $M_{ij} = 1$ if and only if there exists a k , $1 \leq k \leq K$, such that $a_k \leq i \leq b_k$ and $c_k \leq j \leq d_k$?

Reference: [Masek, 1978]. Transformation from 3SAT.

A4.3 DATABASE PROBLEMS

[SR26] MINIMUM CARDINALITY KEY

INSTANCE: A set A of "attribute names," a collection F of ordered pairs of subsets of A (called "functional dependencies" on A), and a positive integer M .

QUESTION: Is there a key of cardinality M or less for the relational system $\langle A, F \rangle$, i.e., a minimal subset $K \subseteq A$ with $|K| \leq M$ such that the ordered pair (K, A) belongs to the "closure" F^* of F defined by (1) $F \subseteq F^*$, (2) $B \subseteq C \subseteq A$ implies $(C, B) \in F^*$, (3) $(B, C), (C, D) \in F^*$ implies $(B, D) \in F^*$, and (4) $(B, C), (B, D) \in F^*$ implies $(B, C \cup D) \in F^*$?

Reference: [Lucchesi and Osborne, 1977], [Lipsky, 1977a]. Transformation from VERTEX COVER. See [Date, 1975] for general background on relational data bases.

[SR27] ADDITIONAL KEY

INSTANCE: A set A of attribute names, a collection F of functional dependencies on A , a subset $R \subseteq A$, and a set K of keys for the relational scheme $\langle R, F \rangle$.

QUESTION: Does R have a key not already contained in K , i.e., is there an $R' \subseteq R$ such that $R' \notin K$, $(R', R) \in F^*$, and for no $R'' \subseteq R'$ is $(R'', R) \in F^*$?

Reference: [Beeri and Bernstein, 1978]. Transformation from HITTING SET.

[SR28] PRIME ATTRIBUTE NAME

INSTANCE: A set A of attribute names, a collection F of functional dependencies on A , and a specified name $x \in A$.

QUESTION: Is x a "prime attribute name" for $\langle A, F \rangle$, i.e., is there a key K for $\langle A, F \rangle$ such that $x \in K$?

Reference: [Lucchesi and Osborne, 1977]. Transformation from MINIMUM CARDINALITY KEY.

[SR29] BOYCE-CODD NORMAL FORM VIOLATION

INSTANCE: A set A of attribute names, a collection F of functional dependencies on A , and a subset $A' \subseteq A$.

QUESTION: Does A' violate Boyce-Codd normal form for the relational system $\langle A, F \rangle$, i.e., is there a subset $X \subseteq A'$ and two attribute names $y, z \in A' - X$ such that $(X, \{y\}) \in F^*$ and $(X, \{z\}) \notin F^*$, where F^* is the closure of F ?

Reference: [Bernstein and Beeri, 1976], [Beeri and Bernstein, 1978]. Transformation from HITTING SET.

Comment: Remains NP-complete even if A' is required to satisfy "third normal form," i.e., if $X \subseteq A'$ is a key for the system $\langle A', F \rangle$ and if two names $y, z \in A' - X$ satisfy $(X, \{y\}) \in F^*$ and $(X, \{z\}) \notin F^*$, then z is a prime attribute for $\langle A', F \rangle$.

[SR30] CONJUNCTIVE QUERY FOLDABILITY

INSTANCE: Finite domain set D , a collection $R = \{R_1, R_2, \dots, R_m\}$ of relations, where each R_i consists of a set of d_i -tuples with entries from D , a set X of distinguished variables, a set Y of undistinguished variables, and two "queries" Q_1 and Q_2 over X, Y, D , and R , where a query Q has the form

$$(x_1, x_2, \dots, x_k)(\exists y_1, y_2, \dots, y_l)(A_1 \wedge A_2 \wedge \dots \wedge A_r)$$

for some k, l , and r , with $X' = \{x_1, x_2, \dots, x_k\} \subseteq X$, $Y' = \{y_1, y_2, \dots, y_l\} \subseteq Y$, and each A_i of the form $R_j(u_1, u_2, \dots, u_{d_j})$ with each $u \in D \cup X' \cup Y'$ (see reference for interpretation of such expressions in terms of data bases).

QUESTION: Is there a function $\sigma: Y \rightarrow X \cup Y \cup D$ such that, if for each $y \in Y$ the symbol $\sigma(y)$ is substituted for every occurrence of y in Q_1 , then the result is query Q_2 ?

Reference: [Chandra and Merlin, 1977]. Transformation from GRAPH 3-COLORABILITY.

Comment: The isomorphism problem for conjunctive queries (with two queries being isomorphic if they are the same up to one-to-one renaming of the variables, reordering of conjuncts, and reordering within quantifications) is polynomially equivalent to graph isomorphism.

[SR31] CONJUNCTIVE BOOLEAN QUERY

INSTANCE: Finite domain set D , a collection $R = \{R_1, R_2, \dots, R_m\}$ of relations, where each R_i consists of a set of d_i -tuples with entries from D , and a conjunctive Boolean query Q over R and D , where such a query Q is of the form

$$(\exists y_1, y_2, \dots, y_l)(A_1 \wedge A_2 \wedge \dots \wedge A_r)$$

with each A_i of the form $R_j(u_1, u_2, \dots, u_{d_j})$ where each $u \in \{y_1, y_2, \dots, y_l\} \cup D$.

QUESTION: Is Q , when interpreted as a statement about R and D , true?

Reference: [Chandra and Merlin, 1977]. Transformation from CLIQUE.

Comment: If we are allowed to replace the conjunctive query Q by an arbitrary first-order sentence involving the predicates in R , then the problem becomes PSPACE-complete, even for $D = \{0, 1\}$.

[SR32] TABLEAU EQUIVALENCE

INSTANCE: A set A of attribute names, a collection F of ordered pairs of subsets of A , a set X of distinguished variables, a set Y of undistinguished variables, a set C_a of constants for each $a \in A$, and two "tableaux" T_1 and T_2 over X, Y , and the C_a . (A tableau is essentially a matrix with a column for each attribute and entries from X, Y , and the C_a , along with a blank symbol. For details and an interpretation in terms of relational expressions, see reference.)

QUESTION: Are T_1 and T_2 "weakly equivalent," i.e., do they represent identical relations under "universal interpretations"?

Reference: [Aho, Sagiv, and Ullman, 1978]. Transformation from 3SAT.

Comment: Remains NP-complete even if the tableaux come from "expressions" that have no "select" operations, or if the tableaux come from expressions that have select operations but F is empty, or if F is empty, the tableaux contain no constants, and the tableaux do not necessarily come from expressions at all. Problem is solvable in polynomial time for "simple" tableaux. The same results hold also for "strong equivalence," where the two tableaux must represent identical relations under *all* interpretations. The problem of tableau "containment," however, is NP-complete even for simple tableaux and for still further restricted tableaux [Sagiv and Yannakakis, 1978].

[SR33] SERIALIZABILITY OF DATABASE HISTORIES

INSTANCE: Set V of database variables, collection T of "transactions" (R_i, W_i) , $1 \leq i \leq n$, where R_i and W_i are both subsets of V (called the "read set" and the "write set," respectively), and a "history" H for T , where a history is simply a permutation of all the R_i and the W_i in which each R_i occurs before the corresponding W_i .

QUESTION: Is there a serial history H' for T (i.e., a history in which each R_i occurs immediately before the corresponding W_i) that is equivalent to H in the sense that (1) both histories have the same set of "live" transactions (where a transaction (R_i, W_i) is live in a history if there is some $v \in V$ such that either W_i is the last write set to contain v or W_i is the last write set to contain v before v appears in the read set of some other live transaction), and (2) for any two live transactions (R_i, W_i) and (R_j, W_j) and any $v \in W_i \cap R_j$, W_i is the last write set to contain v before R_j in H if and only if W_i is the last write set to contain v before R_j in H' ?

Reference: [Papadimitriou, Bernstein, and Rothnie, 1977], [Papadimitriou, 1978c]. Transformation from MONOTONE 3SAT.

Comment: For related polynomial time solvable subcases and variants, see [Papadimitriou, 1978c].

[SR34] SAFETY OF DATABASE TRANSACTION SYSTEMS (*)

INSTANCE: Set V of database variables, and a collection T of "transactions" (R_i, W_i) , $1 \leq i \leq n$, where R_i and W_i are both subsets of V .

QUESTION: Is every history H for T equivalent to some serial history?

Reference: [Papadimitriou, Bernstein, and Rothnie, 1977]. Transformation from HITTING SET.

Comment: Not known either to be in NP or to be in co-NP. Testing whether every history H for T is " D -equivalent" to some serial history can be done in polynomial

time, where two histories are D -equivalent if one can be obtained from the other by a sequence of interchanges of adjacent sets in such a way that at each step the new history is equivalent to the previous one.

[SR35] CONSISTENCY OF DATABASE FREQUENCY TABLES

INSTANCE: Set A of attribute names, domain set D_a for each $a \in A$, set V of objects, collection F of frequency tables for some pairs $a, b \in A$ (where a frequency table for $a, b \in A$ is a function $f_{a,b}: D_a \times D_b \rightarrow Z^+$ with the sum, over all pairs $x \in D_a$ and $y \in D_b$, of $f_{a,b}(x, y)$ equal to $|V|$), and a set K of triples (v, a, x) with $v \in V$, $a \in A$, and $x \in D_a$, representing the known attribute values.

QUESTION: Are the frequency tables in F consistent with the known attribute values in K , i.e., is there a collection of functions $g_a: V \rightarrow D_a$, for each $a \in A$, such that $g_a(v) = x$ if $(v, a, x) \in K$ and such that, for each $f_{a,b} \in F$, $x \in D_a$, and $y \in D_b$, the number of $v \in V$ for which $g_a(v) = x$ and $g_b(v) = y$ is exactly $f_{a,b}(x, y)$?

Reference: [Reiss, 1977b]. Transformation from 3SAT.

Comment: Above result implies that no polynomial time algorithm can be given for "compromising" a data base from its frequency tables by deducing prespecified attribute values, unless $P = NP$ (see reference for details).

[SR36] SAFETY OF FILE PROTECTION SYSTEMS (*)

INSTANCE: Set R of "rights," set O of objects, set $S \subseteq O$ of subjects, set $P(s, o) \subseteq R$ of rights for each ordered pair $s \in S$ and $o \in O$, a finite set C of commands, each having the form "if $r_1 \in P(X_1, Y_1)$ and $r_2 \in P(X_2, Y_2)$ and $\dots$ and $r_m \in P(X_m, Y_m)$, then $\theta_1, \theta_2, \dots, \theta_n$ " for $m, n \geq 0$ and each θ_i of the form "enter r_i into $P(X_i, Y_i)$ " or "delete r_i from $P(X_i, Y_i)$," and a specified right $r' \in R$.

QUESTION: Is there a sequence of commands from C and a way of identifying each r_i, X_i , and Y_i with a particular element of R, S , and O , respectively, such that at some point in the execution of the sequence, the right r' is entered into a set $P(s, o)$ that previously did not contain r' (see reference for details on the execution of such a sequence)?

Reference: [Harrison, Ruzzo, and Ullman, 1976]. Transformation from LINEAR BOUNDED AUTOMATON ACCEPTANCE.

Comment: PSPACE-complete. Undecidable if operations that create or delete "subjects" and "objects" are allowed, even for certain "fixed" systems in which only the initial values of the $P(s, o)$ are allowed to vary. If no command can contain more than one operation, then the problem is NP-complete in general and solvable in polynomial time for fixed systems.

A5 SEQUENCING AND SCHEDULING

A5.1 SEQUENCING ON ONE PROCESSOR

[SS1] SEQUENCING WITH RELEASE TIMES AND DEADLINES

INSTANCE: Set T of tasks and, for each task $t \in T$, a length $l(t) \in \mathbb{Z}^+$, a release time $r(t) \in \mathbb{Z}_0^+$, and a deadline $d(t) \in \mathbb{Z}^+$.

QUESTION: Is there a one-processor schedule for T that satisfies the release time constraints and meets all the deadlines, i.e., a one-to-one function $\sigma: T \rightarrow \mathbb{Z}_0^+$, with $\sigma(t) > \sigma(t')$ implying $\sigma(t) \geq \sigma(t') + l(t')$, such that, for all $t \in T$, $\sigma(t) \geq r(t)$ and $\sigma(t) + l(t) \leq d(t)$?

Reference: [Garey and Johnson, 1977b]. Transformation from 3-PARTITION (see Section 4.2).

Comment: NP-complete in the strong sense. Solvable in pseudo-polynomial time if the number of allowed values for $r(t)$ and $d(t)$ is bounded by a constant, but remains NP-complete (in the ordinary sense) even when each can take on only two values. If all task lengths are 1, or "preemptions" are allowed, or all release times are 0, the general problem can be solved in polynomial time, even under "precedence constraints" [Lawler, 1973], [Lageweg, Lenstra, and Rinnooy Kan, 1976]. Can also be solved in polynomial time even if release times and deadlines are allowed to be arbitrary rationals and there are precedence constraints, so long as all tasks have equal length [Carlier, 1978], [Simons, 1978], [Garey, Johnson, Simons, and Tarjan, 1978], or preemptions are allowed [Blazewicz, 1976].

[SS2] SEQUENCING TO MINIMIZE TARDY TASKS

INSTANCE: Set T of tasks, partial order $<$ on T , for each task $t \in T$ a length $l(t) \in \mathbb{Z}^+$ and a deadline $d(t) \in \mathbb{Z}^+$, and a positive integer $K \leq |T|$.

QUESTION: Is there a one-processor schedule σ for T that obeys the precedence constraints, i.e., such that $t < t'$ implies $\sigma(t) + l(t) < \sigma(t')$, and such that there are at most K tasks $t \in T$ for which $\sigma(t) + l(t) > d(t)$?

Reference: [Garey and Johnson, 1976c]. Transformation from CLIQUE (see Section 3.2.3).

Comment: Remains NP-complete even if all task lengths are 1 and $<$ consists only of "chains" (each task has at most one immediate predecessor and at most one immediate successor) [Lenstra, 1977]. The general problem can be solved in polynomial time if $K=0$ [Lawler, 1973], or if $<$ is empty [Moore, 1968] [Sidney, 1973]. The $<$ empty case remains polynomially solvable if "agreeable" release times (i.e., $r(t) < r(t')$ implies $d(t) \leq d(t')$) are added [Kise, Ibaraki, and Mine, 1978], but is NP-complete for arbitrary release times (see previous problem).

[SS3] SEQUENCING TO MINIMIZE TARDY TASK WEIGHT

INSTANCE: Set T of tasks, for each task $t \in T$ a length $l(t) \in \mathbb{Z}^+$, a weight $w(t) \in \mathbb{Z}^+$, and a deadline $d(t) \in \mathbb{Z}^+$, and a positive integer K .

QUESTION: Is there a one-processor schedule σ for T such that the sum of $w(t)$, taken over all $t \in T$ for which $\sigma(t) + l(t) > d(t)$, does not exceed K ?

Reference: [Karp, 1972]. Transformation from PARTITION.

Comment: Can be solved in pseudo-polynomial time (time polynomial in $|T|$,

$\sum l(t)$, and $\log \sum w(t)$) [Lawler and Moore, 1969]. Can be solved in polynomial time if weights are "agreeable" (i.e., $w(t) < w(t')$ implies $l(t) \geq l(t')$) [Lawler, 1976c].

[SS4] SEQUENCING TO MINIMIZE WEIGHTED COMPLETION TIME

INSTANCE: Set T of tasks, partial order $<$ on T , for each task $t \in T$ a length $l(t) \in \mathbb{Z}^+$ and a weight $w(t) \in \mathbb{Z}^+$, and a positive integer K .

QUESTION: Is there a one-processor schedule σ for T that obeys the precedence constraints and for which the sum, over all $t \in T$, of $(\sigma(t) + l(t)) \cdot w(t)$ is K or less?

Reference: [Lawler, 1978]. Transformation from OPTIMAL LINEAR ARRANGEMENT.

Comment: NP-complete in the strong sense and remains so even if all task lengths are 1 or all task weights are 1. Can be solved in polynomial time for $<$ a "forest" [Horn, 1972], [Adolphson and Hu, 1973], [Garey, 1973], [Sidney, 1975] or if $<$ is "series-parallel" or "generalized series-parallel" [Knuth, 1973], [Lawler, 1978], [Adolphson, 1977], [Monma and Sidney, 1977]. If the partial order $<$ is replaced by individual task deadlines, the resulting problem is NP-complete in the strong sense [Lenstra, 1977], but can be solved in polynomial time if all task weights are equal [Smith, 1956]. If there are individual task release times instead of deadline, the resulting problem is NP-complete in the strong sense, even if all task weights are 1 [Lenstra, Rinnooy Kan, and Brucker, 1977]. The "preemptive" version of this latter problem is NP-complete in the strong sense [Labetoulle, Lawler, Lenstra, and Rinnooy Kan, 1978], but is solvable in polynomial time if all weights are equal [Graham, Lawler, Lenstra, and Rinnooy Kan, 1978].

[SS5] SEQUENCING TO MINIMIZE WEIGHTED TARDINESS

INSTANCE: Set T of tasks, for each task $t \in T$ a length $l(t) \in \mathbb{Z}^+$, a weight $w(t) \in \mathbb{Z}^+$, and a deadline $d(t) \in \mathbb{Z}^+$, and a positive integer K .

QUESTION: Is there a one-processor schedule σ for T such that the sum, taken over all $t \in T$ satisfying $\sigma(t) + l(t) > d(t)$, of $(\sigma(t) + l(t) - d(t)) \cdot w(t)$ is K or less?

Reference: [Lawler, 1977a]. Transformation from 3-PARTITION.

Comment: NP-complete in the strong sense. If all weights are equal, the problem can be solved in pseudo-polynomial time [Lawler, 1977a] and is open as to ordinary NP-completeness. If all lengths are equal (with weights arbitrary), it can be solved in polynomial time by bipartite matching. If precedence constraints are added, the problem is NP-complete even with equal lengths and equal weights [Lenstra and Rinnooy Kan, 1978a]. If release times are added instead, the problem is NP-complete in the strong sense for equal task weights (see SEQUENCING WITH RELEASE TIMES AND DEADLINES), but can be solved by bipartite matching for equal lengths and arbitrary weights [Graham, Lawler, Lenstra, and Rinnooy Kan, 1978].

[SS6] SEQUENCING WITH DEADLINES AND SET-UP TIMES

INSTANCE: Set C of "compilers," set T of tasks, for each $t \in T$ a length $l(t) \in \mathbb{Z}^+$, a deadline $d(t) \in \mathbb{Z}^+$, and a compiler $k(t) \in C$, and for each $c \in C$ a "set-up time" $l(c) \in \mathbb{Z}_0^+$.

QUESTION: Is there a one-processor schedule σ for T that meets all the task deadlines and that satisfies the additional constraint that, whenever two tasks t and t' with $\sigma(t) < \sigma(t')$ are scheduled "consecutively" (i.e., no other task t'' has $\sigma(t) < \sigma(t'') < \sigma(t')$) and have different compilers (i.e., $k(t) \neq k(t')$), then $\sigma(t') \geq \sigma(t) + l(t) + l(k(t'))$?

Reference: [Bruno and Downey, 1978]. Transformation from PARTITION.

Comment: Remains NP-complete even if all set-up times are equal. The related problem in which set-up times are replaced by "changeover costs," and we want to know if there is a schedule that meets all the deadlines and has total changeover cost at most K , is NP-complete even if all changeover costs are equal. Both problems can be solved in pseudo-polynomial time when the number of distinct deadlines is bounded by a constant. If the number of deadlines is unbounded, it is open whether these problems are NP-complete in the strong sense.

[SS7] SEQUENCING TO MINIMIZE MAXIMUM CUMULATIVE COST

INSTANCE: Set T of tasks, partial order $<$ on T , a "cost" $c(t) \in \mathbb{Z}$ for each $t \in T$ (if $c(t) < 0$, it can be viewed as a "profit"), and a constant $K \in \mathbb{Z}$.

QUESTION: Is there a one-processor schedule σ for T that obeys the precedence constraints and which has the property that, for every task $t \in T$, the sum of the costs for all tasks t' with $\sigma(t') \leq \sigma(t)$ is at most K ?

Reference: [Abdel-Wahab, 1976]. Transformation from REGISTER SUFFICIENCY.

Comment: Remains NP-complete even if $c(t) \in \{-1, 0, 1\}$ for all $t \in T$. Can be solved in polynomial time if $<$ is series-parallel [Abdel-Wahab and Kameda, 1978], [Monma and Sidney, 1977].

A5.2 MULTIPROCESSOR SCHEDULING**[SS8] MULTIPROCESSOR SCHEDULING**

INSTANCE: Set T of tasks, number $m \in \mathbb{Z}^+$ of processors, length $l(t) \in \mathbb{Z}^+$ for each $t \in T$, and a deadline $D \in \mathbb{Z}^+$.

QUESTION: Is there an m -processor schedule for T that meets the overall deadline D , i.e., a function $\sigma: T \rightarrow \mathbb{Z}_0^+$ such that, for all $u \geq 0$, the number of tasks $t \in T$ for which $\sigma(t) \leq u < \sigma(t) + l(t)$ is no more than m and such that, for all $t \in T$, $\sigma(t) + l(t) \leq D$?

Reference: Transformation from PARTITION (see Section 3.2.1).

Comment: Remains NP-complete for $m=2$, but can be solved in pseudo-polynomial time for any fixed m . NP-complete in the strong sense for m arbitrary (3-PARTITION is a special case). If all tasks have the same length, then this problem is trivial to solve in polynomial time, even for "different speed" processors.

[SS9] PRECEDENCE CONSTRAINED SCHEDULING

INSTANCE: Set T of tasks, each having length $l(t)=1$, number $m \in \mathbb{Z}^+$ of processors, partial order $<$ on T , and a deadline $D \in \mathbb{Z}^+$.

QUESTION: Is there an m -processor schedule σ for T that meets the overall deadline D and obeys the precedence constraints, i.e., such that $t < t'$ implies $\sigma(t') \geq \sigma(t) + l(t) = \sigma(t) + 1$?

Reference: [Ullman, 1975]. Transformation from 3SAT.

Comment: Remains NP-complete for $D=3$ [Lenstra and Rinnooy Kan, 1978a]. Can be solved in polynomial time if $m=2$ (e.g., see [Coffman and Graham, 1972]) or if m is arbitrary and $<$ is a "forest" [Hu, 1961] or has a chordal graph as complement [Papadimitriou and Yannakakis, 1978b]. Complexity remains open for all fixed $m \geq 3$ when $<$ is arbitrary. The $m=2$ case becomes NP-complete if both task lengths 1 and 2 are allowed [Ullman, 1975]. If each task t can only be executed by a specified processor $p(t)$, the problem is NP-complete for $m=2$ and $<$ arbitrary, and for m arbitrary and $<$ a forest, but can be solved in polynomial time for m arbitrary if $<$ is a "cyclic forest" [Goyal, 1976].

[SS10] RESOURCE CONSTRAINED SCHEDULING

INSTANCE: Set T of tasks, each having length $l(t)=1$, number $m \in \mathbb{Z}^+$ of processors, number $r \in \mathbb{Z}^+$ of resources, resource bounds B_i , $1 \leq i \leq r$, resource requirement $R_i(t)$, $0 \leq R_i(t) \leq B_i$, for each task t and resource i , and an overall deadline $D \in \mathbb{Z}^+$.

QUESTION: Is there an m -processor schedule σ for T that meets the overall deadline D and obeys the resource constraints, i.e., such that for all $u \geq 0$, if $S(u)$ is the set of all $t \in T$ for which $\sigma(t) \leq u < \sigma(t) + l(t)$, then for each resource i the sum of $R_i(t)$ over all $t \in S(u)$ is at most B_i ?

Reference: [Garey and Johnson, 1975]. Transformation from 3-PARTITION.

Comment: NP-complete in the strong sense, even if $r=1$ and $m=3$. Can be solved in polynomial time by matching for $m=2$ and r arbitrary. If a partial order $<$ is added, the problem becomes NP-complete in the strong sense for $r=1$, $m=2$, and $<$ a "forest." If each resource requirement is restricted to be either 0 or B_i , the problem is NP-complete for $m=2$, $r=1$, and $<$ arbitrary [Ullman, 1976].

[SS11] SCHEDULING WITH INDIVIDUAL DEADLINES

INSTANCE: Set T of tasks, each having length $l(t)=1$, number $m \in \mathbb{Z}^+$ of processors, partial order $<$ on T , and for each task $t \in T$ a deadline $d(t) \in \mathbb{Z}^+$.

QUESTION: Is there an m -processor schedule σ for T that obeys the precedence constraints and meets all the deadlines, i.e., $\sigma(t) + l(t) \leq d(t)$ for all $t \in T$?

Reference: [Brucker, Garey, and Johnson, 1977]. Transformation from VERTEX COVER.

Comment: Remains NP-complete even if $<$ is an "out-tree" partial order (no task has more than one immediate predecessor), but can be solved in polynomial time if $<$ is an "in-tree" partial order (no task has more than one immediate successor). Solvable in polynomial time if $m=2$ and $<$ is arbitrary [Garey and Johnson, 1976c], even if individual release times are included [Garey and Johnson, 1977b]. For $<$ empty, can be solved in polynomial time by matching for m arbitrary, even

with release times and with a single resource having 0-1 valued requirements [Blazewicz, 1977b], [Blazewicz, 1978].

[SS12] PREEMPTIVE SCHEDULING

INSTANCE: Set T of tasks, number $m \in \mathbb{Z}^+$ of processors, partial order $<$ on T , length $l(t) \in \mathbb{Z}^+$ for each $t \in T$, and an overall deadline $D \in \mathbb{Z}^+$.

QUESTION: Is there an m -processor "preemptive" schedule for T that obeys the precedence constraints and meets the overall deadline? (Such a schedule σ is identical to an ordinary m -processor schedule, except that we are allowed to subdivide each task $t \in T$ into any number of subtasks $t_1, t_2, \dots, t_k$ such that $\sum_{i=1}^k l(t_i) = l(t)$ and it is required that $\sigma(t_i + 1) \geq \sigma(t_i) + l(t_i)$ for $1 \leq i < k$. The precedence constraints are extended to subtasks by requiring that every subtask of t precede every subtask of t' whenever $t < t'$.)

Reference: [Ullman, 1975]. Transformation from 3SAT.

Comment: Can be solved in polynomial time if $m=2$ [Muntz and Coffman, 1969], if $<$ is a "forest" [Muntz and Coffman, 1970], or if $<$ is empty and individual task deadlines are allowed [Horn, 1974]. If "(uniform) different speed" processors are allowed, the problem can be solved in polynomial time if $m=2$ or if $<$ is empty [Horvath, Lam, and Sethi, 1977], [Gonzalez and Sahni, 1978b] in the latter case even if individual task deadlines are allowed [Sahni and Cho, 1977a]; if both $m=2$ and $<$ is empty, it can be solved in polynomial time, even if both integer release times and deadlines are allowed [Labetoulle, Lawler, Lenstra, and Rinnooy Kan, 1977]. For "unrelated" processors, the case with m fixed and $<$ empty can be solved in polynomial time [Gonzalez, Lawler, and Sahni, 1978], and the case with m arbitrary and $<$ empty can be solved by linear programming [Lawler and Labetoulle, 1978].

[SS13] SCHEDULING TO MINIMIZE WEIGHTED COMPLETION TIME

INSTANCE: Set T of tasks, number $m \in \mathbb{Z}^+$ of processors, for each task $t \in T$ a length $l(t) \in \mathbb{Z}^+$ and a weight $w(t) \in \mathbb{Z}^+$, and a positive integer K .

QUESTION: Is there an m -processor schedule σ for T such that the sum, over all $t \in T$, of $(\sigma(t) + l(t)) \cdot w(t)$ is no more than K ?

Reference: [Lenstra, Rinnooy Kan, and Brucker, 1977]. Transformation from PARTITION.

Comment: Remains NP-complete for $m=2$, and is NP-complete in the strong sense for m arbitrary [Lageweg and Lenstra, 1977]. The problem is solvable in pseudo-polynomial time for fixed m . These results continue to hold if "preemptive" schedules are allowed [McNaughton, 1959]. Can be solved in polynomial time if all lengths are equal (by matching techniques). If instead all weights are equal, it can be solved in polynomial time even for "different speed" processors [Conway, Maxwell, and Miller, 1967] and for "unrelated" processors [Horn, 1973], [Bruno, Coffman, and Sethi, 1974]. The "preemptive" case for different speed processors also can be solved in polynomial time [Gonzalez, 1977]. If precedence constraints are allowed, the original problem is NP-complete in the strong sense even if all weights are equal, $m=2$, and the partial order is either an "in-tree" or an "out-tree" [Sethi, 1977a]. If resources are allowed, the same subcases men-

tioned under RESOURCE CONSTRAINED SCHEDULING are NP-complete, even for equal weights [Blazewicz, 1977a].

A5.3 SHOP SCHEDULING

[SS14] OPEN-SHOP SCHEDULING

INSTANCE: Number $m \in \mathbb{Z}^+$ of processors, set J of jobs, each job $j \in J$ consisting of m tasks $t_1[j], t_2[j], \dots, t_m[j]$ (with $t_i[j]$ to be executed by processor i), a length $l(t) \in \mathbb{Z}_0^+$ for each such task t , and an overall deadline $D \in \mathbb{Z}^+$.

QUESTION: Is there an open-shop schedule for J that meets the deadline, i.e., a collection of one-processor schedules $\sigma_i: J \rightarrow \mathbb{Z}_0^+$, $1 \leq i \leq m$, such that $\sigma_i(j) > \sigma_i(k)$ implies $\sigma_i(j) \geq \sigma_i(k) + l(t_i[k])$, such that for each $j \in J$ the intervals $[\sigma_i(j), \sigma_i(j) + l(t_i[j])]$ are all disjoint, and such that $\sigma_i(j) + l(t_i[j]) \leq D$ for $1 \leq i \leq m$, $1 \leq j \leq |J|$?

Reference: [Gonzalez and Sahni, 1976]. Transformation from PARTITION.

Comment: Remains NP-complete if $m=3$, but can be solved in polynomial time if $m=2$. NP-complete in the strong sense for m arbitrary [Lenstra, 1977]. The general problem is solvable in polynomial time if "preemptive" schedules are allowed [Gonzalez and Sahni, 1976], even if two distinct release times are allowed [Cho and Sahni, 1978]. The $m=2$ preemptive case can be solved in polynomial time even if arbitrary release times are allowed, and the general preemptive case with arbitrary release times and deadlines can be solved by linear programming [Cho and Sahni, 1978].

[SS15] FLOW-SHOP SCHEDULING

INSTANCE: Number $m \in \mathbb{Z}^+$ of processors, set J of jobs, each job $j \in J$ consisting of m tasks $t_1[j], t_2[j], \dots, t_m[j]$, a length $l(t) \in \mathbb{Z}_0^+$ for each such task t , and an overall deadline $D \in \mathbb{Z}^+$.

QUESTION: Is there a flow-shop schedule for J that meets the overall deadline, where such a schedule is identical to an open-shop schedule with the additional constraint that, for each $j \in J$ and $1 \leq i < m$, $\sigma_{i+1}(j) \geq \sigma_i(j) + l(t_i[j])$?

Reference: [Garey, Johnson, and Sethi, 1976]. Transformation from 3-PARTITION.

Comment: NP-complete in the strong sense for $m=3$. Solvable in polynomial time for $m=2$ [Johnson, 1954]. The same results hold if "preemptive" schedules are allowed [Gonzalez and Sahni, 1978a], although if release times are added in this case, the problem is NP-complete in the strong sense, even for $m=2$ [Cho and Sahni, 1978]. If the goal is to meet a bound K on the sum, over all $j \in J$, of $\sigma_m(j) + l(t_m[j])$, then the non-preemptive problem is NP-complete in the strong sense even if $m=2$ [Garey, Johnson, and Sethi, 1976].

[SS16] NO-WAIT FLOW-SHOP SCHEDULING

INSTANCE: (Same as for FLOW-SHOP SCHEDULING).

QUESTION: Is there a flow-shop schedule for J that meets the overall deadline and has the property that, for each $j \in J$ and $1 \leq i < m$, $\sigma_{i+1}(j) = \sigma_i(j) + l(t_i[j])$?

Reference: [Lenstra, Rinnooy Kan, and Brucker, 1977]. Transformation from DIRECTED HAMILTONIAN PATH.

Comment: NP-complete in the strong sense for any fixed $m \geq 4$ [Papadimitriou and Kanellakis, 1978]. Solvable in polynomial time for $m=2$ [Gilmore and Gomory, 1964]. (However, NP-complete in the strong sense for $m=2$ if jobs with no tasks on the first processor are allowed [Sahni and Cho, 1977b].) Open for fixed $m=3$. If the goal is to meet a bound K on the sum, over all $j \in J$, of $\sigma_m(j) + l(t_m[j])$, then the problem is NP-complete in the strong sense for m arbitrary [Lenstra, Rinnooy Kan, and Brucker, 1977] and open for fixed $m \geq 2$. The analogous "no-wait" versions of OPEN-SHOP SCHEDULING and JOB-SHOP SCHEDULING are NP-complete in the strong sense for $m=2$ [Sahni and Cho, 1977b].

[SS17] TWO-PROCESSOR FLOW-SHOP WITH BOUNDED BUFFER

INSTANCE: (Same as for FLOW-SHOP SCHEDULING with $m=2$, with the addition of a "buffer bound" $B \in \mathbb{Z}_0^+$.)

QUESTION: Is there a flow-shop schedule for J that meets the overall deadline and such that, for all $u \geq 0$, the number of jobs $j \in J$ for which both $\sigma_1(j) + l(t_1[j]) \leq u$ and $\sigma_2(j) > u$ does not exceed B ?

Reference: [Papadimitriou and Kanellakis, 1978]. Transformation from NUMERICAL 3-DIMENSIONAL MATCHING.

Comment: NP-complete in the strong sense for any fixed B , $1 \leq B < \infty$. Solvable in polynomial time if $B=0$ [Gilmore and Gomory, 1964] or if $B \geq |J|-1$ [Johnson, 1954].

[SS18] JOB-SHOP SCHEDULING

INSTANCE: Number $m \in \mathbb{Z}^+$ of processors, set J of jobs, each $j \in J$ consisting of an ordered collection of tasks $t_k[j]$, $1 \leq k \leq n_j$, for each such task t a length $l(t) \in \mathbb{Z}_0^+$ and a processor $p(t) \in \{1, 2, \dots, m\}$, where $p(t_k[j]) \neq p(t_{k+1}[j])$ for all $j \in J$ and $1 \leq k < n_j$, and a deadline $D \in \mathbb{Z}^+$.

QUESTION: Is there a job-shop schedule for J that meets the overall deadline, i.e., a collection of one-processor schedules σ_i mapping $\{t: p(t)=i\}$ into $\mathbb{Z}_0^+$, $1 \leq i \leq m$, such that $\sigma_i(t) > \sigma_i(t')$ implies $\sigma_i(t) \geq \sigma_i(t') + l(t)$, such that $\sigma(t_{k+1}[j]) \geq \sigma(t_k[j]) + l(t_k[j])$ (where the appropriate subscripts are to be assumed on σ) for all $j \in J$ and $1 \leq k < n_j$, and such that for all $j \in J$ $\sigma(t_{n_j}[j]) + l(t_{n_j}[j]) \leq D$ (again assuming the appropriate subscript on σ)?

Reference: [Garey, Johnson, and Sethi, 1976]. Transformation from 3-PARTITION.

Comment: NP-complete in the strong sense for $m=2$. Can be solved in polynomial time if $m=2$ and $n_j \leq 2$ for all $j \in J$ [Jackson, 1956]. NP-complete (in the ordinary sense) if $m=2$ and $n_j \leq 3$ for all $j \in J$, or if $m=3$ and $n_j \leq 2$ for all $j \in J$ [Gonzalez and Sahni, 1978a]. All the above results continue to hold if "preemptive" schedules are allowed [Gonzalez and Sahni, 1978a]. If in the nonpreemptive case all tasks have the same length, the problem is NP-complete for $m=3$ and open for $m=2$ [Lenstra and Rinnooy Kan, 1978b].

A5.4 MISCELLANEOUS

XX[SS19] TIMETABLE DESIGN

INSTANCE: Set H of "work periods," set C of "craftsmen," set T of "tasks," a subset $A(c) \subseteq H$ of "available hours" for each craftsman $c \in C$, a subset $A(t) \subseteq H$ of "available hours" for each task $t \in T$, and, for each pair $(c, t) \in C \times T$, a number $R(c, t) \in \mathbb{Z}_0^+$ of "required work periods."

QUESTION: Is there a timetable for completing all the tasks, i.e., a function $f: C \times T \times H \rightarrow \{0, 1\}$ (where $f(c, t, h) = 1$ means that craftsman c works on task t during period h) such that (1) $f(c, t, h) = 1$ only if $h \in A(c) \cap A(t)$, (2) for each $h \in H$ and $c \in C$ there is at most one $t \in T$ for which $f(c, t, h) = 1$, (3) for each $h \in H$ and $t \in T$ there is at most one $c \in C$ for which $f(c, t, h) = 1$, and (4) for each pair $(c, t) \in C \times T$ there are exactly $R(c, t)$ values of h for which $f(c, t, h) = 1$?

Reference: [Even, Itai, and Shamir, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete even if $|H| = 3$, $A(t) = H$ for all $t \in T$, and each $R(c, t) \in \{0, 1\}$. The general problem can be solved in polynomial time if $|A(c)| \leq 2$ for all $c \in C$ or if $A(c) = A(t) = H$ for all $c \in C$ and $t \in T$.

[SS20] STAFF SCHEDULING

INSTANCE: Positive integers m and k , a collection C of m -tuples, each having k 1's and $m-k$ 0's (representing possible worker schedules), a "requirement" m -tuple $\bar{R}$ of non-negative integers, and a number n of workers.

QUESTION: Is there a schedule $f: C \rightarrow \mathbb{Z}_0^+$ such that $\sum_{\bar{c} \in C} f(\bar{c}) \leq n$ and such that $\sum_{\bar{c} \in C} f(\bar{c}) \cdot \bar{c} \geq \bar{R}$?

Reference: [Garey and Johnson, —] Transformation from X3C.

Comment: Solvable in polynomial time if every $\bar{c} \in C$ has the cyclic one's property, i.e., has all its 1's occurring in consecutive positions with position 1 regarded as following position m [Bartholdi, Orlin, and Ratliff, 1977]. (This corresponds to workers who are available only for consecutive hours of the day, or days of the week.)

[SS21] PRODUCTION PLANNING

INSTANCE: Number $n \in \mathbb{Z}^+$ of periods, for each period i , $1 \leq i \leq n$, a demand $r_i \in \mathbb{Z}_0^+$, a production capacity $c_i \in \mathbb{Z}_0^+$, a production set-up cost $b_i \in \mathbb{Z}_0^+$, an incremental production cost coefficient $p_i \in \mathbb{Z}_0^+$, and an inventory cost coefficient $h_i \in \mathbb{Z}_0^+$, and an overall bound $B \in \mathbb{Z}^+$.

QUESTION: Do there exist production amounts $x_i \in \mathbb{Z}_0^+$ and associated inventory levels $I_i = \sum_{j=1}^i (x_j - r_j)$, $1 \leq i \leq n$, such that all $x_i \leq c_i$, all $I_i \geq 0$, and

$$\sum_{i=1}^n (p_i x_i + h_i I_i) + \sum_{x_i > 0} b_i \leq B?$$

Reference: [Lenstra, Rinnooy Kan, and Florian, 1978]. Transformation from PARTITION.

Comment: Solvable in pseudo-polynomial time, but remains NP-complete even if all demands are equal, all set-up costs are equal, and all inventory costs are 0. If all capacities are equal, the problem can be solved in polynomial time [Florian and Klein, 1971]. The cited algorithms can be generalized to allow for arbitrary mono-

tone non-decreasing concave cost functions, if these can be computed in polynomial time.

[SS22] DEADLOCK AVOIDANCE

INSTANCE: Set $\{P_1, P_2, \dots, P_m\}$ of "process flow diagrams" (directed acyclic graphs), set Q of "resources," state S of system giving current "active" vertex in each process and "allocation" of resources (see references for details).

QUESTION: Is S "unsafe," i.e., are there control flows for the various processes from state S such that no sequence of resource allocations and deallocations can enable the system to reach a "final" state?

Reference: [Araki, Sugiyama, Kasami, and Okui, 1977], [Sugiyama, Araki, Okui, and Kasami, 1977]. Transformation from 3SAT.

Comment: Remains NP-complete even if allocation calls are "properly nested" and no allocation call involves more than two resources. See references for additional complexity results. See also [Gold, 1978] for results and algorithms for a related model of the deadlock problem.

A6 MATHEMATICAL PROGRAMMING

[MP1] INTEGER PROGRAMMING

INSTANCE: Finite set X of pairs $(\bar{x}, b)$, where $\bar{x}$ is an m -tuple of integers and b is an integer, an m -tuple $\bar{c}$ of integers, and an integer B .

QUESTION: Is there an m -tuple $\bar{y}$ of integers such that $\bar{x} \cdot \bar{y} \leq b$ for all $(\bar{x}, b) \in X$ and such that $\bar{c} \cdot \bar{y} \geq B$ (where the dot-product $\bar{u} \cdot \bar{v}$ of two m -tuples $\bar{u} = (u_1, u_2, \dots, u_m)$ and $\bar{v} = (v_1, v_2, \dots, v_m)$ is given by $\sum_{i=1}^m u_i v_i$)?

Reference: [Karp, 1972], [Borosh and Treybig, 1976]. Transformation from 3SAT. The second reference proves membership in NP.

Comment: NP-complete in the strong sense. Variant in which all components of $\bar{y}$ are required to belong to $\{0, 1\}$ (ZERO-ONE INTEGER PROGRAMMING) is also NP-complete, even if each b , all components of each $\bar{x}$, and all components of $\bar{c}$ are required to belong to $\{0, 1\}$. Also NP-complete are the questions of whether a $\bar{y}$ with non-negative integer entries exists such that $\bar{x} \cdot \bar{y} = b$ for all $(\bar{x}, b) \in X$, and the question of whether there exists any $\bar{y}$ with integer entries such that $\bar{x} \cdot \bar{y} \geq 0$ for all $(\bar{x}, b) \in X$ [Sahni, 1974].

[MP2] QUADRATIC PROGRAMMING (*)

INSTANCE: Finite set X of pairs $(\bar{x}, b)$, where $\bar{x}$ is an m -tuple of rational numbers and b is a rational number, two m -tuples $\bar{c}$ and $\bar{d}$ of rational numbers, and a rational number B .

QUESTION: Is there an m -tuple $\bar{y}$ of rational numbers such that $\bar{x} \cdot \bar{y} \leq b$ for all $(\bar{x}, b) \in X$ and such that $\sum_{i=1}^m (c_i y_i^2 + d_i y_i) \geq B$, where c_i , y_i , and d_i denote the i^{th} components of $\bar{c}$, $\bar{y}$, and $\bar{d}$ respectively?

Reference: [Sahni, 1974]. Transformation from PARTITION.

Comment: Not known to be in NP, unless the c_i 's are all non-negative [Klee, 1978]. If the constraints are quadratic and the objective function is linear (the reverse of the situation above), then the problem is also NP-hard [Sahni, 1974]. If we add to this last problem the requirement that all entries of $\bar{y}$ be integers, then the problem becomes undecidable [Jeroslow, 1973].

[MP3] COST-PARAMETRIC LINEAR PROGRAMMING

INSTANCE: Finite set X of pairs $(\bar{x}, b)$, where $\bar{x}$ is an m -tuple of integers and b is an integer, a set $J \subseteq \{1, 2, \dots, m\}$, and a positive rational number q .

QUESTION: Is there an m -tuple $\bar{c}$ with rational entries such that $(\bar{c} \cdot \bar{c})^{1/2} \leq q$ and such that, if Y is the set of all m -tuples $\bar{y}$ with non-negative rational entries satisfying $\bar{x} \cdot \bar{y} \geq b$ for all $(\bar{x}, b) \in X$, then the minimum of $\sum_{j \in J} c_j y_j$ over all $\bar{y} \in Y$ exceeds

$$\frac{1}{2} \max \{ |c_j| : j \in J \} + \sum_{j \in J} \min \{ 0, c_j \} ?$$

Reference: [Jeroslow, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete for any fixed $q > 0$. The problem arises from first order error analysis for linear programming.

[MP4] FEASIBLE BASIS EXTENSION

INSTANCE: An $m \times n$ integer matrix A , $m < n$, a column vector $\bar{a}$ of length m , and a subset S of the columns of A with $|S| < m$.

QUESTION: Is there a *feasible basis* B for $A\bar{x} = \bar{a}$, $\bar{x} \geq 0$, i.e., a nonsingular $m \times m$ submatrix B of A such that $B^{-1}\bar{a} \geq 0$, and such that B contains all the columns in S ?

Reference: [Murty, 1972]. Transformation from HAMILTONIAN CIRCUIT.

[MP5] MINIMUM WEIGHT SOLUTION TO LINEAR EQUATIONS

INSTANCE: Finite set X of pairs $(\bar{x}, b)$, where $\bar{x}$ is an m -tuple of integers and b is an integer, and a positive integer $K \leq m$.

QUESTION: Is there an m -tuple $\bar{y}$ with rational entries such that $\bar{y}$ has at most K non-zero entries and such that $\bar{x} \cdot \bar{y} = b$ for all $(\bar{x}, b) \in X$?

Reference: [Garey and Johnson, —]. Transformation from X3C.

Comment: NP-complete in the strong sense. Solvable in polynomial time if $K = m$.

[MP6] OPEN HEMISPHERE

INSTANCE: Finite set X of m -tuples of integers, and a positive integer $K \leq |X|$.

QUESTION: Is there an m -tuple $\bar{y}$ of rational numbers such that $\bar{x} \cdot \bar{y} > 0$ for at least K m -tuples $\bar{x} \in X$?

Reference: [Johnson and Preparata, 1978]. Transformation from MAXIMUM 2-SATISFIABILITY.

Comment: NP-complete in the strong sense, but solvable in polynomial time for any fixed m , even in a “weighted” version of the problem. The same results hold for the related CLOSED HEMISPHERE problem in which we ask that $\bar{y}$ satisfy $\bar{x} \cdot \bar{y} \geq 0$ for at least K m -tuples $\bar{x} \in X$ [Johnson and Preparata, 1978]. If $K = 0$ or $K = |X|$, both problems are polynomially equivalent to linear programming [Reiss and Dobkin, 1976].

[MP7] K-RELEVANCY

INSTANCE: Finite set X of pairs $(\bar{x}, b)$, where $\bar{x}$ is an m -tuple of integers and b is an integer, and a positive integer $K \leq |X|$.

QUESTION: Is there a subset $X' \subseteq X$ with $|X'| \leq K$ such that, for all m -tuples $\bar{y}$ of rational numbers, if $\bar{x} \cdot \bar{y} \leq b$ for all $(\bar{x}, b) \in X'$, then $\bar{x} \cdot \bar{y} \leq b$ for all $(\bar{x}, b) \in X$?

Reference: [Reiss and Dobkin, 1976]. Transformation from X3C.

Comment: NP-complete in the strong sense. Equivalent to linear programming if $K = |X| - 1$ [Reiss and Dobkin, 1976]. Other NP-complete problems of this form, where a standard linear programming problem is modified by asking that the desired property hold for some subset of K constraints, can be found in the reference.

[MP8] TRAVELING SALESMAN POLYTOPE NON-ADJACENCY

INSTANCE: Graph $G = (V, E)$, two Hamiltonian circuits C and C' for G .

QUESTION: Do C and C' correspond to non-adjacent vertices of the “traveling salesman polytope” for G ?

Reference: [Papadimitriou, 1978a]. Transformation from 3SAT.

Comment: Result also holds for the “non-symmetric” case where G is a directed graph and C and C' are directed Hamiltonian circuits. Analogous polytope non-adjacency problems for graph matching and CLIQUE can be solved in polynomial time [Chvátal, 1975].

[MP9] KNAPSACK

INSTANCE: Finite set U , for each $u \in U$ a size $s(u) \in \mathbb{Z}^+$ and a value $v(u) \in \mathbb{Z}^+$, and positive integers B and K .

QUESTION: Is there a subset $U' \subseteq U$ such that $\sum_{u \in U'} s(u) \leq B$ and such that $\sum_{u \in U'} v(u) \geq K$?

Reference: [Karp, 1972]. Transformation from PARTITION.

Comment: Remains NP-complete if $s(u) = v(u)$ for all $u \in U$ (SUBSET SUM). Can be solved in pseudo-polynomial time by dynamic programming (e.g., see [Dantzig, 1957] or [Lawler, 1976a]).

[MP10] INTEGER KNAPSACK

INSTANCE: Finite set U , for each $u \in U$ a size $s(u) \in \mathbb{Z}^+$ and a value $v(u) \in \mathbb{Z}^+$, and positive integers B and K .

QUESTION: Is there an assignment of a non-negative integer $c(u)$ to each $u \in U$ such that $\sum_{u \in U} c(u) \cdot s(u) \leq B$ and such that $\sum_{u \in U} c(u) \cdot v(u) \geq K$?

Reference: [Lueker, 1975]. Transformation from SUBSET SUM.

Comment: Remains NP-complete if $s(u) = v(u)$ for all $u \in U$. Solvable in pseudo-polynomial time by dynamic programming. Solvable in polynomial time if $|U| = 2$ [Hirschberg and Wong, 1976].

[MP11] CONTINUOUS MULTIPLE CHOICE KNAPSACK

INSTANCE: Finite set U , for each $u \in U$ a size $s(u) \in \mathbb{Z}^+$ and a value $v(u) \in \mathbb{Z}^+$, a partition of U into disjoint sets $U_1, U_2, \dots, U_m$, and positive integers B and K .

QUESTION: Is there a choice of a unique element $u_i \in U_i$, $1 \leq i \leq m$, and an assignment of rational numbers r_i , $0 \leq r_i \leq 1$, to these elements, such that $\sum_{i=1}^m r_i \cdot s(u_i) \leq B$ and $\sum_{i=1}^m r_i \cdot v(u_i) \geq K$?

Reference: [Ibaraki, 1978]. Transformation from PARTITION.

Comment: Solvable in pseudo-polynomial time, but remains NP-complete even if $|U_i| \leq 2$, $1 \leq i \leq m$. Solvable in polynomial time by “greedy” algorithms if $|U_i| = 1$, $1 \leq i \leq m$, or if we only require that the $r_i \geq 0$ but place no upper bound on them. [Ibaraki, Hasegawa, Teranaka, and Iwase, 1978].

[MP12] PARTIALLY ORDERED KNAPSACK

INSTANCE: Finite set U , partial order $<$ on U , for each $u \in U$ a size $s(u) \in \mathbb{Z}^+$ and a value $v(u) \in \mathbb{Z}^+$, positive integers B and K .

QUESTION: Is there a subset $U' \subseteq U$ such that if $u \in U'$ and $u' < u$, then $u' \in U'$, and such that $\sum_{u \in U'} s(u) \leq B$ and $\sum_{u \in U'} v(u) \geq K$?

Reference: [Garey and Johnson, —]. Transformation from CLIQUE. Problem is discussed in [Ibarra and Kim, 1975b].

Comment: NP-complete in the strong sense, even if $s(u) = v(u)$ for all $u \in U$. General problem is solvable in pseudo-polynomial time if $<$ is a "tree" partial order [Garey and Johnson, —].

[MP13] COMPARATIVE VECTOR INEQUALITIES

INSTANCE: Sets $X = \{\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k\}$ and $Y = \{\bar{y}_1, \bar{y}_2, \dots, \bar{y}_l\}$ of m -tuples of integers.

QUESTION: Is there an m -tuple $\bar{z}$ of integers such that the number of m -tuples $\bar{x}_i$ satisfying $\bar{x}_i \geq \bar{z}$ is at least as large as the number of m -tuples $\bar{y}_j$ satisfying $\bar{y}_j \geq \bar{z}$, where two m -tuples $\bar{u}$ and $\bar{v}$ satisfy $\bar{u} \geq \bar{v}$ if and only if no component of $\bar{u}$ is less than the corresponding component of $\bar{v}$?

Reference: [Plaisted, 1976]. Transformation from COMPARATIVE CONTAINMENT (with equal weights).

Comment: Remains NP-complete even if all components of the $\bar{x}_i$ and $\bar{y}_j$ are required to belong to $\{0, 1\}$.

A7 ALGEBRA AND NUMBER THEORY

A7.1 DIVISIBILITY PROBLEMS

[AN1] QUADRATIC CONGRUENCES

INSTANCE: Positive integers a , b , and c .

QUESTION: Is there a positive integer $x < c$ such that $x^2 \equiv a \pmod{b}$?

Reference: [Manders and Adleman, 1978]. Transformation from 3SAT.

Comment: Remains NP-complete even if the instance includes a prime factorization of b and solutions to the congruence modulo all prime powers occurring in the factorization. Solvable in polynomial time if $c = \infty$ (i.e., there is no upper bound on x) and the prime factorization of b is given. Assuming the Extended Riemann Hypothesis, the problem is solvable in polynomial time when b is prime. The general problem is trivially solvable in pseudo-polynomial time.

[AN2] SIMULTANEOUS INCONGRUENCES

INSTANCE: Collection $\{(a_1, b_1), \dots, (a_n, b_n)\}$ of ordered pairs of positive integers, with $a_i \leq b_i$ for $1 \leq i \leq n$.

QUESTION: Is there an integer x such that, for $1 \leq i \leq n$, $x \not\equiv a_i \pmod{b_i}$?

Reference: [Stockmeyer and Meyer, 1973]. Transformation from 3SAT.

[AN3] SIMULTANEOUS DIVISIBILITY OF LINEAR POLYNOMIALS (*)

INSTANCE: Vectors $a_i = (a_i[0], \dots, a_i[m])$ and $b_i = (b_i[0], \dots, b_i[m])$, $1 \leq i \leq n$, with positive integer entries.

QUESTION: Do there exist positive integers $x_1, x_2, \dots, x_m$ such that, for $1 \leq i \leq n$, $a_i[0] + \sum_{j=1}^m (a_i[j] \cdot x_j)$ divides $b_i[0] + \sum_{j=1}^m (b_i[j] \cdot x_j)$?

Reference: [Lipshitz, 1977], [Lipshitz, 1978]. Transformation from QUADRATIC DIOPHANTINE EQUATIONS.

Comment: Not known to be in NP, but belongs to NP for any fixed n . NP-complete for any fixed $n \geq 5$. General problem is undecidable if the vector entries and the x_j are allowed to range over the ring of "integers" in a real quadratic extension of the rationals. See reference for related decidability and undecidability results.

[AN4] COMPARATIVE DIVISIBILITY

INSTANCE: Sequences $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_m$ of positive integers.

QUESTION: Is there a positive integer c such that the number of i for which c divides a_i is more than the number of j for which c divides b_j ?

Reference: [Plaisted, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete even if all a_i are different and all b_j are different [Garey and Johnson, —].

[AN5] EXPONENTIAL EXPRESSION DIVISIBILITY (*)

INSTANCE: Sequences $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_m$ of positive integers, and an integer q .

QUESTION: Does $\prod_{i=1}^n (q^{a_i} - 1)$ divide $\prod_{j=1}^m (q^{b_j} - 1)$?

Reference: [Plaisted, 1976]. Transformation from 3SAT.

Comment: Not known to be in NP or co-NP, but solvable in pseudo-polynomial time using standard greatest common divisor algorithms. Remains NP-hard for any fixed value of q with $|q| > 1$, even if the a_i and b_j are restricted to being products of distinct primes.

[AN6] NON-DIVISIBILITY OF A PRODUCT POLYNOMIAL

INSTANCE: Sequences $A_i = \langle (a_i[1], b_i[1]), \dots, (a_i[k], b_i[k]) \rangle$, $1 \leq i \leq m$, of pairs of integers, with each $b_i[j] \geq 0$, and an integer N .

QUESTION: Is $\prod_{i=1}^m (\sum_{j=1}^k a_i[j] \cdot z^{b_i[j]})$ not divisible by $z^N - 1$?

Reference: [Plaisted, 1977a], [Plaisted, 1977b]. Transformation from 3SAT. Proof of membership in NP is non-trivial and appears in the second reference.

Comment: The related problem in which we are given two sequences $\langle a_1, a_2, \dots, a_m \rangle$ and $\langle b_1, b_2, \dots, b_n \rangle$ of positive integers and are asked whether $\prod_{i=1}^m (z^{a_i} - 1)$ does not divide $\prod_{j=1}^n (z^{b_j} - 1)$ is also NP-complete [Plaisted, 1976].

[AN7] NON-TRIVIAL GREATEST COMMON DIVISOR (*)

INSTANCE: Sequences $A_i = \langle (a_i[1], b_i[1]), \dots, (a_i[k], b_i[k]) \rangle$, $1 \leq i \leq m$, of pairs of integers, with each $b_i[j] \geq 0$.

QUESTION: Does the greatest common divisor of the polynomials $\sum_{j=1}^k a_i[j] \cdot z^{b_i[j]}$, $1 \leq i \leq m$, have degree greater than zero?

Reference: [Plaisted, 1977a]. Transformation from 3SAT.

Comment: Not known to be in NP or co-NP. Remains NP-hard if each $a_i[j]$ is either -1 or $+1$ [Plaisted, 1976] or if $m=2$ [Plaisted, 1977b]. The analogous problem in which the instance also includes a positive integer K , and we are asked if the least common multiple of the given polynomials has degree less than K , is NP-hard under the same restrictions. Both problems can be solved in pseudo-polynomial time using standard algorithms.

47.2 SOLVABILITY OF EQUATIONS

[AN8] QUADRATIC DIOPHANTINE EQUATIONS

INSTANCE: Positive integers a, b , and c .

QUESTION: Are there positive integers x and y such that $ax^2 + by = c$?

Reference: [Manders and Adleman, 1978]. Transformation from 3SAT.

Comment: Diophantine equations of the forms $ax^k = c$ and $\sum_{i=1}^k a_i \cdot x_i = c$ are solvable in polynomial time for arbitrary values of k . The general Diophantine problem, "Given a polynomial with integer coefficients in k variables, does it have an integer solution?" is undecidable, even for $k=13$ [Matijasevic and Robinson, 1975]. However, the given problem can be generalized considerably (to simultaneous equations in many variables) while remaining in NP, so long as only one variable enters into the equations in a non-linear way (see [Gurari and Ibarra, 1978]).

[AN9] ALGEBRAIC EQUATIONS OVER GF[2]

INSTANCE: Polynomials $P_i(x_1, x_2, \dots, x_n)$, $1 \leq i \leq m$, over GF[2], i.e., each polynomial is a sum of terms, where each term is either the integer 1 or a product of distinct x_j .

QUESTION: Do there exist $u_1, u_2, \dots, u_n \in \{0, 1\}$ such that, for $1 \leq i \leq m$, $P_i(u_1, u_2, \dots, u_n) = 0$, where arithmetic operations are as defined in GF[2], with $1+1=0$ and $1 \cdot 1 = 1$?

Reference: [Fraenkel and Yesha, 1977]. Transformation from X3C.

Comment: Remains NP-complete even if none of the polynomials has a term involving more than two variables [Valiant, 1977c]. Easily solved in polynomial time if no term involves more than one variable or if there is just one polynomial. Variant in which the u_j are allowed to range over the algebraic closure of GF[2] is NP-hard, even if no term involves more than two variables [Fraenkel and Yesha, 1977].

[AN10] ROOT OF MODULUS 1 (*)

INSTANCE: Ordered pairs $(a[i], b[i])$, $1 \leq i \leq n$, of integers, with each $b[i] \geq 0$.

QUESTION: Does the polynomial $\sum_{i=1}^n a[i] \cdot z^{b[i]}$ have a root on the complex unit circle, i.e., is there a complex number q with $|q|=1$ such that $\sum_{i=1}^n a[i] \cdot q^{b[i]} = 0$?

Reference: [Plaisted, 1977b]. Transformation from 3SAT.

Comment: Not known to be in NP or co-NP.

[AN11] NUMBER OF ROOTS FOR A PRODUCT POLYNOMIAL (*)

INSTANCE: Sequences $A_i = \langle (a_i[1], b_i[1]), \dots, (a_i[k], b_i[k]) \rangle$, $1 \leq i \leq m$, of pairs of integers, with each $b_i[j] \geq 0$, and a positive integer K .

QUESTION: Does the polynomial $\prod_{i=1}^m (\sum_{j=1}^k a_i[j] \cdot z^{b_i[j]})$ have fewer than K distinct complex roots?

Reference: [Plaisted, 1977a]. Transformation from 3SAT.

Comment: Not known to be in NP or co-NP. Remains NP-hard if each $a_i[j]$ is either -1 or $+1$, as does the variant in which the instance also includes an integer M and we are asked whether the product polynomial has fewer than K complex roots of multiplicity M [Plaisted, 1976].

[AN12] PERIODIC SOLUTION RECURRENCE RELATION (*)

INSTANCE: Ordered pairs (c_i, b_i) , $1 \leq i \leq m$, of integers, with all b_i positive.

QUESTION: Is there a sequence $a_0, a_1, \dots, a_{n-1}$ of integers, with $n \geq \max\{b_i\}$, such that the infinite sequence $a_0, a_1, \dots$ defined by the recurrence relation

$$a_i = \sum_{j=1}^m c_j \cdot a_{(i-b_j)}$$

satisfies $a_i \equiv a_{i \pmod n}$, for all $i \geq n$?

Reference: [Plaisted, 1977b]. Transformation from 3SAT

Comment: Not known to be in NP or co-NP. See reference for related results.

A7.3 MISCELLANEOUS

[AN13] PERMANENT EVALUATION (*)

INSTANCE: An $n \times n$ matrix M of 0's and 1's, and a positive integer $K \leq n!$.

QUESTION: Is the value of the permanent of M equal to K ?

Reference: [Valiant, 1977a]. Transformation from 3SAT.

Comment: The problem is NP-hard but not known to be in NP, as is the case for the variants in which we ask whether the value of the permanent is " K or less" or " K or more." The problem of computing the value of the permanent of M is #P-complete.

[AN14] COSINE PRODUCT INTEGRATION

INSTANCE: Sequence $(a_1, a_2, \dots, a_n)$ of integers.

QUESTION: Does $\int_0^{2\pi} \left(\prod_{i=1}^n \cos(a_i \theta) \right) d\theta = 0$?

Reference: [Plaisted, 1976]. Transformation from PARTITION.

Comment: Solvable in pseudo-polynomial time. See reference for related complexity results concerning integration.

[AN15] EQUILIBRIUM POINT

INSTANCE: Set $x = \{x_1, x_2, \dots, x_n\}$ of variables, collection $\{F_i: 1 \leq i \leq n\}$ of product polynomials over X and the integers, and a finite "range-set" $M_i \subseteq \mathbb{Z}$ for $1 \leq i \leq n$.

QUESTION: Does there exist a sequence $y_1, y_2, \dots, y_n$ of integers, with $y_i \in M_i$, such that for $1 \leq i \leq n$ and all $y \in M_i$,

$$F_i(y_1, y_2, \dots, y_{i-1}, y_i, y_{i+1}, \dots, y_n) \geq F_i(y_1, y_2, \dots, y_{i-1}, y, y_{i+1}, \dots, y_n) ?$$

Reference: [Sahni, 1974]. Transformation from 3SAT.

Comment: Remains NP-complete even if $M_i = \{0, 1\}$ for $1 \leq i \leq n$.

[AN16] UNIFICATION WITH COMMUTATIVE OPERATORS

INSTANCE: Set V of variables, set C of constants, ordered pairs (e_i, f_i) , $1 \leq i \leq n$, of "expressions," where an expression is either a variable from V , a constant from C , or $(e + f)$ where e and f are expressions.

QUESTION: Is there an assignment to each $v \in V$ of a variable-free expression $I(v)$ such that, if $I(e)$ denotes the expression obtained by replacing each occurrence of each variable v in e by $I(v)$, then $I(e_i) \equiv I(f_i)$ for $1 \leq i \leq n$, where $e \equiv f$ if $e = f$ or if $e = (a + b)$, $f = (c + d)$, and either $a \equiv c$ and $b \equiv d$ or $a \equiv d$ and $b \equiv c$?

Reference: [Sethi, 1977b]. Transformation from 3SAT.

Comment: Remains NP-complete even if no e_j or f_i contains more than 7 occurrences of constants and variables. The variant in which the operator is non-commutative (and hence $e \equiv f$ only if $e = f$) is solvable in polynomial time [Pater-son and Wegman, 1976].

[AN17] UNIFICATION FOR FINITELY PRESENTED ALGEBRAS

INSTANCE: Finite presentation of an algebra A in terms of a set G of generators, a collection O of operators of various finite dimensions, and a collection Γ of defining relations on well-formed formulas over G and O ; two well-formed expressions e and f over G, O , and a variable set V (see reference for details).

QUESTION: Is there an assignment to each $v \in V$ of a unique "term" $I(v)$ over G and O such that, if $I(e)$ and $I(f)$ denote the expressions obtained by replacing all variables in e and f by their corresponding terms, then $I(e)$ and $I(f)$ represent the same element in A ?

Reference: [Kozen, 1977a], [Kozen, 1976]. Transformation from 3SAT. Proof of membership in NP is non-trivial and appears in the second reference.

Comment: Remains NP-complete if only one of e and f contains variable symbols, but is solvable in polynomial time if neither contains variable symbols. See [Kozen, 1977b] for quantified versions of this problem that are complete for PSPACE and for the various levels of the polynomial hierarchy.

[AN18] INTEGER EXPRESSION MEMBERSHIP

INSTANCE: Integer expression e over the operations $\cup$ and $+$, where if $n \in \mathbb{Z}^+$, the binary representation of n is an integer expression representing n , and if f and g are integer expressions representing the sets F and G , then $f \cup g$ is an integer expression representing the set $F \cup G$ and $f + g$ is an integer expression representing the set $\{m + n: m \in F \text{ and } n \in G\}$, and a positive integer K .

QUESTION: Is K in the set represented by e ?

Reference: [Stockmeyer and Meyer, 1973]. Transformation from SUBSET SUM.

Comment: The related INTEGER EXPRESSION INEQUIVALENCE problem, "given two integer expressions e and f , do they represent different sets?" is NP-hard and in fact complete for Σ_2^P in the polynomial hierarchy ([Stockmeyer and Meyer, 1973], [Stockmeyer, 1976a], see also Section 7.2). If the operator " $\neg$ " is allowed, with $\neg e$ representing the set of all positive integers not represented by e , then both the membership and inequivalence problems become PSPACE-complete [Stockmeyer and Meyer, 1973].

A8 GAMES AND PUZZLES

[GP1] GENERALIZED HEX (*)

INSTANCE: Graph $G = (V, E)$ and two specified vertices $s, t \in V$.

QUESTION: Does player 1 have a forced win in the following game played on G ? The players alternate choosing a vertex from $V - \{s, t\}$, with those chosen by player 1 being colored "blue" and those chosen by player 2 being colored "red." Play continues until all such vertices have been colored, and player 1 wins if and only if there is a path from s to t in G that passes through only blue vertices.

Reference: [Even and Tarjan, 1976]. Transformation from QBF.

Comment: PSPACE-complete. The variant in which players alternate choosing an edge instead of a vertex, known as "the Shannon switching game on edges," can be solved in polynomial time [Bruno and Weinberg, 1970]. If G is a directed graph and player 1 wants a "blue" directed path from s to t , both the vertex selection game and the arc selection game are PSPACE-complete [Even and Tarjan, 1976].

[GP2] GENERALIZED GEOGRAPHY (*)

INSTANCE: Directed graph $G = (V, A)$ and a specified vertex $v_0 \in V$.

QUESTION: Does player 1 have a forced win in the following game played on G ? Players alternate choosing a new arc from A . The first arc chosen must have its tail at v_0 and each subsequently chosen arc must have its tail at the vertex that was the head of the previous arc. The first player unable to choose such a new arc loses.

Reference: [Schaefer, 1978a]. Transformation from QBF.

Comment: PSPACE-complete, even if G is bipartite, planar, and has no in- or out-degree exceeding 2 and no degree exceeding 3 (PLANAR GEOGRAPHY) [Lichtenstein and Sipser, 1978]. This game is a generalization of the "Geography" game in which players alternate choosing countries, each name beginning with the same letter that ends the previous country's name.

[GP3] GENERALIZED KAYLES (*)

INSTANCE: Graph $G = (V, E)$.

QUESTION: Does player 1 have a forced win in the following game played on G ? Players alternate choosing a vertex in the graph, removing that vertex and all vertices adjacent to it from the graph. Player 1 wins if and only if player 2 is the first player left with no vertices to choose from.

Reference: [Schaefer, 1978a]. Transformation from QBF.

Comment: PSPACE-complete. The variant in which $G = (V_1 \cup V_2, E)$ is bipartite, with each edge involving one vertex from V_1 and one from V_2 , and player i can only choose vertices from the set V_i (but still removes all adjacent vertices as before) is also PSPACE-complete. For a description of the game Kayles upon which this generalization is based, see [Conway, 1976].

[GP4] SEQUENTIAL TRUTH ASSIGNMENT (*)

INSTANCE: A sequence $U = \langle u_1, u_2, \dots, u_n \rangle$ of variables and a collection C of clauses over U (as in an instance of SATISFIABILITY).

QUESTION: Does player 1 have a forced win in the following game played on U

and C ? Players alternate assigning truth values to the variables in U , with player 1 assigning a value to u_{2i-1} and player 2 assigning a value to u_{2i} on their i^{th} turns. Player 1 wins if and only if the resulting truth assignment satisfies all clauses in C .

Reference: [Stockmeyer and Meyer, 1973]. Transformation from QBF.

Comment: PSPACE-complete, even if each clause in C has only three literals. Solvable in polynomial time if no clause has more than two literals [Schaefer, 1978b].

[GP5] VARIABLE PARTITION TRUTH ASSIGNMENT (*)

INSTANCE: A set U of variables and a collection C of clauses over U .

QUESTION: Does player 1 have a forced win in the following game played on U and C ? Players alternate choosing a variable from U until all variables have been chosen. Player 1 wins if and only if a satisfying truth assignment for C is obtained by setting "true" all variables chosen by player 1 and setting "false" all variables chosen by player 2.

Reference: [Schaefer, 1978a]. Transformation from QBF.

Comment: PSPACE-complete, even if each clause consists only of un-negated literals (i.e., contains no literals of the form $\bar{u}$ for $u \in U$). Analogous results for several other games played on logical expressions can be found in the reference.

[GP6] SIFT (*)

INSTANCE: Two collections A and B of subsets of a finite set X , with A and B having no subsets in common.

QUESTION: Does player 1 have a forced win in the following game played on A , B , and X ? Players alternate choosing an element from X until the set X' of all elements chosen so far either intersects all the subsets in A or intersects all the subsets in B . Player 1 wins if and only if the final set X' of chosen elements intersects all the subsets in B and, if player 1 made the last move, does *not* intersect all subsets in A .

Reference: [Schaefer, 1978a]. Transformation from QBF.

Comment: PSPACE-complete.

[GP7] ALTERNATING HITTING SET (*)

INSTANCE: A collection C of subsets of a basic set B .

QUESTION: Does player 1 have a forced win in the following game played on C and B ? Players alternate choosing a new element of B until, for each $c \in C$, some member of c has been chosen. The player whose choice causes this to happen loses.

Reference: [Schaefer, 1978a]. Transformation from QBF.

Comment: PSPACE-complete even if no set in C contains more than two elements, a subcase of the original HITTING SET problem that can be solved in polynomial time. If the roles of winner and loser are reversed, the problem is PSPACE-complete even if no set in C contains more than three elements.

[GP8] ALTERNATING MAXIMUM WEIGHTED MATCHING (*)

INSTANCE: Graph $G=(V,E)$, a weight $w(e) \in \mathbb{Z}^+$ for each $e \in E$, and a bound $B \in \mathbb{Z}^+$.

QUESTION: Does player 1 have a forced win in the following game played on G ? Players alternate choosing a new edge from E , subject to the constraint that no edge can share an endpoint with any of the already chosen edges. If the sum of the weights of the edges chosen ever exceeds B , player 1 wins.

Reference: [Dobkin and Ladner, 1978]. Transformation from QBF.

Comment: PSPACE-complete, even though the corresponding weighted matching problem can be solved in polynomial time (e.g., see [Lawler, 1976a]).

[GP9] ANNIHILATION (*)

INSTANCE: Directed acyclic graph $G=(V,A)$, collection $\{A_i; 1 \leq i \leq r\}$ of (not necessarily disjoint) subsets of A , function f_0 mapping V into $\{0,1,2,\dots,r\}$, where $f_0(v)=i>0$ means that a "token" of type i is "on" vertex v and $f_0(v)=0$ means that v is unoccupied.

QUESTION: Does player 1 have a forced win in the following game played on G ? A position is a function $f: V \rightarrow \{0,1,\dots,r\}$ with f_0 being the initial position and players alternating moves. A player moves by selecting a vertex $v \in V$ with $f(v)>0$ and an arc $(v,w) \in A_{f(v)}$, and the move corresponds to moving the token on vertex v to vertex w . The new position f' is the same as f except that $f'(v)=0$ and $f'(w)$ is either 0 or $f(v)$, depending, respectively, on whether $f(w)>0$ or $f(w)=0$. (If $f(w)>0$, then both the token moved to w and the token already there are "annihilated.") Player 1 wins if and only if player 2 is the first player unable to move.

Reference: [Fraenkel and Yesha, 1977]. Transformation from VERTEX COVER.

Comment: NP-hard and in PSPACE, but not known to be PSPACE-complete. Remains NP-hard even if $r=2$ and $A_1 \cap A_2$ is empty. Problem can be solved in polynomial time if $r=1$ [Fraenkel and Yesha, 1976]. Related NP-hardness results for other token-moving games on directed graphs (REMOVE, CONTRAJUNCTIVE, CAPTURE, BLOCKING, TARGET) can be found in [Fraenkel and Yesha, 1977].

[GP10] N×N CHECKERS (*)

INSTANCE: Positive integer N , a partition of the black squares of an $N \times N$ Checkerboard into those that are empty, those that are occupied by "Black kings," and those that are occupied by "Red kings," and the identity of the player (Red or Black) whose turn it is.

QUESTION: Does Black have a forced win from the given position in a game of Checkers played according to the standard rules, modified only to take into account the expanded board and number of pieces?

Reference: [Fraenkel, Garey, Johnson, Schaefer, and Yesha, 1978]. Transformation from PLANAR GEOGRAPHY.

Comment: PSPACE-hard, and PSPACE-complete for certain drawing rules. The related problem in which we ask whether Black can jump all of Red's pieces in one turn is solvable in polynomial time.

[GP11] N×N GO (*)

INSTANCE: Positive integer N , a partition of the "points" on an $N \times N$ Go board into those that are empty, those that are occupied by White stones and those that are occupied by Black stones, and the name (Black or White) of the player whose turn it is.

QUESTION: Does White have a forced win from the given position in a game of Go played according to the standard rules, modified only to take into account the expanded board?

Reference: [Lichtenstein and Sipser, 1978]. Transformation from PLANAR GEOGRAPHY.

Comment: PSPACE-hard.

[GP12] LEFT-RIGHT HACKENBUSH FOR REDWOOD FURNITURE

INSTANCE: A piece of "redwood furniture," i.e., a connected graph $G=(V,E)$ with a specified "ground" vertex $v \in V$ and a partition of the edges into sets L and R , where L is the set of all edges containing v (the set of "feet"), $R=E-L$, and each "foot" in L shares a vertex with at most one edge in R , which is its corresponding "leg" (not all edges in R need to be legs however), and a positive integer K .

QUESTION: Is the "value" of the Left-Right Hackenbush game played on G less than or equal to 2^{-K} (see [Conway, 1976] for the definition of the game, there called Hackenbush Restrained, and for the definition of "value")?

Reference: [Berlekamp, 1976]. Transformation from SET COVERING.

Comment: Remains NP-complete even for "bipartite" redwood furniture, but can be solved in polynomial time for the subclass of redwood furniture known as "redwood trees." As a consequence of this result, the problem of determining if player 1 has a win in an arbitrary game of Left-Right Hackenbush is NP-hard.

[GP13] SQUARE-TILING

INSTANCE: Set C of "colors," collection $T \subseteq C^4$ of "tiles" (where $\langle a,b,c,d \rangle$ denotes a tile whose top, right, bottom, and left sides are colored a,b,c , and d , respectively), and a positive integer $N \leq |C|$.

QUESTION: Is there a tiling of an $N \times N$ square using the tiles in T , i.e., an assignment of a tile $A(i,j) \in T$ to each ordered pair i,j , $1 \leq i \leq N$, $1 \leq j \leq N$, such that (1) if $f(i,j) = \langle a,b,c,d \rangle$ and $f(i+1,j) = \langle a',b',c',d' \rangle$, then $a=c'$, and (2) if $f(i,j) = \langle a,b,c,d \rangle$ and $f(i,j+1) = \langle a',b',c',d' \rangle$, then $b=d'$?

Reference: [Garey, Johnson, and Papadimitriou, 1977]. Transformation from DIRECTED HAMILTONIAN PATH.

Comment: Variant in which we ask if T can be used to tile the entire plane ($\mathbb{Z} \times \mathbb{Z}$) "periodically" with period less than N is also NP-complete. In general, the problem of whether a set of tiles can be used to tile the plane is undecidable [Berger, 1966], as is the problem of whether a set of tiles can be used to tile the plane periodically.

[GP14] CROSSWORD PUZZLE CONSTRUCTION

INSTANCE: A finite set $W \subseteq \Sigma^*$ of words and an $n \times n$ matrix A of 0's and 1's.

QUESTION: Can an $n \times n$ crossword puzzle be built up from the words in W and blank squares corresponding to the 0's of A , i.e., if E is the set of pairs (i, j) such that $A_{ij} = 0$, is there an assignment $f: E \rightarrow \Sigma$ such that the letters assigned to any maximal horizontal or vertical contiguous sequence of members of E form, in order, a word of W ?

Reference: [Lewis and Papadimitriou, 1978]. Transformation from X3C.

Comment: Remains NP-complete even if all entries in A are 0.

[GP15] GENERALIZED INSTANT INSANITY

INSTANCE: Finite set C of "colors" and a set Q of cubes, with $|Q| = |C|$ and with each side of each cube in Q having some assigned color from C .

QUESTION: Can the cubes in Q be stacked in one vertical column such that each of the colors in C appears exactly once on each of the four sides of the column?

Reference: [Robertson and Munro, 1978]. Transformation from EXACT COVER.

Comment: The associated two-person game, in which players alternate placing a new cube on the stack, with player 1 trying to construct a stack as specified above and player 2 trying to prevent this, is PSPACE-complete with respect to whether the first player has a forced win. INSTANT INSANITY is a trade name of Parker Brothers, Inc.

A9 LOGIC**A9.1 PROPOSITIONAL LOGIC****[LO1] SATISFIABILITY**

INSTANCE: Set U of variables, collection C of clauses over U (see Section 2.6 for definitions).

QUESTION: Is there a satisfying truth assignment for C ?

Reference: [Cook, 1971a]. Generic transformation.

Comment: Remains NP-complete even if each $c \in C$ satisfies $|c| = 3$ (3SAT), or if each $c \in C$ satisfies $|c| \leq 3$ and, for each $u \in U$, there are at most 3 clauses in C that contain either u or $\bar{u}$. Also remains NP-complete if each $c \in C$ has $|c| \leq 3$ and the bipartite graph $G = (V, E)$, where $V = U \cup C$ and E contains exactly those pairs $\{u, c\}$ such that either u or $\bar{u}$ belongs to the clause c , is planar (PLANAR 3SAT) [Lichtenstein, 1977]. The general problem is solvable in polynomial time if each $c \in C$ has $|c| \leq 2$ (e.g., see [Even, Itai, and Shamir, 1976]).

[LO2] 3-SATISFIABILITY (3SAT)

INSTANCE: Set U of variables, collection C of clauses over U such that each clause $c \in C$ has $|c| = 3$.

QUESTION: Is there a satisfying truth assignment for C ?

Reference: [Cook, 1971a]. Transformation from SATISFIABILITY.

Comment: Remains NP-complete even if each clause contains either only negated variables or only un-negated variables (MONOTONE 3SAT) [Gold, 1974], or if for each $u \in U$ there are at most 5 clauses in C that contain either u or $\bar{u}$.

[LO3] NOT-ALL-EQUAL 3SAT

INSTANCE: Set U of variables, collection C of clauses over U such that each clause $c \in C$ has $|c| = 3$.

QUESTION: Is there a truth assignment for U such that each clause in C has at least one true literal and at least one false literal?

Reference: [Schaefer, 1978b]. Transformation from 3SAT.

[LO4] ONE-IN-THREE 3SAT

INSTANCE: Set U of variables, collection C of clauses over U such that each clause $c \in C$ has $|c| = 3$.

QUESTION: Is there a truth assignment for U such that each clause in C has exactly one true literal?

Reference: [Schaefer, 1978b]. Transformation from 3SAT.

Comment: Remains NP-complete even if no $c \in C$ contains a negated literal.

[LO5] MAXIMUM 2-SATISFIABILITY

INSTANCE: Set U of variables, collection C of clauses over U such that each clause $c \in C$ has $|c| = 2$, positive integer $K \leq |C|$.

QUESTION: Is there a truth assignment for U that simultaneously satisfies at least K of the clauses in C ?

Reference: [Garey, Johnson, and Stockmeyer, 1976]. Transformation from 3SAT.

Comment: Solvable in polynomial time if $K = |C|$ (e.g., see [Even, Itai, and Shamir, 1976]).

[LO6] GENERALIZED SATISFIABILITY

INSTANCE: Positive integers $k_1, k_2, \dots, k_m$, sequence $S = \langle R_1, R_2, \dots, R_m \rangle$ of subsets $R_i \subseteq \{T, F\}^{k_i}$, set U of variables, and, for $1 \leq i \leq m$, a collection C_i of k_i -tuples of variables from U .

QUESTION: Is there a truth assignment $t: U \rightarrow \{T, F\}$ such that for all i , $1 \leq i \leq m$, and for all k_i -tuples $(u[1], u[2], \dots, u[k_i])$ in C_i , we have

$$(t(u[1]), t(u[2]), \dots, t(u[k_i])) \in R_i ?$$

Reference: [Schaefer, 1978b]. Transformation from 3SAT.

Comment: For any fixed sequence S , the problem is NP-complete unless one of the following six alternatives holds, in which case the problem with that S is solvable in polynomial time:

- (1) Each R_i contains $\{T\}^{k_i}$,
- (2) each R_i contains $\{F\}^{k_i}$,
- (3) each R_i is logically "equivalent" to some conjunctive normal form expression having at most one negated literal per clause,
- (4) each R_i is logically "equivalent" to some conjunctive normal form expression having at most one un-negated literal per clause,
- (5) each R_i is logically "equivalent" to some conjunctive normal form expression having at most 2 literals per clause, or
- (6) each R_i is the "solution set" for some system of linear equations over $\text{GF}[2]$.

The NP-completeness of 3SAT, ONE-IN-THREE 3SAT, and NOT-ALL-EQUAL 3SAT all follow from this classification. If the tuples in each C_i are allowed to be in $(U \cup \{T, F\})^{k_i}$ ("formulas with constants"), the problem is NP-complete even if (1) or (2) holds, but is still polynomially solvable if (3), (4), (5), or (6) holds. The quantified version of the problem "with constants," where we are also given a sequence $Q_1, Q_2, \dots, Q_n$ of quantifiers (each Q_i being either $\forall$ or $\exists$) and ask if

$$(Q_1 u_1)(Q_2 u_2) \cdots (Q_n u_n) [c \in R_i \text{ for all } c \in C_i, 1 \leq i \leq m]$$

is PSPACE-complete, even for fixed S , so long as S does not meet any of (3), (4), (5), or (6), and is solvable in polynomial time for any fixed S that does meet one of (3), (4), (5), or (6).

[LO7] SATISFIABILITY OF BOOLEAN EXPRESSIONS

INSTANCE: Variable set U , a subset B of the set of 16 possible binary Boolean connectives, and a well-formed Boolean expression E over U and B .

QUESTION: Is there a truth assignment for U that satisfies E ?

Reference: [Cook, 1971a]. Generic transformation.

Comment: Remains NP-complete if B is restricted to $\{\wedge, \vee, \neg, \neg\}$, or any other

truth-functionally complete set of connectives. Also NP-complete for any truth-functionally incomplete set of connectives containing $\{\neg\}$, $\{\neg, \vee\}$, or $\{\neg, \wedge\}$ as a subset [Lewis, 1978]. Problem is solvable in polynomial time for any truth-functionally incomplete set of connectives *not* containing one of these four sets as a subset.

[LO8] NON-TAUTOLOGY

INSTANCE: Boolean expression E over a set U of variables, using the connectives " $\neg$ " (not), " $\vee$ " (or), " $\wedge$ " (and), and " $\rightarrow$ " (implies).

QUESTION: Is E *not* a tautology, i.e., is there a truth assignment for U that makes E false?

Reference: [Cook, 1971a]. Transformation from SATISFIABILITY.

Comment: Remains NP-complete even if E is in "disjunctive normal form" with at most 3 literals per disjunct.

[LO9] MINIMUM DISJUNCTIVE NORMAL FORM

INSTANCE: Set $U = \{u_1, u_2, \dots, u_n\}$ of variables, set $A \subseteq \{T, F\}^n$ of "truth assignments," and a positive integer K .

QUESTION: Is there a disjunctive normal form expression E over U , having no more than K disjuncts, such that E is true for precisely those truth assignments in A , and no others?

Reference: [Gimpel, 1965]. Transformation from MINIMUM COVER.

Comment: Variant in which the instance contains a *complete* truth table, i.e., disjoint sets A and $B \subseteq \{T, F\}^n$ such that $A \cup B = \{T, F\}^n$, and E must be true for all truth assignments in A and false for all those in B , is also NP-complete, despite the possibly much larger instance size [Masek, 1978].

[LO10] TRUTH-FUNCTIONALLY COMPLETE CONNECTIVES

INSTANCE: Set U of variables, collection C of well-formed Boolean expressions over U .

QUESTION: Is C truth-functionally complete, i.e., is there a truth-functionally complete set of logical connectives (unary and binary operators) $D = \{\theta_1, \theta_2, \dots, \theta_k\}$ such that for each $\theta_i \in D$ there is an expression $E \in C$ and a substitution $s: U \rightarrow \{a, b\}$ for which $s(E) \equiv a\theta_i b$ or $s(E) \equiv \theta_i a$ (depending on whether θ_i is binary or unary)?

Reference: [Statman, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete even if $|C| = 2$.

A9.2 MISCELLANEOUS

[LO11] QUANTIFIED BOOLEAN FORMULAS (QBF) (*)

INSTANCE: Set $U = \{u_1, u_2, \dots, u_n\}$ of variables, well-formed quantified Boolean formula $F = (Q_1 u_1)(Q_2 u_2) \cdots (Q_n u_n)E$, where E is a Boolean expression and each Q_i is either $\forall$ or $\exists$.

QUESTION: Is F true?

Reference: [Stockmeyer and Meyer, 1973]. Generic transformation.

Comment: PSPACE-complete, even if E is in conjunctive normal form with three literals per clause (QUANTIFIED 3SAT), but solvable in polynomial time when there are at most two literals per clause [Schaefer, 1978b]. If F is restricted to at most k alternations of quantifiers (i.e., there are at most k indices i such that $Q_i \neq Q_{i+1}$), then the restricted problem is complete for some class in the polynomial hierarchy, depending on k and the allowed values for Q_1 (see Section 7.2).

[LO12] FIRST ORDER THEORY OF EQUALITY (*)

INSTANCE: Finite set $U = \{u_1, u_2, \dots, u_n\}$ of variables, sentence S over U in the first order theory of equality. (Such sentences can be defined inductively as follows: An "expression" is of the form " $u = v$ " where $u, v \in U$, or of the form " $\neg E$," " $(E \vee F)$," " $(E \wedge F)$," or " $(E \rightarrow F)$ " where E and F are expressions. A sentence is of the form $(Q_1 u_1)(Q_2 u_2) \dots (Q_n u_n) E$ where E is an expression and each Q_i is either $\forall$ or $\exists$.)

QUESTION: Is S true in all models of the theory?

Reference: [Stockmeyer and Meyer, 1973]. Generic transformation.

Comment: PSPACE-complete. The analogous problem for any fixed first order theory that has a model in which some predicate symbol is interpreted as a relation that holds sometimes but not always is PSPACE-hard [Hunt, 1977].

[LO13] MODAL LOGIC S5-SATISFIABILITY

INSTANCE: Well-formed modal formula A over a finite set U of variables, where a modal formula is either a variable $u \in U$ or is of the form " $(A \wedge B)$," " $\neg A$," or " $\Box A$," where A and B are modal formulas.

QUESTION: Is A "S5-satisfiable," i.e., is there a model (W, R, V) , where W is a set, R is a reflexive, transitive, and symmetric binary relation on W , and V is a mapping from $U \times W$ into $\{T, F\}$ such that, for some $w \in W$, $V(A, w) = T$, where V is extended to formulas by $V(A \wedge B, w) = T$ if and only if $V(A, w) = V(B, w) = T$, $V(\neg A, w) = T$ if and only if $V(A, w) = F$, and $V(\Box A, w) = T$ if and only if $V(A, w') = T$ for all $w' \in W$ satisfying $(w, w') \in R$?

Reference: [Ladner, 1977]. Transformation from 3SAT. Nontrivial part is proving membership in NP.

[LO14] MODAL LOGIC PROVABILITY (*)

INSTANCE: Well-formed modal formula A , modal system $S \in \{K, T, S4\}$ (see reference or [Hughes and Cresswell, 1968] for details of K , T , and $S4$).

QUESTION: Is A provable in system S ?

Reference: [Ladner, 1977]. Transformation from QBF.

Comment: PSPACE-complete for fixed $S \in \{K, T, S4\}$ or for any fixed modal system S in which everything provable in K , but nothing not provable in $S4$, can be proved.

[LO15] PREDICATE LOGIC WITHOUT NEGATION

INSTANCE: Sets $U = \{u_1, u_2, \dots, u_n\}$ of variables, $F = \{f_1^{m_1}, f_2^{m_2}, \dots, f_k^{m_k}\}$ of function symbols, and $R = \{R_1^{r_1}, R_2^{r_2}, \dots, R_j^{r_j}\}$ of relation symbols ($m_i \geq 0$ and $r_i \geq 0$ being the dimensions of the corresponding functions and relations), and a well-formed predicate logic sentence A without negations over U , F , and R . (Such a sentence can be defined inductively as follows: A term is a variable $u \in U$ or of the form " $f_i^{m_i}(t_1, t_2, \dots, t_{m_i})$ " where each t_j is a term. A formula is of the form " $t_1 = t_2$ " where t_1 and t_2 are terms, " $R_i^{r_i}(t_1, t_2, \dots, t_{r_i})$ " where each t_j is a term, or " $(A \wedge B)$," " $(A \vee B)$," " $\forall u_i (A)$," or " $\exists u_i (A)$ " where A and B are formulas and $u_i \in U$. A sentence is a formula in which all variables are quantified before they occur.)

QUESTION: Is A true under all interpretations of F and R ?

Reference: [Kozen, 1977c]. Transformation from 3SAT. Nontrivial part is proving membership in NP.

Comment: Remains NP-complete even if there are no universal quantifiers, no relation symbols, and only two functions, both with dimension 0 (and hence constants).

[LO16] CONJUNCTIVE SATISFIABILITY WITH FUNCTIONS AND INEQUALITIES

INSTANCE: Set U of variables, set F of univariate function symbols, and a collection C of "clauses" of the form $U * V$ where $*$ is either " $\leq$," " $>$," " $=$," or " $\neq$," and U and V are either " 0 ," " 1 ," " u ," " $f(0)$," " $f(1)$," or " $f(u)$," for some $f \in F$ and $u \in U$.

QUESTION: Is there an assignment of integer values to all the variables $u \in U$ and to all $f(u)$, for $u \in U$ and $f \in F$, such that all the clauses in C are satisfied under the usual interpretations of $\leq$, $>$, $=$, and $\neq$?

Reference: [Pratt, 1977]. Transformation from 3SAT.

Comment: Remains NP-complete even if $=$ and $\neq$ are not used. Solvable in polynomial time if $\leq$ and $>$ are not used [Nelson and Oppen, 1977], or if $=$ and $\neq$ are not used and no function symbols are allowed [Litvintchouk and Pratt, 1977]. Variant in which W and V are either of the form " u " or " $u+c$ " for some $u \in U$ and $c \in \mathbb{Z}$ is NP-complete if all four relations are allowed, but solvable in polynomial time if only $\leq$ and $>$ or only $=$ and $\neq$ are allowed [Chan, 1977].

[LO17] MINIMUM AXIOM SET

INSTANCE: Finite set S of "sentences," subset $T \subseteq S$ of "true sentences," an "implication relation" R consisting of pairs (A, s) where $A \subseteq S$ and $s \in S$, and a positive integer $K \leq |S|$.

QUESTION: Is there a subset $S_0 \subseteq T$ with $|S_0| \leq K$ and a positive integer n such that, if we define S_i , $1 \leq i \leq n$, to consist of exactly those $s \in S$ for which either $s \in S_{i-1}$ or there exists a $U \subseteq S_{i-1}$ such that $(U, s) \in R$, then $S_n = T$?

Reference: [Pudlák, 1975]. Transformation from X3C.

Comment: Remains NP-complete even if $T = S$.

[LO18] FIRST ORDER SUBSUMPTION

INSTANCE: Finite set U of "variable symbols," finite set C of "function symbols," collection $E = \{E_1, E_2, \dots, E_m\}$ of expressions over $U \cup C$, collection $F = \{F_1, F_2, \dots, F_n\}$ of expressions over C .

QUESTION: Is there a substitution mapping s that assigns to each $u \in U$ an expression $s(u)$ over C such that, if $s(E_i)$ denotes the result of substituting for each occurrence in E_i of each $u \in U$ the corresponding expression $s(u)$, then $\{s(E_1), s(E_2), \dots, s(E_m)\}$ is a subset of $\{F_1, F_2, \dots, F_n\}$?

Reference: [Baxter, 1976], [Baxter, 1977]. Transformation from 3SAT.

Comment: Remains NP-complete for any fixed $n \geq 3$, but is solvable in polynomial time for any fixed m .

[LO19] SECOND ORDER INSTANTIATION

INSTANCE: Two "second order logic expressions" E_1 and E_2 , the second of which contains no variables (in a second order expression, functions can be variables; see references for details).

QUESTION: Is there a substitution for the variables of E_1 that yields an expression identical to E_2 ?

Reference: [Baxter, 1976]. Transformation from 3SAT. Proof of membership in NP is nontrivial.

Comment: The more general SECOND ORDER UNIFICATION problem, where both E_1 and E_2 can contain variables and we ask if there is a substitution for the variables in E_1 and E_2 that results in identical expressions, is not known to be decidable. THIRD ORDER UNIFICATION is undecidable [Huet, 1973], whereas FIRST ORDER UNIFICATION can be solved in polynomial time [Baxter, 1975], [Paterson and Wegman, 1978].

A10 AUTOMATA AND LANGUAGE THEORY**A10.1 AUTOMATA THEORY****[AL1] FINITE STATE AUTOMATON INEQUIVALENCE (*)**

INSTANCE: Two nondeterministic finite state automata A_1 and A_2 having the same input alphabet Σ (where such an automaton $A = (Q, \Sigma, \delta, q_0, F)$ consists of a finite set Q of states, input alphabet Σ , transition function δ mapping $Q \times \Sigma$ into subsets of Q , initial state q_0 , and a set $F \subseteq K$ of "accept" states, e.g., see [Hopcroft and Ullman, 1969]).

QUESTION: Do A_1 and A_2 recognize different languages?

Reference: [Kleene, 1956]. Transformation from REGULAR EXPRESSION NON-UNIVERSALITY.

Comment: PSPACE-complete, even if $|\Sigma| = 2$ and A_2 is the trivial automaton recognizing Σ^* . The general problem is NP-complete if $|\Sigma| = 1$, or if A_1 and A_2 both recognize finite languages (a property that can be checked in polynomial time, e.g., see [Hopcroft and Ullman, 1969]). Problem is solvable in polynomial time if A_1 and A_2 are deterministic finite state automata, e.g., see [Hopcroft and Ullman, 1969].

[AL2] TWO-WAY FINITE STATE AUTOMATON NON-EMPTYNESS (*)

INSTANCE: A two-way nondeterministic finite state automaton $A = (Q, \Sigma, \delta, q_0, F)$ (where Q , Σ , q_0 , and F are the same as for a one-way nondeterministic finite state automaton, but the transition function δ maps $Q \times \Sigma$ into subsets of $Q \times \{-1, 0, 1\}$, e.g., see [Hopcroft and Ullman, 1969]).

QUESTION: Is there an $x \in \Sigma^*$ such that A accepts x ?

Reference: [Hunt, 1973b]. Transformation from LINEAR BOUNDED AUTOMATON ACCEPTANCE.

Comment: PSPACE-complete, even if $|\Sigma| = 2$ and A is deterministic. If $|\Sigma| = 1$ the general problem is NP-complete [Galil, 1976]. If A is a one-way nondeterministic finite state automaton, the general problem can be solved in polynomial time (e.g., see [Hopcroft and Ullman, 1969]). Analogous results for the question of whether A recognizes an infinite language can be found in the above references.

[AL3] LINEAR BOUNDED AUTOMATON ACCEPTANCE (*)

INSTANCE: A "linear bounded automaton" A with input alphabet Σ (see [Hopcroft and Ullman, 1969] for definition), and a string $x \in \Sigma^*$.

QUESTION: Does A accept x ?

Reference: [Karp, 1972]. Generic transformation.

Comment: PSPACE-complete, even if A is deterministic (the LINEAR SPACE ACCEPTANCE problem of Section 7.4). Moreover, there exist fixed deterministic linear bounded automata for which the problem is PSPACE-complete.

[AL4] QUASI-REALTIME AUTOMATON ACCEPTANCE

INSTANCE: A multi-tape nondeterministic Turing machine M (Turing machine program, in our terminology), whose input tape read-head must move right at each

step, and which must halt whenever the read-head sees a blank, and a string x over the input alphabet Σ of M . (For a more complete description of this type of machine and its equivalent formulations, see [Book and Greibach, 1970].)

QUESTION: Does M accept x ?

Reference: [Book, 1972]. Generic transformation.

Comment: Remains NP-complete even if M has only a single work tape in addition to its input tape. See also QUASI-REALTIME LANGUAGE MEMBERSHIP (the languages accepted by quasi-realtime automata are the same as the quasi-realtime languages defined in that entry).

[AL5] NON-ERASING STACK AUTOMATON ACCEPTANCE (*)

INSTANCE: A "one-way nondeterministic non-erasing stack automaton" (a 1NESA) A with input alphabet Σ (see [Hopcroft and Ullman, 1969] for definition), and a string $x \in \Sigma^*$.

QUESTION: Does A accept x ?

Reference: [Galil, 1976], [Hopcroft and Ullman, 1967]. Transformation from LINEAR BOUNDED AUTOMATON ACCEPTANCE. The second reference proves membership in PSPACE.

Comment: PSPACE-complete, even if $x \in \Sigma^*$ is fixed and A is restricted to be a "checking stack automaton" (as defined in [Greibach, 1969]). If x is the empty string and A is further restricted to be a checking stack automaton with a single stack symbol, the problem becomes NP-complete [Galil, 1976]. If instead x is allowed to vary and A is fixed, the problem is in NP for each 1NESA and remains so if A is allowed to be a general "nested stack automaton" [Rounds, 1973]. There exist particular 1NESAs for which the problem is NP-complete [Rounds, 1973], and these particular 1NESAs can be chosen to be checking stack automata [Shamir and Beeri, 1974] that are also "reading pushdown automata" [Hunt, 1976]. However, if A is restricted to be a "one-way nondeterministic pushdown automaton," then the problem can be solved in polynomial time (even with A allowed to vary), as indeed is the case for "two-way nondeterministic pushdown automata" [Aho, Hopcroft, and Ullman, 1968].

[AL6] FINITE STATE AUTOMATA INTERSECTION (*)

INSTANCE: Sequence $A_1, A_2, \dots, A_n$ of deterministic finite state automata having the same input alphabet Σ .

QUESTION: Is there a string $x \in \Sigma^*$ accepted by each of the A_i , $1 \leq i \leq n$?

Reference: [Kozen, 1977d]. Transformation from LINEAR SPACE ACCEPTANCE.

Comment: PSPACE-complete. Solvable in polynomial time for any fixed n (e.g., see [Hopcroft and Ullman, 1969]).

[AL7] REDUCTION OF INCOMPLETELY SPECIFIED AUTOMATA

INSTANCE: An incompletely specified deterministic finite state automaton $A = (Q, \Sigma, \delta, q_0, F)$, where Q is the set of states, Σ is the input alphabet, δ is a "partial" transition function mapping a subset of $Q \times \Sigma$ into Q , $q_0 \in Q$ is the initial state, and $F \subseteq Q$ is the set of "accept" states, and a positive integer K .

QUESTION: Can the transition function δ be extended to a total function from $Q \times \Sigma$ into Q in such a way that the resulting completely specified automaton has an equivalent "reduced automaton" with K or fewer states?

Reference: [Pfleeger, 1973]. Transformation from GRAPH 3-COLORABILITY.

Comment: Remains NP-complete for any fixed $K \geq 6$. Related question in which "state-splitting" (as used in [Paull and Unger, 1959]) is allowed is also NP-complete for any fixed $K \geq 6$ [Pfleeger, 1973]. If both "state-splitting" and "symbol-splitting" (as used in [Grasselli and Luccio, 1966]) are allowed, the analogous problem in which the corresponding reduced automaton is to have the sum of the number of states and the number of symbols be no more than K is also NP-complete [Pfleeger, 1974]. The problem of determining the minimum state deterministic finite state automaton equivalent to a given completely specified one can be solved in polynomial time (e.g., see [Hopcroft, 1971] or [Aho and Ullman, 1972]). The corresponding problem for completely specified nondeterministic finite state automata is PSPACE-complete (see FINITE STATE AUTOMATA INEQUIVALENCE).

[AL8] MINIMUM INFERRED FINITE STATE AUTOMATON

INSTANCE: Finite alphabet Σ , two finite subsets $S, T \subseteq \Sigma^*$, positive integer K .

QUESTION: Is there a K -state deterministic finite automaton A that recognizes a language $L \subseteq \Sigma^*$ such that $S \subseteq L$ and $T \subseteq \Sigma^* - L$?

Reference: [Gold, 1974]. Transformation from MONOTONE 3SAT.

Comment: Can be solved in polynomial time if $S \cup T = \Sigma^{(n)}$ for some n , where $\Sigma^{(n)}$ is the set of all strings of length n or less over Σ [Trakhtenbrot and Barzdin, 1973]. However, for any fixed $\epsilon > 0$, the problem remains NP-complete if restricted to instances for which $(S \cup T) \subseteq \Sigma^{(n)}$ and $|\Sigma^{(n)} - (S \cup T)| \leq |\Sigma^{(n)}|^\epsilon$ [Angluin, 1977].

A10.2 FORMAL LANGUAGES

[AL9] REGULAR EXPRESSION INEQUIVALENCE (*)

INSTANCE: Regular expressions E_1 and E_2 over the operators $\{\cup, \cdot, *\}$ and the alphabet Σ (see Section 7.4 for definition).

QUESTION: Do E_1 and E_2 represent different languages?

Reference: [Stockmeyer and Meyer, 1973], [Stockmeyer, 1974a]. Generic transformation. The second reference proves membership in PSPACE.

Comment: PSPACE-complete, even if $|\Sigma| = 2$ and $E_2 = \Sigma^*$ (REGULAR EXPRESSION NON-UNIVERSALITY, see Section 7.4). In fact, PSPACE-complete if E_2 is any fixed expression representing an "unbounded" language [Hunt, Rosenkrantz, and Szymanski, 1976a]. NP-complete for fixed E_2 representing any infinite "bounded" language, but solvable in polynomial time for fixed E_2 representing any finite language. The general problem remains PSPACE-complete if E_1 and E_2 both have "star height" k for a fixed $k \geq 1$ [Hunt, Rosenkrantz, and Szymanski, 1976a], but is NP-complete for $k = 0$ ("star free") [Stockmeyer and Meyer, 1973], [Hunt, 1973a]. Also NP-complete if one or both of E_1 and E_2 represent bounded languages (a property that can be checked in polynomial time) [Hunt, Rosenkrantz, and Szymanski, 1976a] or if $|\Sigma| = 1$ [Stockmeyer and Meyer, 1973]. For related

results and intractable generalizations, see cited references, [Hunt, 1973b], and [Hunt and Rosenkrantz, 1978].

[AL10] MINIMUM INFERRED REGULAR EXPRESSION

INSTANCE: Finite alphabet Σ , two finite subsets $S, T \subseteq \Sigma^*$, positive integer K .
 QUESTION: Is there a regular expression E over Σ that has K or fewer occurrences of symbols from Σ and such that, if $L \subseteq \Sigma^*$ is the language represented by E , then $S \subseteq L$ and $T \subseteq \Sigma^* - L$?

Reference: [Angluin, 1977]. Transformation from 3SAT.

Comment: Remains NP-complete even if E is required to contain no “ $\cup$ ” operations or to be “star-free” (contain no “ $*$ ” operations) [Angluin, 1976].

[AL11] REYNOLDS COVERING FOR CONTEXT-FREE GRAMMARS

INSTANCE: Context-free grammars $G_1 = (N_1, \Sigma, \Pi_1, S_1)$ and $G_2 = (N_2, \Sigma, \Pi_2, S_2)$, where Σ is a finite set of “terminal” symbols, N_i is a finite set of “nonterminal” symbols, $S_i \in N_i$ is the “initial” symbol, and Π_i is a set of “productions” of the form “ $A \rightarrow w$,” where $A \in N_i$ and $w \in (N_i \cup \Sigma)^*$.

QUESTION: Does G_2 “Reynolds cover” G_1 , i.e., is there a function f mapping $N_1 \cup \Sigma$ into $N_2 \cup \Sigma$ such that $f(x) = x$ for all $x \in \Sigma$, $f(A) \in N_2$ for all $A \in N_1$, $f(S_1) = S_2$, and for each production $A \rightarrow x_1 x_2 \cdots x_n$ in Π_1 , the image $f(A) \rightarrow f(x_1) f(x_2) \cdots f(x_n)$ of that production is in Π_2 ?

Reference: [Hunt and Rosenkrantz, 1977]. Transformation from 3SAT.

Comment: Remains NP-complete even if G_1 and G_2 are restricted to “regular” grammars. The same results hold for the related questions of whether G_2 “weakly Reynolds covers” G_1 or whether G_2 is a “homomorphic image” of G_1 . The problem “Given G is there an $LL(k)$ context-free grammar H such that H Reynolds covers G ?” is solvable in polynomial time, as are the related problems where $LL(k)$ is replaced by $LR(k)$ or one of a number of other grammar classes (see [Hunt and Rosenkrantz, 1977]).

[AL12] COVERING FOR LINEAR GRAMMARS (*)

INSTANCE: Two linear context-free grammars $G_1 = (N_1, \Sigma, \Pi_1, S_1)$ and $G_2 = (N_2, \Sigma, \Pi_2, S_2)$, where no production in such a grammar is allowed to have more than one nonterminal symbol on its right hand side.

QUESTION: Is there a function $h: P_1 \rightarrow P_2 \cup \{\lambda\}$ (where λ denotes the empty production) such that G_1 covers G_2 under h , i.e., such that for all strings $w \in \Sigma^*$ (1) if w is derivable from S_1 under the sequence of productions $p_1, p_2, \dots, p_n$, then w is derivable from S_2 under the sequence $h(p_1), h(p_2), \dots, h(p_n)$, and (2) if w is derivable from S_2 under the sequence of productions $q_1, q_2, \dots, q_n$ from Π_2 , then there exists a sequence of productions $p_1, p_2, \dots, p_m$ that is a derivation of w in G_1 such that $h(p_1), h(p_2), \dots, h(p_m)$ equals $q_1, q_2, \dots, q_n$?

Reference: [Hunt, Rosenkrantz, and Szymanski, 1976a], [Hunt, Rosenkrantz, and Szymanski, 1976b]. Transformation from REGULAR EXPRESSION NON-UNIVERSALITY. The second reference proves membership in PSPACE.

Comment: PSPACE-complete, even for “regular” grammars. Undecidable for ar-

bitrary context-free grammars. See [Hunt and Rosenkrantz, 1977] for related results.

[AL13] STRUCTURAL INEQUIVALENCE FOR LINEAR GRAMMARS (*)

INSTANCE: Two linear context-free grammars $G_1 = (N_1, \Sigma, \Pi_1, S_1)$ and $G_2 = (N_2, \Sigma, \Pi_2, S_2)$.

QUESTION: Are G_1 and G_2 “structurally inequivalent,” i.e., do the parenthesized grammars obtained from G_1 and G_2 by replacing each production $A \rightarrow w$ by $A \rightarrow (w)$ (where “(” and “)” are new terminal symbols) generate different languages?

Reference: [Hunt, Rosenkrantz, and Szymanski, 1976a]. Transformation from REGULAR EXPRESSION NON-UNIVERSALITY.

Comment: PSPACE-complete, even if G_1 and G_2 are regular and $|\Sigma| = 2$. NP-complete if $|\Sigma| = 1$. For arbitrary context-free grammars, problem is decidable but not known to be in PSPACE.

[AL14] REGULAR GRAMMAR INEQUIVALENCE (*)

INSTANCE: Regular grammars $G_1 = (N_1, \Sigma, \Pi_1, S_1)$ and $G_2 = (N_2, \Sigma, \Pi_2, S_2)$, where a regular grammar is a context-free grammar in which each production has the form $A \rightarrow aB$ or $A \rightarrow a$ with $A, B \in N$ and $a \in \Sigma$.

QUESTION: Do G_1 and G_2 generate different languages?

Reference: [Chomsky and Miller, 1958]. Transformation from FINITE STATE AUTOMATON INEQUIVALENCE.

Comment: PSPACE-complete, even if $|\Sigma| = 2$ and G_2 is a fixed grammar generating Σ^* (REGULAR GRAMMAR NON-UNIVERSALITY). The general problem is NP-complete if $|\Sigma| = 1$ or if both grammars generate finite languages (a property that can be checked in polynomial time, e.g., see [Hopcroft and Ullman, 1969]). If G_1 is allowed to be an arbitrary linear grammar and G_2 is a fixed grammar generating Σ^* (LINEAR GRAMMAR NON-UNIVERSALITY), the problem is undecidable [Hunt, Rosenkrantz, and Szymanski, 1976a].

[AL15] NON-LR(K) CONTEXT-FREE GRAMMAR

INSTANCE: Context-free grammar G , positive integer K written in unary notation.

QUESTION: Is G not an $LR(K)$ grammar (see reference for definition)?

Reference: [Hunt, Szymanski, and Ullman, 1975]. Generic transformation.

Comment: Solvable in polynomial time for any fixed K . If K is written in binary (as in our standard encodings), then the problem is complete for NEXP-TIME and hence intractable. Determining whether there exists an integer K such that G is an $LR(K)$ grammar is undecidable [Hunt and Szymanski, 1976a]. The same results hold if “ $LR(K)$ ” is replaced by “ $LL(K)$,” “ $LC(K)$,” “ $SLR(K)$,” or any one of a number of other properties (see above references). However, in the case of $LL(K)$, if it is known that there is some K' for which G is $LR(K')$, then one can decide whether there exists a K for which G is $LL(K)$ in polynomial time [Hunt and Szymanski, 1978].

[AL16] ETOL GRAMMAR NON-EMPTINESS (*)

INSTANCE: An ETOL grammar $G = (N, \Sigma, \{\delta_1, \delta_2, \dots, \delta_n\}, S)$, where N is a finite set of "nonterminal" symbols, Σ is a finite set of "terminal" symbols, $S \in N$ is the "initial" symbol, and each δ_i is a "table" of productions that take symbols in $N \cup \Sigma$ to strings in $(N \cup \Sigma)^*$ (at each step of a derivation, every symbol in the current string is replaced according to some production from a particular chosen table, e.g., see [Herman and Rozenberg, 1975]).

QUESTION: Is the language generated by G non-empty?

Reference: [Jones and Skyum, 1976]. Transformation from REGULAR EXPRESSION NON-UNIVERSALITY.

Comment: PSPACE-complete, even if G is an " ϵ -free EDTOL grammar"; NP-complete if G contains just one table and exactly one production for each symbol (G is an "EDOL grammar"). The same results hold for the question of whether G generates an infinite language. These problems are solvable in polynomial time for context-free grammars, but undecidable for context-sensitive grammars, e.g., see [Hopcroft and Ullman, 1969].

[AL17] CONTEXT-FREE PROGRAMMED LANGUAGE MEMBERSHIP

INSTANCE: An ϵ -free, context-free programmed grammar $G = (N, \Sigma, \Pi, S)$ and a string $x \in \Sigma^*$. (In such a grammar, the productions in Π are of the form $A \rightarrow w(T)(F)$, where $A \in N$, $w \in (N \cup \Sigma)^* - \epsilon$, and T and F are subsets of Π indicating where the next production to be applied must be chosen from, depending on whether the last production chosen was applicable or not; see [Rosenkrantz, 1969].)

QUESTION: Is x in the language generated by G ?

Reference: [Shamir and Beeri, 1974]. Transformation from 3SAT. The ϵ -free property ensures membership in NP.

Comment: Remains NP-complete even if all productions $A \rightarrow w(T)(F)$ in Π are required to have $T = F$ [van Leeuwen, 1975]. If $T = F = \Pi$ for all productions in Π and if productions to the empty string ϵ are permitted, we obtain the membership problem for context-free languages, which can be solved in polynomial time, e.g., see [Hopcroft and Ullman, 1969].

[AL18] QUASI-REAL-TIME LANGUAGE MEMBERSHIP

INSTANCE: Context-free grammars G_1, G_2 , and G_3 having the same terminal alphabet Σ , a second finite alphabet Γ , a function $f: \Sigma \rightarrow \Gamma$, and a string $w \in \Gamma^*$.

QUESTION: Is w in the "quasi-real-time language" determined by G_1, G_2, G_3 , and h , i.e., the language $L = h(L_1 \cap L_2 \cap L_3)$ consisting of all strings from Γ^* of the form $h(x_1)h(x_2) \dots h(x_k)$ such that $x_1x_2 \dots x_k \in (L_1 \cap L_2 \cap L_3)$, where L_1, L_2 , and L_3 are the languages generated by G_1, G_2 , and G_3 respectively?

Reference: [Hunt, 1973b], [Greibach, 1973b]. Transformation from 3SAT.

Comment: Solvable in polynomial time if h is one-to-one (by standard context-free parsing techniques, e.g., see [Hopcroft and Ullman, 1969]). The problem remains NP-complete if $L_3 = \Sigma^*$, i.e., $L = h(L_1 \cap L_2)$ [Greibach, 1973a], but is polynomially solvable if both L_2 and L_3 equal Σ^* , i.e., $L = h(L_1)$.

[AL19] ETOL LANGUAGE MEMBERSHIP (*)

INSTANCE: An ETOL grammar $G = (N, \Sigma, \{\delta_1, \delta_2, \dots, \delta_n\}, S)$ and a string $w \in \Sigma^*$.

QUESTION: Is w in the language generated by G ?

Reference: [van Leeuwen, 1975]. Transformation from 3SAT.

Comment: PSPACE-complete, even if G is an " ϵ -free EDTOL grammar" [Jones and Skyum, 1976]. If G is fixed, the problem is in NP and there exist particular grammars for which it is NP-complete, even if G is a "TOL grammar" (has no nonterminals) and is ϵ -free [Van Leeuwen, 1975]. The problem is solvable in polynomial time for fixed G if G is an "EDTOL grammar" [Jones and Skyum, 1977] or if G is an "EOL grammar" (has only one table) [Opatrny and Culik, 1975].

[AL20] CONTEXT-SENSITIVE LANGUAGE MEMBERSHIP (*)

INSTANCE: Context-sensitive grammar $G = (N, \Sigma, \Pi, S)$ and a string $w \in \Sigma^*$. (In a context-sensitive grammar, each production has the form $x \rightarrow y$ where x and y are nonempty strings over $N \cup \Sigma$ and $|y| \geq |x|$).

QUESTION: Is w in the language generated by G ?

Reference: [Kuroda, 1964]. Transformation from LINEAR BOUNDED AUTOMATON ACCEPTANCE.

Comment: PSPACE-complete, even for deterministic context-sensitive grammars. Moreover, there exist fixed context-sensitive grammars for which the problem is PSPACE-complete, and a fixed "linear time" context-sensitive grammar for which the problem is NP-complete [Book, 1978]. (For any fixed linear time context-sensitive grammar the problem is in NP.)

[AL21] TREE TRANSDUCER LANGUAGE MEMBERSHIP (*)

INSTANCE: A "top-down finite-state tree transducer" M with output alphabet Γ , a context-free grammar G , and a string $w \in \Gamma^*$ (see references for detailed definitions).

QUESTION: Is w in the "yield" of the "surface set" determined by M and G ?

Reference: [Reiss, 1977a]. Generic transformation.

Comment: PSPACE-complete. Problem is in NP for fixed M and G , and there exist particular choices for M and G for which the problem is NP-complete [Rounds, 1973]. The general problem is solvable in polynomial time if M is required to be "linear", while for fixed M the problem is solvable in polynomial time if M is "deterministic" [Reiss, 1977b].

A11 PROGRAM OPTIMIZATION

A11.1 CODE GENERATION

[PO1] REGISTER SUFFICIENCY

INSTANCE: Directed acyclic graph $G = (V, A)$, positive integer K .

QUESTION: Is there a computation for G that uses K or fewer registers, i.e., an ordering $v_1, v_2, \dots, v_n$ of the vertices in V , where $n = |V|$, and a sequence $S_0, S_1, \dots, S_n$ of subsets of V , each satisfying $|S_i| \leq K$, such that S_0 is empty, S_n contains all vertices with in-degree 0 in G , and, for $1 \leq i \leq n$, $v_i \in S_i$, $S_i - \{v_i\} \subseteq S_{i-1}$, and S_{i-1} contains all vertices u for which $(v_i, u) \in A$?

Reference: [Sethi, 1975]. Transformation from 3SAT.

Comment: Remains NP-complete even if all vertices of G have out-degree 2 or less. The variant in which "recomputation" is allowed (i.e., we ask for sequences $v_1, v_2, \dots, v_m$ and $S_0, S_1, \dots, S_m$, where no a priori bound is placed on m and the vertex sequence can contain repeated vertices, but all other properties stated above must hold) is NP-hard and is not known to be in NP.

[PO2] FEASIBLE REGISTER ASSIGNMENT

INSTANCE: Directed acyclic graph $G = (V, A)$, positive integer K , and a register assignment $f: V \rightarrow \{R_1, R_2, \dots, R_k\}$.

QUESTION: Is there a computation for G using the given register assignment, i.e., an ordering $v_1, v_2, \dots, v_n$ of V and a sequence $S_0, S_1, \dots, S_p$ of subsets of V that satisfies all the properties given in REGISTER SUFFICIENCY and that in addition satisfies, for $1 \leq j \leq K$ and $1 \leq i \leq n$, there is at most one vertex $u \in S_i$ for which $f(u) = R_j$?

Reference: [Sethi, 1975]. Transformation from 3SAT.

Comment: Remains NP-complete even if all vertices of G have out-degree 2 or less.

[PO3] REGISTER SUFFICIENCY FOR LOOPS

INSTANCE: Set V of loop variables, a loop length $N \in \mathbb{Z}^+$, for each variable $v \in V$ a start time $s(v) \in \mathbb{Z}_0^+$ and a duration $l(v) \in \mathbb{Z}^+$, and a positive integer K .

QUESTION: Can the loop variables be safely stored in K registers, i.e., is their an assignment $f: V \rightarrow \{1, 2, \dots, K\}$ such that if $f(v) = f(u)$ for some $u \neq v \in V$, then $s(u) \leq s(v)$ implies $s(u) + l(u) \leq s(v)$ and $s(v) + l(v) \pmod{N} \leq s(u)$?

Reference: [Garey, Johnson, Miller, and Papadimitriou, 1978]. Transformation from permutation generation.

Comment: Solvable in polynomial time for any fixed K .

[PO4] CODE GENERATION ON A ONE-REGISTER MACHINE

INSTANCE: Directed acyclic graph $G = (V, A)$ in which no vertex has out-degree larger than 2, and a positive integer K .

QUESTION: Is there a program with K or fewer instructions for computing all the root vertices of G (i.e., those with in-degree 0) on a one-register machine, starting with all the leaves of G (i.e., those with out-degree 0) in memory and using only

LOAD, STORE, and OP instructions? (A LOAD instruction copies a specified vertex into the register. A STORE instruction copies the vertex in the register into memory. A new vertex v can be computed by an OP instruction if the vertex u in the register is such that $(v, u) \in A$ and, if there is another vertex u' such that $(v, u') \in A$, then u' is in memory. Execution of the OP instruction replaces u by v in the register. The computation of a new vertex is not completed until it is copied into memory by a STORE instruction.)

Reference: [Bruno and Sethi, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete even if all vertices having in-degree larger than one have arcs only to leaves of G [Aho, Johnson, and Ullman, 1977a]. Solvable in polynomial time if G is a directed forest [Sethi and Ullman, 1970].

[PO5] CODE GENERATION WITH UNLIMITED REGISTERS

INSTANCE: Directed acyclic graph $G = (V, A)$ in which no vertex has out-degree larger than 2, partition of A into disjoint sets L and R such that two arcs leaving the same vertex always belong to different sets, and a positive integer K .

QUESTION: Is there a program with K or fewer instructions for computing all the root vertices of G , starting with all the leaves of G stored in registers and using only instructions of the form " $r_i \leftarrow r_j$ " or " $r_i \leftarrow r_j \text{ op } r_j$," $i, j \in \mathbb{Z}^+$, where a vertex v with out-degree 2 and outgoing arcs $(v, u) \in L$ and $(v, w) \in R$ can be computed only by an instruction $r_i \leftarrow r_i \text{ op } r_j$ when r_i contains u and r_j contains w ?

Reference: [Aho, Johnson, and Ullman, 1977a]. Transformation from FEEDBACK VERTEX SET.

Comment: Remains NP-complete even if only leaves of G have in-degree exceeding 1. The "commutative" variant in which instructions of the form " $r_i \leftarrow r_j \text{ op } r_i$ " are also allowed is NP-complete [Aho, Johnson, and Ullman, 1977b]. Both problems can be solved in polynomial time if G is a forest or if 3-address instructions " $r_i \leftarrow r_j \text{ op } r_k$ " are allowed [Aho, Johnson, and Ullman, 1977a].

[PO6] CODE GENERATION FOR PARALLEL ASSIGNMENTS

INSTANCE: Set $V = \{v_1, v_2, \dots, v_n\}$ of variables, set $A = \{A_1, A_2, \dots, A_n\}$ of assignments, each A_i of the form " $v_i \leftarrow \text{op}(B_i)$ " for some subset $B_i \subseteq V$, and a positive integer K .

QUESTION: Is there an ordering $v_{\pi(1)}, v_{\pi(2)}, \dots, v_{\pi(n)}$ of V such that there are at most K values of i , $1 \leq i \leq n$, for which $v_{\pi(i)} \in B_{\pi(j)}$ for some $j > i$?

Reference: [Sethi, 1973]. Transformation from FEEDBACK VERTEX SET.

Comment: Remains NP-complete even if each B_i satisfies $|B_i| \leq 2$.

[PO7] CODE GENERATION WITH ADDRESS EXPRESSIONS

INSTANCE: Sequence $I = (I_1, I_2, \dots, I_n)$ of instructions, for each $I_i \in I$ an expression $g(I_i)$ of the form " I_j ," " $I_j + k$," " $I_j - k$," or " k " where $I_j \in I$ and $k \in \mathbb{Z}^+$, and positive integers B, C , and M .

QUESTION: Can the instructions in I be stored as one- and two-byte instructions so that the total memory required is at most M , i.e., is there a one-to-one function $f: I \rightarrow \{1, 2, \dots, M\}$ such that $f(I_i) < f(I_j)$ whenever $i < j$ and such that, if $h(I_i)$ is defined to be $f(I_j)$, $f(I_i) \pm k$, or k depending on whether $g(I_i)$ is I_j , $I_j \pm k$, or

k , then for each i , $1 \leq i \leq n$, either $-C < f(I_i) - h(I_i) < B$ or $f(I_i) + 1$ is not in the range of f ?

Reference: [Szymanski, 1978]. Transformation from 3SAT.

Comment: Remains NP-complete for certain fixed values of B and C , e.g., 128 and 127 (much smaller values also are possible). Solvable in polynomial time if no "pathological" expressions occur (see reference for details).

[PO8] CODE GENERATION WITH UNFIXED VARIABLE LOCATIONS

INSTANCE: Sequence $I = (I_1, I_2, \dots, I_n)$ of instructions, finite set V of variables, assignment $g: I \rightarrow I \cup V$, positive integers B and M .

QUESTION: Can the instructions in I be stored as one- and two-byte instructions and the variables stored among them so that the total memory required is at most M , i.e., is there a one-to-one function $f: I \cup V \rightarrow \{1, 2, \dots, M\}$ such that $f(I_i) < f(I_j)$ whenever $i < j$ and such that, for $1 \leq i \leq n$, either $|f(I_i) - f(g(I_i))| < B$ or $f(I_i) + 1$ is not in the range of f ?

Reference: [Robertson, 1977]. Transformation from 3SAT.

Comment: Remains NP-complete even for certain fixed values of B , e.g., $B = 31$. Solvable in polynomial time if V is empty.

[PO9] ENSEMBLE COMPUTATION

INSTANCE: Collection C of subsets of a finite set A , positive integer J .

QUESTION: Is there a sequence $S = (z_1 \leftarrow x_1 \cup y_1, z_2 \leftarrow x_2 \cup y_2, \dots, z_j \leftarrow x_j \cup y_j)$ of $j \leq J$ union operations, where each x_i and y_i is either $\{a\}$ for some $a \in A$ or z_k for some $k < i$, such that x_i and y_i are disjoint, $1 \leq i \leq j$, and such that for every subset $c \in C$ there exists some z_i , $1 \leq i \leq j$, that is identical to c ?

Reference: [Garey and Johnson, —]. Transformation from VERTEX COVER (see Section 3.2.2).

Comment: Remains NP-complete even if each $c \in C$ satisfies $|c| \leq 3$. The analogous problem in which x_i and y_i need not be disjoint for $1 \leq i \leq j$ is also NP-complete under the same restriction.

[PO10] MICROCODE BIT OPTIMIZATION

INSTANCE: Finite set A of "micro-commands," collection $C = \{C_1, C_2, \dots, C_m\}$ of subsets of A called "micro-instructions," and a positive integer K .

QUESTION: Is there a K -bit instruction format for the given micro-instructions, i.e., is there a partition of A into disjoint subsets $A_1, A_2, \dots, A_n$ such that no pair A_i, C_j have more than one element in common and such that $\sum_{i=1}^n [\log_2(|A_i| + 1)] \leq K$?

Reference: [Robertson, 1978]. Transformation from 3DM.

A11.2 PROGRAMS AND SCHEMES

[PO11] INEQUIVALENCE OF PROGRAMS WITH ARRAYS

INSTANCE: Finite sets X , Θ , and R of variables, operators, and array variables, two programs P_1 and P_2 made up of "operate" ($x_0 \leftarrow \theta x_1 x_2 \dots x_r$), "update" ($\alpha[x_i] \leftarrow x_j$), and "select" ($x_i \leftarrow \alpha[x_j]$) commands, where each $x_i \in X$, $\theta \in \Theta$, r is the "arity" of θ , and $\alpha \in R$, a finite value set V , and an interpretation of each operator $\theta \in \Theta$ as a specific function from V^r to V .

QUESTION: Is there an initial assignment of a value from V to each variable in X such that the two programs yield different final values for some variable in X (see reference for details on the execution of such programs)?

Reference: [Downey and Sethi, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete even if there are no operate commands and only one array variable. Solvable in polynomial time if there are no update commands, or no select commands, or no array variables.

[PO12] INEQUIVALENCE OF PROGRAMS WITH ASSIGNMENTS

INSTANCE: Finite set X of variables, two programs P_1 and P_2 , each a sequence of assignments of the form " $x_0 \leftarrow \text{if } x_1 = x_2 \text{ then } x_3 \text{ else } x_4$ " where the x_i are in X , and a value set V .

QUESTION: Is there an initial assignment of a value from V to each variable in X such that the two programs yield different final values for some variable in X (see reference for details on the execution of such programs)?

Reference: [Downey and Sethi, 1976]. Transformation from 3SAT.

Comment: Remains NP-complete for $V = \{0, 1\}$. This problem can be embedded in many inequivalence problems for simple programs, thus rendering them NP-hard [Downey and Sethi, 1976], [van Leeuwen, 1977].

[PO13] INEQUIVALENCE OF FINITE MEMORY PROGRAMS (*)

INSTANCE: Finite set X of variables, finite alphabet Σ , two programs P_1 and P_2 , each a sequence $I_1, I_2, \dots, I_m$ of instructions (not necessarily of the same length m) of the form "read x_i ," "write v_j ," " $x_i \leftarrow v_j$," "if $v_j = v_k$ goto I_l ," "accept," or "halt," where each $x_i \in X$, each $v_j \in X \cup \Sigma \cup \{\$, \}$, and I_m is either "halt" or "accept."

QUESTION: Is there a string $w \in \Sigma^*$ such that the two programs yield different outputs for input w (see reference for details on the execution of such programs)?

Reference: [Jones and Muchnik, 1977]. Transformation from LINEAR BOUNDED AUTOMATON ACCEPTANCE.

Comment: PSPACE-complete, even if P_2 is a fixed program with no write instructions and hence no output. See reference for a number of other special cases and variants that are PSPACE-complete or harder.

[PO14] INEQUIVALENCE OF LOOP PROGRAMS WITHOUT NESTING

INSTANCE: Finite set X of variables, subset $Y \subseteq X$ of input variables, specified output variable x_0 , two loop programs P_1 and P_2 without nested loops, i.e., sequences of instructions of the form " $x \leftarrow y$," " $x \leftarrow x + 1$," " $x \leftarrow 0$," "loop x ," and

“end,” where $x, y \in X$ and each *loop* instruction is followed by a corresponding *end* instruction before any further *loop* instructions occur.

QUESTION: Is there an initial assignment $f: Y \rightarrow Z^+$ of integers to the input variables such that the two programs halt with different values for the output variable x_0 (see references for details on the execution of such programs)?

Reference: [Constable, Hunt, and Sahni, 1974], [Tsichritzis, 1970]. Transformation from 3SAT. The second reference proves membership in NP.

Comment: Problem becomes undecidable if nested loops are allowed (even for nesting of only depth 2) [Meyer and Ritchie, 1967]. Solvable in polynomial time if *loop* statements are not allowed [Tsichritzis, 1970]. See [Hunt, 1977] for a generalization of the main result.

[PO15] INEQUIVALENCE OF SIMPLE FUNCTIONS

INSTANCE: Finite set X of variables, two expressions f and g over X , each being a composition of functions from the collection “ $s(x) = x+1$,” “ $p(x) = \max\{x-1, 0\}$,” “ $plus(x, y) = x+y$,” “ $div(x, t) = \lfloor x/t \rfloor$,” “ $mod(x, t) = x - t \cdot \lfloor x/t \rfloor$,” “ $w(x, y) = \text{if } y=0 \text{ then } x \text{ else } 0$,” and “ $select_i^t(x_1, x_2, \dots, x_n) = x_i$ ” where $x, y, x_i \in X$, $i, n, t \in \mathbb{Z}^+$, and $i \leq n$.

QUESTION: Is there an assignment of non-negative integer values to the variables in X for which the values of f and g differ?

Reference: [Tsichritzis, 1970]. Transformation from INEQUIVALENCE OF LOOP PROGRAMS WITHOUT NESTING.

Comment: Remains NP-complete even if f and g are defined only in terms of $w(x, y)$, in terms of *plus* and *mod*, or in terms of *plus* and *p* [Lieberherr, 1977]. Variants in which f and g are defined in terms of *plus* and “ $sub1(x) = \max\{0, 1-x\}$,” or solely in terms of “ $minus(x, y) = \max\{0, x-y\}$,” (where in both cases $x, y \in X \cup \mathbb{Z}^+$) are also NP-complete [Constable, Hunt, and Sahni, 1974].

[PO16] STRONG INEQUIVALENCE OF IANOV SCHEMES

INSTANCE: Finite sets F and P of function and predicate symbols, single variable x , and two Ianov schemes over F, P , and x , each a sequence $I_1, I_2, \dots, I_m$ of instructions of the form “ $x \leftarrow f(x)$,” “if $p(x)$ then goto I_j else goto I_k ,” and “halt,” where $f \in F$ and $p \in P$.

QUESTION: Are the two given Ianov schemes not strongly equivalent, i.e., is there a domain set D , an interpretation of each $f \in F$ as a function $f: D \rightarrow D$, an interpretation of each $p \in P$ as a function $p: D \rightarrow \{T, F\}$, and an initial value $x_0 \in D$ for x , such that either both schemes halt with different final values for x or one halts and the other doesn't?

Reference: [Constable, Hunt, and Sahni, 1974], [Rutledge, 1964]. Transformation from 3SAT. Membership in NP follows from the second reference.

Comment: Remains NP-complete even if neither program contains any loops and P_2 is the trivial program that leaves the value of x unchanged. The strong inequality problem for Ianov schemes with two variables is undecidable, even if $|F| = |P| = 1$ [Luckham, Park, and Paterson, 1970]. See references, [Hunt, 1978], and [Hunt and Szymanski, 1976b] for analogous results for other properties, such as “weak equivalence,” “divergence,” “halting,” etc. Strong equivalence can be

tested in polynomial time for Ianov schemes that are “strongly free,” i.e., in which at least one function application occurs between every two successive predicate tests [Constable, Hunt, and Sahni, 1974]. Strong equivalence is open for “free” Ianov schemes.

[PO17] STRONG INEQUIVALENCE FOR MONADIC RECURSION SCHEMES

INSTANCE: Finite sets F and P of function and predicate symbols, set G of “defined” function symbols disjoint from F , specified symbol $f_0 \in G$, and two linear monadic recursion schemes S_1 and S_2 , each consisting of a defining statement for each $f \in G$ of the form “ $f x = \text{if } p x \text{ then } \alpha x \text{ else } \beta x$ ” where $p \in P$, $\alpha, \beta \in (F \cup G)^*$, and α and β each contain at most one occurrence of a symbol from G .

QUESTION: Is there a domain set D , an interpretation of each $f \in F$ as a function $f: D \rightarrow D$, an interpretation of each $p \in P$ as a function $p: D \rightarrow \{T, F\}$, and an initial value $x_0 \in D$ such that, as defined by the recursion schemes S_1 and S_2 , either the two values for $f_0(x_0)$ differ or one is defined and the other isn't?

Reference: [Constable, Hunt, and Sahni, 1974]. Transformation from STRONG INEQUIVALENCE OF IANOV SCHEMES. Proof of membership in NP is non-trivial.

Comment: Remains NP-complete even if one scheme trivially sets $f_0(x) = x$ and the other is “right linear,” i.e., each α and β only contains a defined symbol as its rightmost character. See reference for other NP-completeness and NP-hardness results concerning linear monadic recursion schemes.

[PO18] NON-CONTAINMENT FOR FREE B-SCHEMES

INSTANCE: Two free B -schemes S_1 and S_2 , where a free B -scheme is a rooted, directed acyclic graph $G = (V, A)$, all of whose vertices have out-degree 0 (leaves) or 2 (tests), with the two arcs leaving a test vertex labeled L and R respectively, together with a set B of Boolean variable symbols and a label $l(v) \in B$ for each test vertex, such that no two test vertices on the same directed path get the same label, and a set F of function symbols along with a label $l(v) \in F \cup \{\Omega\}$ for each leaf in V .

QUESTION: Is S_1 not “contained” in S_2 , i.e., is there an assignment $t: B_1 \cup B_2 \rightarrow \{L, R\}$ such that if the paths from the roots of G_1 and G_2 to leaf vertices determined by always leaving a test vertex v by the arc labeled $t(l(v))$ terminate at leaves labeled f_1 and f_2 respectively, then $f_1 \neq f_2$ and $f_1 \neq \Omega$?

Reference: [Fortune, Hopcroft, and Schmidt, 1977]. Transformation from 3SAT.

Comment: The “strong inequality” problem for free B -schemes (same as above, only all that we now require is that $f_1 \neq f_2$) is open, but can be solved in polynomial time if one of S_1 and S_2 is an “ordered” B -scheme. The open version is Turing equivalent to the strong inequality problem for free Ianov schemes (see STRONG INEQUIVALENCE OF IANOV SCHEMES).

[PO19] NON-FREEDOM FOR LOOP-FREE PROGRAM SCHEMES

INSTANCE: Finite sets F and P of function and predicate symbols, set X of variables, and a loop-free monadic program scheme S over F, P , and X , where such a scheme consists of a sequence $I_1, I_2, \dots, I_m$ of instructions of the form " $x \leftarrow f(y)$," " $\text{if } p(x) \text{ then goto } I_j \text{ else goto } I_k$," and " halt ," with $x \in X$, $f \in F$, and $p \in P$, and must be such that no directed cycles occur in the corresponding "flow graph."

QUESTION: Is S non-free, i.e., is there a directed path in the flow graph for S that can never be followed in any computation, no matter what the interpretation of the functions and predicates in F and P and the initial values for the variables in X ?

Reference: [Constable, Hunt, and Sahni, 1974]. Transformation from 3SAT.

Comment: Remains NP-complete for $|X|=2$. If $|X|=1$, the problem is solvable in polynomial time. If loops are allowed and $|X|$ is arbitrary, the problem is undecidable [Paterson, 1967].

[PO20] PROGRAMS WITH FORMALLY RECURSIVE PROCEDURES

INSTANCE: Finite set A of procedure identifiers, ALGOL-like program P involving procedure declarations and procedure calls for procedures in A (see reference for details).

QUESTION: Is any of the procedures in A "formally recursive" in program P (in the sense of [Langmaack, 1973])?

Reference: [Winklmann, 1977]. Transformation from 3SAT.

Comment: See reference for related results concerning deciding whether P has the "formal most-recent property," "formal parameter correctness," the "formal macro-property," and others.

A12 MISCELLANEOUS

[MS1] BETWEENNESS

INSTANCE: Finite set A , collection C of ordered triples (a, b, c) of distinct elements from A .

QUESTION: Is there a one-to-one function $f: A \rightarrow \{1, 2, \dots, |A|\}$ such that for each $(a, b, c) \in C$, we have either $f(a) < f(b) < f(c)$ or $f(c) < f(b) < f(a)$?

Reference: [Opatrný, 1978]. Transformation from SET SPLITTING.

[MS2] CYCLIC ORDERING

INSTANCE: Finite set A , collection C of ordered triples (a, b, c) of distinct elements from A .

QUESTION: Is there a one-to-one function $f: A \rightarrow \{1, 2, \dots, |A|\}$ such that, for each $(a, b, c) \in A$, we have either $f(a) < f(b) < f(c)$ or $f(b) < f(c) < f(a)$ or $f(c) < f(a) < f(b)$?

Reference: [Galil and Megiddo, 1977]. Transformation from 3SAT.

[MS3] NON-LIVENESS OF FREE CHOICE PETRI NETS

INSTANCE: Petri net $P = (n, M_0, T)$, where $n \in \mathbb{Z}^+$, M_0 is an n -tuple of non-negative integers, and T is a set of transitions $\langle a, b \rangle$ in which both a and b are n -tuples of 0's and 1's, such that P has the "free choice" property, i.e., for each $\langle a, b \rangle \in T$, either a contains exactly one 1 or in every other transition $\langle c, d \rangle \in T$, c has a 0 in every position where a has a 1.

QUESTION: Is P not "live," i.e., is there a transition $t \in T$ and a sequence σ of transitions from T such that, for every sequence τ of transitions from T , the sequence $\sigma \tau t$ is not "fireable" at M_0 , where $\langle a_1, b_1 \rangle \langle a_2, b_2 \rangle \dots \langle a_m, b_m \rangle$ is fireable at M_0 if and only if the sequence $M_0, M_1, \dots, M_{2m}$ in which $M_{2i+1} = M_{2i} - a_i$ and $M_{2i+2} = M_{2i+1} + b_i$, $0 \leq i < m$, contains no vector with a negative component?

Reference: [Jones, Landweber, and Lien, 1977]. Transformation from 3SAT. Proof of membership in NP is nontrivial and is based on a result of [Hack, 1972].

[MS4] REACHABILITY FOR 1-CONSERVATIVE PETRI NETS (*)

INSTANCE: Petri net $P = (n, M_0, T)$ that is "1-conservative," i.e., for each $\langle a, b \rangle \in T$, a and b have the same number of 1's, and an n -tuple M of nonnegative integers.

QUESTION: Is M reachable from M_0 in P , i.e., is there a sequence $\langle a_1, b_1 \rangle \langle a_2, b_2 \rangle \dots \langle a_m, b_m \rangle$ of transitions from T such that the sequence $M_0, M_1, \dots, M_{2m}$ obtained as in the preceding problem contains no vector with a negative component and satisfies $M_{2m} = M$?

Reference: [Jones, Landweber, and Lien, 1977]. Transformation from LINEAR BOUNDED AUTOMATON ACCEPTANCE.

Comment: PSPACE-complete, even if P is also a free choice Petri net. Problem is not known to be decidable for arbitrary Petri nets, but is known to require at least exponential space [Lipton, 1975]. Analogous results hold for the "coverability" problem: Is there an M' having each of its components no smaller than the

corresponding component of M such that M' is reachable from M_0 ? The related “ K -boundedness” problem (given P and an integer K , is there no vector that exceeds K in every component that is reachable from M_0 ?) is PSPACE-complete for arbitrary Petri nets, as well as for 1-conservative free choice Petri nets. See [Jones, Landweber, and Lien, 1977] and [Hunt, 1977] for additional details and related results.

[MS5] FINITE FUNCTION GENERATION (*)

INSTANCE: Finite set A , a collection F of functions $f: A \rightarrow A$, and a specified function $h: A \rightarrow A$.

QUESTION: Can h be generated from the functions in F by composition?

Reference: [Kozen, 1977d]. Transformation from FINITE STATE AUTOMATA INTERSECTION.

Comment: PSPACE-complete.

[MS6] PERMUTATION GENERATION

INSTANCE: Permutation σ of the integers $\{1, 2, \dots, N\}$, and a sequence $S_1, S_2, \dots, S_m$ of subsets of $\{1, 2, \dots, N\}$.

QUESTION: Can σ be expressed as a composition $\sigma = \sigma_1 \sigma_2 \dots \sigma_m$, where for each i , $1 \leq i \leq m$, σ_i is a permutation of $\{1, 2, \dots, N\}$ that leaves all elements in $\{1, 2, \dots, N\} - S_i$ fixed?

Reference: [Garey, Johnson, Miller, Papadimitriou, 1978]. Transformation from X3C.

Comment: Solvable in polynomial time for any fixed N .

[MS7] DECODING OF LINEAR CODES

INSTANCE: An $n \times m$ matrix $A = (a_{ij})$ of 0's and 1's, a vector $\bar{y} = (y_1, y_2, \dots, y_m)$ of 0's and 1's, and a positive integer K .

QUESTION: Is there a 0-1 vector $\bar{x} = (x_1, x_2, \dots, x_n)$ with no more than K 1's such that, for $1 \leq j \leq m$, $\sum_{i=1}^n x_i \cdot a_{ij} \equiv y_j \pmod{2}$?

Reference: [Berlekamp, McEliece, and van Tilborg, 1978]. Transformation from 3DM.

Comment: If $\bar{y}$ is the all zero vector, and hence we are asking for a “codeword” of Hamming weight K or less, the problem is open. The variant in which we ask for an $\bar{x}$ with exactly K 1's is NP-complete, even for fixed $\bar{y} = (0, 0, \dots, 0)$.

[MS8] SHAPLEY-SHUBIK VOTING POWER

INSTANCE: Ordered set $V = \{v_1, v_2, \dots, v_n\}$ of voters, number of votes $w_i \in \mathbb{Z}^+$ for each $v_i \in V$, and a quota $q \in \mathbb{Z}^+$.

QUESTION: Does voter v_1 have non-zero “Shapley-Shubik voting power,” where the voting power $p(v)$ for a voter $v \in V$ is defined to be $(1/n!)$ times the number of permutations π of $\{1, 2, \dots, n\}$ for which $\sum_{i=1}^{j-1} w_{\pi(i)} < q$, $\sum_{i=1}^j w_{\pi(i)} \geq q$, and $v = v_{\pi(j)}$?

Reference: [Garey and Johnson, —]. Transformation from PARTITION. The definition of voting power is from [Shapley and Shubik, 1954].

Comment: Determining the value of the Shapley-Shubik voting power for a given voter is #P-complete, but that value can be computed in pseudo-polynomial time by dynamic programming.

[MS9] CLUSTERING

INSTANCE: Finite set X , a distance $d(x, y) \in \mathbb{Z}_0^+$ for each pair $x, y \in X$, and two positive integers K and B .

QUESTION: Is there a partition of X into disjoint sets $X_1, X_2, \dots, X_k$ such that, for $1 \leq i \leq k$ and all pairs $x, y \in X_i$, $d(x, y) \leq B$?

Reference: [Brucker, 1978]. Transformation from GRAPH 3-COLORABILITY.

Comment: Remains NP-complete even for fixed $K=3$ and all distances in $\{0, 1\}$. Solvable in polynomial time for $K=2$. Variants in which we ask that the sum, over all X_i , of $\max\{d(x, y) : x, y \in X_i\}$ or of $\sum_{x, y \in X_i} d(x, y)$ be at most B , are similarly NP-complete (with the last one NP-complete even for $K=2$).

[MS10] RANDOMIZATION TEST FOR MATCHED PAIRS (*)

INSTANCE: Sequence $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ of ordered pairs of integers, nonnegative integer K .

QUESTION: Are there at least K subsets $S \subseteq \{1, 2, \dots, n\}$ for which

$$\sum_{i \in S} |x_i - y_i| \leq \sum_{x_i > y_i} (x_i - y_i) \quad ?$$

Reference: [Shamos, 1976]. Transformation from PARTITION.

Comment: Not known to be in NP. The corresponding enumeration problem is #P-complete, but solvable in pseudo-polynomial time by dynamic programming.

[MS11] MAXIMUM LIKELIHOOD RANKING

INSTANCE: An $n \times n$ matrix $A = (a_{ij})$ with integer entries satisfying $a_{ij} + a_{ji} = 0$ for all $i, j \in \{1, 2, \dots, n\}$, positive integer B .

QUESTION: Is there a matrix $B = (b_{ij})$ obtained from A by simultaneous row and column permutations such that

$$\sum_{1 \leq i < j \leq n} \min\{b_{ij}, 0\} \geq -B \quad ?$$

Reference: [Rafsky, 1977]. Transformation from FEEDBACK ARC SET.

Comment: NP-complete in the strong sense.

[MS12] MATRIX DOMINATION

INSTANCE: An $n \times n$ matrix M with entries from $\{0, 1\}$, and a positive integer K .

QUESTION: Is there a set of K or fewer non-zero entries in M that dominate all others, i.e., a subset $C \subseteq \{1, 2, \dots, n\} \times \{1, 2, \dots, n\}$ with $|C| \leq K$ such that $M_{ij} = 1$ for all $(i, j) \in C$ and such that, whenever $M_{ij} = 1$, there exists an $(i', j') \in C$ for which either $i = i'$ or $j = j'$?

Reference: [Yannakakis and Gavril, 1978]. Transformation from MINIMUM MAXIMAL MATCHING.

Comment: Remains NP-complete even if M is upper triangular.

[MS13] MATRIX COVER

INSTANCE: An $n \times n$ matrix $A = (a_{ij})$ with nonnegative integer entries, and an integer K .

QUESTION: Is there a function $f: \{1, 2, \dots, n\} \rightarrow \{-1, +1\}$ such that

$$\sum_{1 \leq i, j \leq n} a_{ij} \cdot f(i) \cdot f(j) \geq K?$$

Reference: [Garey and Johnson, —]. Transformation from MAX CUT.

Comment: NP-complete in the strong sense and remains so if A is required to be positive definite.

[MS14] SIMPLY DEVIATED DISJUNCTION

INSTANCE: Collection M of m -tuples $(M_i[1], M_i[2], \dots, M_i[m])$, $1 \leq i \leq n$, with each $M_i[j]$ being either 0, 1, or x .

QUESTION: Is there a partition of $\{1, 2, \dots, m\}$ into disjoint sets I, J and an assignment $f: \{1, 2, \dots, m\} \rightarrow \{0, 1\}$ such that, if Φ is the formula $\bigvee_{j \in I} (M[j] = f(j))$

and Ψ is the formula $\bigvee_{j \in J} (M[j] = f(j))$, then Φ and Ψ are simply deviated in M , i.e., the number of $M_i \in M$ such that Φ and Ψ are both true for M_i times the number of $M_i \in M$ such that Φ and Ψ are both false for M_i is larger than the number of $M_i \in M$ such that Φ is true and Ψ is false for M_i times the number of $M_i \in M$ such that Φ is false and Ψ is true for M_i ? (The definition of “simply deviated” is from [Havránek, 1975].)

Reference: [Pudlák and Springsteel, 1975]. Transformation from MAX CUT.

Comment: Remains NP-complete even if $f(j) = 1$ for $1 \leq j \leq m$. Solvable in polynomial time if each $M_i[j]$ is either 0 or 1. See reference for additional related results.

[MS15] DECISION TREE

INSTANCE: Finite set X of objects, collection $T = \{T_1, T_2, \dots, T_m\}$ of binary tests $T_j: X \rightarrow \{0, 1\}$, positive integer K .

QUESTION: Is there a decision tree for X using the tests in T that has total external path length K or less? (A decision tree is a binary tree in which each non-leaf vertex is labelled by a test from T , each leaf is labelled by an object from X , the edge from a non-leaf vertex to its left son is labelled 0 and the one to its right son is labelled 1, and, if $T_{i_1}, O_{i_1}, T_{i_2}, O_{i_2}, \dots, T_{i_k}, O_{i_k}$ is the sequence of vertex and edge labels on the path from the root to a leaf labelled by $x \in X$, then x is the unique object for which $T_{i_j}(x) = O_{i_j}$ for all j , $1 \leq j \leq k$. The total external path length of such a tree is the sum, over all leaves, of the number of edges on the path from the root to that leaf.)

Reference: [Hyafil and Rivest, 1976]. Transformation from X3C.

Comment: Remains NP-complete even if for each $T_i \in T$ there are at most three distinct objects $x \in X$ for which $T_i(x) = 1$.

[MS16] MINIMUM WEIGHT AND/OR GRAPH SOLUTION

INSTANCE: Directed acyclic graph $G = (V, A)$ with a single vertex $s \in V$ having in-degree 0, assignment $f(v) \in \{and, or\}$ for each $v \in V$ having nonzero out-degree weight $w(a) \in \mathbb{Z}^+$ for each $a \in A$, and a positive integer K .

QUESTION: Is there a subgraph $G' = (V', A')$ of G such that $s \in V'$, such that if $v \in V'$ and $f(v) = and$ then all arcs leaving v in A belong to A' , such that if $v \in V$ and $f(v) = or$ then at least one of the arcs leaving v in A belongs to A' , and such that the sum of the weights of the arcs in A' does not exceed K ?

Reference: [Sahni, 1974]. Transformation from X3C.

Comment: Remains NP-complete even if $w(a) = 1$ for all $a \in A$ [Garey and Johnson, —]. The general problem is solvable in polynomial time for rooted directed trees by dynamic programming.

[MS17] FAULT DETECTION IN LOGIC CIRCUITS

INSTANCE: Directed acyclic graph $G = (V, A)$ with a single vertex $v^* \in V$ having out-degree 0, an assignment $f: (V - \{v^*\}) \rightarrow \{I, and, or, not\}$ such that $f(v) = I$ implies v has in-degree 0, $f(v) = not$ implies v has in-degree 1, and $f(v) = and$ or $f(v) = or$ implies v has in-degree 2, and a subset $V' \subseteq V$.

QUESTION: Can all single faults occurring at vertices of V' be detected by input output experiments, i.e., regarding G as a logic circuit with input vertices I , output vertex v^* , and logic gates for the functions “and,” “or,” and “not” at the specified vertices, is there for each $v \in V'$ and $x \in \{T, F\}$ an assignment of a value to each vertex in I of a value in $\{T, F\}$ such that the output of the circuit for those input values differs from the output of the same circuit with the output of the gate v “stuck-at” x ?

Reference: [Ibarra and Sahni, 1975]. Transformation from 3SAT.

Comment: Remains NP-complete even if $V' = V$ or if V' contains just a single vertex v with $f(v) = I$.

[MS18] FAULT DETECTION IN DIRECTED GRAPHS

INSTANCE: Directed acyclic graph $G = (V, A)$, with $I \subseteq V$ denoting those vertices with in-degree 0 and $O \subseteq V$ denoting those vertices with out-degree 0, and positive integer K .

QUESTION: Is there a “test set” of size K or less that can detect every “single fault” in G , i.e., is there a subset $T \subseteq I \times O$ with $|T| \leq K$ such that, for every $v \in V$, there exists some pair $(u_1, u_2) \in T$ such that v is on a directed path from u_1 to u_2 in G ?

Reference: [Ibaraki, Kameda, and Toida, 1977]. Transformation from X3C.

Comment: Remains NP-complete even if $|O| = 1$. Variant in which we ask that T be sufficient for “locating” any single fault, i.e., that for every pair $v, v' \in V$ there some $(u_1, u_2) \in T$ such that v is on a directed path from u_1 to u_2 but v' is on no such path, is also NP-complete for $|O| = 1$. Both problems can be solved in polynomial time if $K \geq |I| \cdot |O|$.

[MS19] FAULT DETECTION WITH TEST POINTS

INSTANCE: Directed acyclic graph $G=(V,A)$ having exactly one vertex $s \in V$ with in-degree 0 and exactly one vertex $t \in V$ with out-degree 0, and a positive integer K .

QUESTION: Can all "single faults" in G be located by attaching K or fewer "test points" to arcs in A , i.e., is there a subset $A' \subseteq A$ with $|A'| \leq K$ such that the test set

$$T = \left\{ \{s\} \cup \{u_1: (u_1, u_2) \in A'\} \right\} \times \left\{ \{t\} \cup \{u_2: (u_1, u_2) \in A'\} \right\}$$

has the property that, for each pair $v, v' \in V - \{s, t\}$, there is some $(u_1, u_2) \in T$ such that v is on a directed path from u_1 to u_2 but v' is on no such path?

Reference: [Ibaraki, Kameda, and Toida, 1977]. Transformation from X3C.

Comment: Variants in which we are asked to locate all single faults by using K or fewer "test connections" or "blocking gates" are also NP-complete, as are the problems of finding a test set T with $|T| \leq K$ in the presence of a fixed set of "test points," "test connections," or "blocking gates." See reference for more details.

A13 OPEN PROBLEMS**[OPEN1] GRAPH ISOMORPHISM**

INSTANCE: Two graphs $G_1=(V_1, E_1)$ and $G_2=(V_2, E_2)$.

QUESTION: Are G_1 and G_2 isomorphic, i.e., is there a one-to-one onto function $f: V_1 \rightarrow V_2$ such that $\{u, v\} \in E_1$ if and only if $\{f(u), f(v)\} \in E_2$?

Comment: The problem remains open even if G_1 and G_2 are restricted to regular graphs, bipartite graphs, line graphs, comparability graphs, chordal graphs, or undirected path graphs (i.e., intersection graphs for the set of paths in an undirected tree), [Hirschberg and Edelman, 1973], [Babai, 1976], [Booth, 1978], [Miller, 1977]. Solvable in polynomial time for planar graphs (e.g., see [Hopcroft and Wong, 1974]) and for interval graphs [Booth and Lueker, 1975]. The problem is in $NP \cap co-NP$ for "arc transitive" cubic graphs [Miller, 1977]. Problems polynomially equivalent to GRAPH ISOMORPHISM include directed graph isomorphism, context-free grammar isomorphism [Hunt and Rosenkrantz, 1977], finitely presented algebra isomorphism [Kozen, 1977a], semi-group isomorphism [Booth, 1978], conjunctive query isomorphism [Chandra and Merlin, 1977], the problem of determining whether a graph is isomorphic to its complement [Colbourne and Colbourne, 1978], and the problem of counting the number of distinct isomorphisms between G_1 and G_2 [Babai, 1977], [Mathon, 1978]. A special case of CLIQUE that is polynomially equivalent to GRAPH ISOMORPHISM is described in [Kozen, 1978]. Isomorphism problems that are perhaps easier than GRAPH ISOMORPHISM include group isomorphism and Latin square isomorphism both of which can be solved in time $O(n^{\log n})$ [Miller, 1978].

[OPEN2] SUBGRAPH HOMEOMORPHISM (FOR A FIXED GRAPH H)

INSTANCE: Graph $G=(V, E)$.

QUESTION: Does G contain a subgraph homeomorphic to H , i.e., a subgraph $G'=(V', E')$ that can be converted to a graph isomorphic to H by repeatedly removing any vertex of degree 2 and adding the edge joining its two neighbors?

Comment: If H is allowed to vary as part of the instance, the problem is NP-complete, since it contains HAMILTONIAN CIRCUIT as a special case. Solvable in polynomial time for certain fixed graphs H , such as a triangle. Is there any fixed graph H for which this problem is NP-complete? If not, is there any fixed graph $H=(U, F)$ for which the following related problem is NP-complete: Given a graph $G=(V, E)$ and a one-to-one function $f: U \rightarrow V$, is there a subgraph $G'=(V', E')$ that can be converted to a graph isomorphic to H as above and such that f provides the required isomorphism? This latter problem is also known to be NP-complete if H is allowed to vary as part of the instance, since it contains DISJOINT CONNECTING PATHS as a special case. Several complicated polynomial time algorithms have been found for particular values of H , such as a triangle [LaPaugh and Rivest, 1978] and two disjoint edges [Shiloach, 1978]. Is there any fixed integer K such that the problem is NP-complete for H consisting of K disjoint edges?

[OPEN3] GRAPH GENUS

INSTANCE: Graph $G = (V, E)$ and a non-negative integer K .

QUESTION: Can G be embedded on a surface of genus K such that no two edges cross one another?

Comment: Solvable in polynomial time for $K=0$, i.e., if the question is whether G is planar (e.g., see [Hopcroft and Tarjan, 1974]). A polynomial time algorithm for $K=1$ and cubic graphs is announced in [Filotti, 1978], and, in [Reif, 1978b], polynomial time algorithms for arbitrary graphs and *any* fixed value of K are presented. In addition, for some restricted classes of graphs, such as cliques, cubes, and complete bipartite graphs, simple closed formulas for the genus have been derived (e.g., see [Harary, 1969]). Although the problem for general G and K is open, the closely related GENUS EXTENSION problem (given G , K , and an embedding of a subgraph of G into a surface of genus K , can the embedding be extended to one for all of G ?) is NP-complete [Reif, 1978a]. Open problems for other generalizations of planarity include "Does G have crossing number K or less, i.e., can G be embedded in the plane with K or fewer pairs of edges crossing one another?" and "Does G have thickness K or less, i.e., can E be partitioned into K disjoint sets $E_1, E_2, \dots, E_k$ such that each subgraph $G_i = (V, E_i)$ is planar?" Related NP-complete problems include PLANAR SUBGRAPH and PLANAR INDUCED SUBGRAPH (see INDUCED SUBGRAPH WITH PROPERTY II).

[OPEN4] CHORDAL GRAPH COMPLETION

INSTANCE: Graph $G = (V, E)$ and a positive integer K .

QUESTION: Is there a superset E' containing E of unordered pairs of vertices from V that satisfies $|E' - E| \leq K$ and such that $G' = (V, E')$ is chordal, i.e., such that for every simple cycle of more than 3 vertices in G' , there is some edge in E' that is not involved in the cycle but that joins two vertices in the cycle?

Comment: This problem is equivalent to the undirected version of DIRECTED ELIMINATION ORDERING and corresponds to the problem of minimizing "fill-in" when applying Gaussian elimination to symmetric matrices (e.g., see [Rose, Tarjan, and Lueker, 1976]). See [Gavril, 1974b] for an alternative characterization of chordal graphs.

[OPEN5] CHROMATIC INDEX

INSTANCE: Graph $G = (V, E)$ and a positive integer K .

QUESTION: Does G have chromatic index K or less, i.e., can E be partitioned into disjoint sets $E_1, E_2, \dots, E_k$, with $k \leq K$, such that, for $1 \leq i \leq k$, no two edges in E_i share a common endpoint in G ?

Comment: By Vizing's Theorem (e.g., see [Berge, 1973]), the chromatic index for G is either h or $h+1$, where h is the maximum vertex degree in G , so the above question may be restated as "Given G , is the chromatic index of G equal to its maximum vertex degree?" The answer is always "yes" for bipartite graphs (e.g., see [Berge, 1973]), and there exist polynomial time algorithms for constructing the desired partition in this case (e.g., see [Gabow, 1976]). A particular case that is open is that for cubic graphs (i.e., regular of degree 3), in which case the problem can be restated as "Given G , can the vertices of G be covered by disjoint simple cycles, each involving an even number of vertices?" This latter problem is one of a number of open problems involving parity. Another such problem is: "Given a col-

lection C of subsets of a finite set X , is there a nonempty subcollection $C' \subseteq C$ such that each $x \in X$ belongs to an even number (possibly 0) of sets in C' ?" which is equivalent to the open problem mentioned in the comments for DECODING OF LINEAR CODES.

[OPEN6] SPANNING TREE PARITY PROBLEM

INSTANCE: Graph $G = (V, E)$ and a partition of E into disjoint 2-element sets $E_1, E_2, \dots, E_m$.

QUESTION: Is there a spanning tree $T = (V, E')$ for G such that for each E_i , $1 \leq i \leq m$, either $E_i \subseteq E'$ or $E_i \cap E' = \emptyset$?

Comment: This is a typical special case of the general "matroid parity problem" (e.g., see [Lawler, 1976a]), which is itself a generalization of graph matching and the two matroid intersection problem, both of which can be solved in polynomial time (assuming, in the matroid case, that there exist polynomial time algorithms for telling whether a set is an independent set of the matroids in question). The related "multiple choice spanning tree" problem, where at *most* one member of each E_i can be in E' , is a special case of the two matroid intersection problem and hence can be solved in polynomial time (see MULTIPLE CHOICE BRANCHING).

[OPEN7] PARTIAL ORDER DIMENSION

INSTANCE: Directed acyclic graph $G = (V, A)$ that is transitive, i.e., whenever $(u, v) \in A$ and $(v, w) \in A$, then $(u, w) \in A$, and a positive integer $K \leq |V|^2$.

QUESTION: Does there exist a collection of $k \leq K$ linear orderings of V such that $(u, v) \in A$ if and only if u is less than v in each of the orderings?

Comment: Solvable in polynomial time for $K=2$ [Lawler, 1976d]. Open for arbitrary K and for any fixed $K \geq 3$.

[OPEN8] PRECEDENCE CONSTRAINED 3-PROCESSOR SCHEDULING

INSTANCE: Set T of unit length tasks, partial order $<$ on T , and a deadline $D \in \mathbb{Z}^+$.

QUESTION: Can T be scheduled on 3 processors so as to satisfy the precedence constraints and meet the overall deadline D , i.e., is there a schedule $\sigma: T \rightarrow \{0, 1, \dots, D-1\}$ such that $t < t'$ implies $\sigma(t) < \sigma(t')$ and such that for each integer i , $0 \leq i \leq D-1$, there are at most 3 tasks $t \in T$ for which $\sigma(t) = i$?

Comment: The corresponding problem for 2 processors is solvable in polynomial time [Fujii, Kasami, and Ninomiya, 1969], [Coffman and Graham, 1972], even with individual task deadlines and release times [Garey and Johnson, 1977b]. If the number of processors is allowed to vary as part of the instance, the problem is NP-complete [Ullman, 1975]. See PRECEDENCE CONSTRAINED SCHEDULING for more details. Is there any fixed value of K for which the K -processor version of the above problem is NP-complete?

[OPEN9] LINEAR PROGRAMMING

INSTANCE: Integer-valued vectors $V_i = (v_i[1], v_i[2], \dots, v_i[n])$, $1 \leq i \leq m$, $D = (d_1, d_2, \dots, d_m)$, and $C = (c_1, c_2, \dots, c_n)$, and an integer B .

QUESTION: Is there a vector $X = (x_1, x_2, \dots, x_n)$ of rational numbers such that, for $1 \leq i \leq m$, $V_i \cdot X \leq d_i$ and such that $C \cdot X \geq B$?

Comment: The problem is in $NP \cap co-NP$ (membership in $co-NP$ follows from the fundamental duality theorem of linear programming). For any fixed value of m , the problem can be solved in polynomial time. There are many variants of LINEAR PROGRAMMING that are polynomially equivalent to it (e.g., see [Reiss and Dobkin, 1976]). One such variant is that in which we drop the vector C from the instance and drop the requirement that $C \cdot X \geq B$ (see also [Papadimitriou, 1978b]). Examples of network flow problems polynomially equivalent to LINEAR PROGRAMMING are mentioned in the comments for UNDIRECTED FLOW WITH LOWER BOUNDS, PATH CONSTRAINED NETWORK FLOW, and TWO COMMODITY INTEGRAL FLOW. A generalization of LINEAR PROGRAMMING that is also open though still in NP is the “linear complementarity” problem (see [Murty, 1976]).

[OPEN10] TOTAL UNIMODULARITY

INSTANCE: An $m \times n$ matrix M with entries from the set $\{-1, 0, 1\}$.

QUESTION: Is M *not* totally unimodular, i.e., is there a square submatrix of M whose determinant is *not* in the set $\{-1, 0, 1\}$?

Comment: The problem remains open even if all entries in M are from $\{0, 1\}$. The significance of totally unimodular matrices for integer programming is discussed, for example, in [Lawler, 1976] and [Garfinkel and Nemhauser, 1972].

[OPEN11] COMPOSITE NUMBER

INSTANCE: Positive integer N .

QUESTION: Are there positive integers $m, n > 1$ such that $N = m \cdot n$?

Comment: The problem is in $NP \cap co-NP$ [Pratt, 1975]. Although no polynomial time algorithm is known, there is an algorithm for the problem that runs in polynomial time if the “Extended Riemann Hypothesis” holds [Miller, 1976]. However, there is no such algorithm known for determining the prime factors of N , and this latter problem may be harder than the basic decision problem. Of course, all these problems are easily solved in pseudo-polynomial time.

[OPEN12] MINIMUM LENGTH TRIANGULATION

INSTANCE: Collection $C = \{(a_i, b_i) : 1 \leq i \leq n\}$ of pairs of integers, giving the coordinates of n points in the plane, and a positive integer B .

QUESTION: Is there a triangulation of the set of points represented by C that has total “discrete-Euclidean” length B or less? Here a triangulation is a collection of non-intersecting line segments, each joining two points in C , that divides the interior of the convex hull into triangular regions. The discrete-Euclidean length of a line segment joining (a_i, b_i) and (a_j, b_j) is given by $\lceil [(a_i - a_j)^2 + (b_i - b_j)^2]^{1/2} \rceil$, and the total length of a triangulation is the sum of the lengths of its constituent line segments.

Comment: The analogous problem for the rectilinear metric is also open. [Lloyd, 1977] presents counterexamples to a number of conjectured polynomial time algorithms for the problem and proves that the related CONSTRAINED TRIANGULATION problem is NP-complete.

Symbol Index

$O(f(n))$	order of $f(n)$	6
Z	the set of all integers	207
Z^+	the set of all positive integers	18
Z_0^+	the set of all non-negative integers	206
$\lceil \alpha \rceil$	least integer not less than α	11
$\lfloor \alpha \rfloor$	greatest integer not exceeding α	135
$ x $	length of string x	21
$ S $	cardinality of set S	11
α	polynomially transforms to	34
α_γ	gamma-reduces	159
α_T	Turing-reduces	113
α_c^{NP}	conjunctive truth-table reduces	164
α_{LOG}	log-space reduces	178
ϵ	the empty string	19
Σ^*	set of all finite strings over alphabet Σ	19
Σ^+	$\Sigma^* - \{\epsilon\}$	110
Π	a problem	4
Π^c	complement of decision problem Π	114
$\Delta_k^P, \Pi_k^P, \Sigma_k^P$	classes in the polynomial hierarchy	162
$\#P$	number-P	168
$\leq$	partial order	73